MATLAB Apps for Chemical Reaction Engineering

Source Code — Michaël Grimsberg (MathWorks FileExchange #155172)

Batch reactor 1

batch1.mlapp

```
classdef batch1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Batchreactor1UIFigure           matlab.ui.Figure
        Label_3                         matlab.ui.control.Label
        Label_2                         matlab.ui.control.Label
        Label                           matlab.ui.control.Label
        AdiabaticCheckBox               matlab.ui.control.CheckBox
        hSlider                         matlab.ui.control.Slider
        heightanddiametercmSliderLabel  matlab.ui.control.Label
        TaSlider                        matlab.ui.control.Slider
        heattransferfluidtemperatureKSliderLabel  matlab.ui.control.Label
        rubrik                          matlab.ui.control.Label
        UIAxes                          matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.Batchreactor1UIFigure,'off');
                if app.AdiabaticCheckBox.Value
                    U=0;
                    app.TaSlider.Visible=0;
                    app.heattransferfluidtemperatureKSliderLabel.Visible=0;
                else
                    app.TaSlider.Visible=1;
                    app.heattransferfluidtemperatureKSliderLabel.Visible=1;
                    U=1.8;
                end
                h=app.hSlider.Value;
                Ta=app.TaSlider.Value;
                d=h;
                Cp=125;
                T0=323;
                tf=60;
                DeltaH=-25000;
                Ca0=0.015;
                V=pi/4*d^2*h;
                Na0=Ca0*V;
                A=pi/4*d^2+(pi*d)*h;
                [t,y]=ode23s(@modell,[0 tf],[Na0 T0]);
                plot(app.UIAxes,t,y(:,2),'k','LineWidth',1.5)
                app.UIAxes.YLim=[300 550];
                app.UIAxes.XLim=[0 tf];
                app.Label.Text=sprintf('area to volume = %5.3f cm\x207b\x00b9',A/V);
                app.Label_2.Text= sprintf('    surface area = %4.0f dm\x00b2',A/100);
                app.Label_3.Text= sprintf('             volume = %4.0f dm\x00b3',V/1000);
                toogleUIC(app.Batchreactor1UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

            function dy = modell(~,y)
                dy=zeros(size(y));
                nA=y(1);T=y(2);
                k=3000*exp(-30000/8.314/T);
                ra=-k*nA;
```

```
            dy(1)=ra;
            dy(2)=(DeltaH*ra+U*A*(Ta-T))/Na0/Cp;
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        draw(app);
    end

    % Value changed function: AdiabaticCheckBox, TaSlider, hSlider
    function AdiabaticCheckBoxValueChanged(app, event)
        draw(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Batchreactor1UIFigure and hide until all components are created
        app.Batchreactor1UIFigure = uifigure('Visible', 'off');
        app.Batchreactor1UIFigure.Color = [0.94 0.94 0.94];
        app.Batchreactor1UIFigure.Position = [100 100 737 604];
        app.Batchreactor1UIFigure.Name = 'Batch reactor 1';
        app.Batchreactor1UIFigure.Scrollable = 'on';

        % Create UIAxes
        app.UIAxes = uiaxes(app.Batchreactor1UIFigure);
        xlabel(app.UIAxes, 'Time [min]')
        ylabel(app.UIAxes, 'Reactor tempoerature [K]')
        app.UIAxes.FontName = 'Helvetica';
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.XGrid = 'on';
        app.UIAxes.YGrid = 'on';
        app.UIAxes.Position = [74 33 418 260];

        % Create rubrik
        app.rubrik = uilabel(app.Batchreactor1UIFigure);
        app.rubrik.FontSize = 24;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [205 552 332 30];
        app.rubrik.Text = {'Scale-Up of a Batch Reactor'; ''};

        % Create heattransferfluidtemperatureKSliderLabel
        app.heattransferfluidtemperatureKSliderLabel = uilabel(app.Batchreactor1UIFigure);
        app.heattransferfluidtemperatureKSliderLabel.HorizontalAlignment = 'right';
        app.heattransferfluidtemperatureKSliderLabel.Position = [73 462 184 22];
        app.heattransferfluidtemperatureKSliderLabel.Text = 'heat transfer fluid temperature [K]';

        % Create TaSlider
        app.TaSlider = uislider(app.Batchreactor1UIFigure);
        app.TaSlider.Limits = [300 350];
        app.TaSlider.ValueChangedFcn = createCallbackFcn(app, @AdiabaticCheckBoxValueChanged, true);
        app.TaSlider.MinorTicks = [];
        app.TaSlider.Position = [278 471 150 3];
        app.TaSlider.Value = 325;

        % Create heightanddiametercmSliderLabel
        app.heightanddiametercmSliderLabel = uilabel(app.Batchreactor1UIFigure);
        app.heightanddiametercmSliderLabel.HorizontalAlignment = 'right';
        app.heightanddiametercmSliderLabel.Position = [119 385 138 22];
```

```
            app.heightanddiametercmSliderLabel.Text = 'height and diameter [cm]';

            % Create hSlider
            app.hSlider = uislider(app.Batchreactor1UIFigure);
            app.hSlider.Limits = [10 200];
            app.hSlider.MajorTicks = [10 50 100 150 200];
            app.hSlider.ValueChangedFcn = createCallbackFcn(app, @AdiabaticCheckBoxValueChanged, true);
            app.hSlider.MinorTicks = [25 75 125 175];
            app.hSlider.Position = [278 394 150 3];
            app.hSlider.Value = 80;

            % Create AdiabaticCheckBox
            app.AdiabaticCheckBox = uicheckbox(app.Batchreactor1UIFigure);
            app.AdiabaticCheckBox.ValueChangedFcn = createCallbackFcn(app, @AdiabaticCheckBoxValueChanged, true);
            app.AdiabaticCheckBox.Text = 'Adiabatic';
            app.AdiabaticCheckBox.Position = [36 499 71 22];

            % Create Label
            app.Label = uilabel(app.Batchreactor1UIFigure);
            app.Label.FontSize = 14;
            app.Label.FontWeight = 'bold';
            app.Label.Position = [479 384 249 22];
            app.Label.Text = '';

            % Create Label_2
            app.Label_2 = uilabel(app.Batchreactor1UIFigure);
            app.Label_2.FontSize = 14;
            app.Label_2.FontWeight = 'bold';
            app.Label_2.Position = [483 334 241 22];
            app.Label_2.Text = '';

            % Create Label_3
            app.Label_3 = uilabel(app.Batchreactor1UIFigure);
            app.Label_3.FontSize = 14;
            app.Label_3.FontWeight = 'bold';
            app.Label_3.Position = [480 313 249 22];
            app.Label_3.Text = '';

            % Show the figure after all components are created
            app.Batchreactor1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = batch1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Batchreactor1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Batchreactor1UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Batch reactor 2

batch2.mlapp

```
classdef batch2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Batchreactor2UIFigure       matlab.ui.Figure
        nuSpinner                   matlab.ui.control.Spinner
        productstoichiometriccoeffficientLabel  matlab.ui.control.Label
        nSpinner                    matlab.ui.control.Spinner
        reactionorderSpinnerLabel   matlab.ui.control.Label
        xe                          matlab.ui.control.Slider
        finalconversionSliderLabel  matlab.ui.control.Label
        rubtik                      matlab.ui.control.Label
        UIAxes                      matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function displayOrder(app)
            app.UIAxes.Title.String=sprintf('A\\rightarrow%3.1fB\t\t\t r=1.0C_A^{%3.1f} [mol/(s m^3)]',app.nuSpinner.Value,app.n
        end


        function simulera(app)
            try
                toogleUIC(app.Batchreactor2UIFigure,'off');
                V0=0.15;
                k=1;
                T=140;
                R=0.73;
                P0=1.5;
                order=app.nSpinner.Value;
                stoich=app.nuSpinner.Value;
                finalX=app.xe.Value;
                NA0=P0*V0/R/T;
                CA0=NA0/V0;
                [t1,C]=ode45(@constV,[0 1000],[CA0 0],odeset('Events',@stopV));
                [t2,N]=ode45(@constP,[0 1000],[NA0 0],odeset('Events',@stopP));
                plot(app.UIAxes,...
                    t1,(CA0-C(:,1))./CA0,'-',...
                    t2,(NA0-N(:,1))./NA0,'--');
                legend(app.UIAxes,'Constant volume','Constant pressure','Location',"northwest");
                toogleUIC(app.Batchreactor2UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

            function dy = constV(~,C)
                dy=zeros(size(C));
                rate=k*C(1).^order;
                dy(1)=-rate;
                dy(2)=stoich*rate;
            end
            function [value,isterminal,direction]=stopV(~,C)
                value=(CA0-C(1))/CA0-finalX;
                isterminal=1;
                direction=1;
            end
            function dy=constP(~,N)
                dy=zeros(size(N));
                V=V0*sum(N)/NA0;
                Ca=N(1)/sum(N)*P0/R/T;
                rate=k*Ca.^order;
                dy(1)=-rate*V;
                dy(2)=stoich*rate*V;
            end
            function [value,isterminal,direction]=stopP(~,N)
```

```
            value=(NA0-N(1))/NA0-finalX;
            isterminal=1;
            direction=1;
        end

    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.UIAxes.TitleFontWeight='B';
        app.UIAxes.TitleFontSizeMultiplier=1.5;
        displayOrder(app);
        simulera(app);
    end

    % Callback function
    function nValueChanged(app, event)
        value=round(app.n.Value*2);
        app.n.Value=value/2;
        displayOrder(app);
        simulera(app);
    end

    % Callback function
    function nuValueChanged(app, event)
        value=round(app.nu.Value*2);
        app.nu.Value=value/2;
        displayOrder(app);
        simulera(app);
    end

    % Value changed function: xe
    function xeValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: nSpinner
    function nSpinnerValueChanged(app, event)
        displayOrder(app);
        simulera(app);
    end

    % Value changed function: nuSpinner
    function nuSpinnerValueChanged(app, event)
        displayOrder(app);
        simulera(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Batchreactor2UIFigure and hide until all components are created
        app.Batchreactor2UIFigure = uifigure('Visible', 'off');
        app.Batchreactor2UIFigure.Color = [0.94 0.94 0.94];
        app.Batchreactor2UIFigure.Position = [100 100 640 559];
        app.Batchreactor2UIFigure.Name = 'Batch reactor 2';

        % Create UIAxes
        app.UIAxes = uiaxes(app.Batchreactor2UIFigure);
        title(app.UIAxes, 'A \rightarrow B')
        xlabel(app.UIAxes, 'Time')
        ylabel(app.UIAxes, 'Conversion')
```

```
            app.UIAxes.PlotBoxAspectRatio = [1.92996108949416 1 1];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [46 32 543 340];

            % Create rubtik
            app.rubtik = uilabel(app.Batchreactor2UIFigure);
            app.rubtik.FontSize = 18;
            app.rubtik.FontWeight = 'bold';
            app.rubtik.Position = [94 518 508 22];
            app.rubtik.Text = 'Batch Reactors at Constant Volume or Constant Pressure';

            % Create finalconversionSliderLabel
            app.finalconversionSliderLabel = uilabel(app.Batchreactor2UIFigure);
            app.finalconversionSliderLabel.HorizontalAlignment = 'right';
            app.finalconversionSliderLabel.Position = [163 403 89 22];
            app.finalconversionSliderLabel.Text = 'final conversion';

            % Create xe
            app.xe = uislider(app.Batchreactor2UIFigure);
            app.xe.Limits = [0 0.8];
            app.xe.MajorTicks = [0 0.2 0.4 0.6 0.8 0.9999];
            app.xe.MajorTickLabels = {'0', '0.2', '0.4', '0.6', '0.8', '1.0'};
            app.xe.ValueChangedFcn = createCallbackFcn(app, @xeValueChanged, true);
            app.xe.MinorTicks = [];
            app.xe.Position = [283 413 165 3];
            app.xe.Value = 0.5;

            % Create reactionorderSpinnerLabel
            app.reactionorderSpinnerLabel = uilabel(app.Batchreactor2UIFigure);
            app.reactionorderSpinnerLabel.HorizontalAlignment = 'right';
            app.reactionorderSpinnerLabel.Position = [163 482 80 22];
            app.reactionorderSpinnerLabel.Text = 'reaction order';

            % Create nSpinner
            app.nSpinner = uispinner(app.Batchreactor2UIFigure);
            app.nSpinner.Step = 0.5;
            app.nSpinner.Limits = [0.5 2];
            app.nSpinner.ValueChangedFcn = createCallbackFcn(app, @nSpinnerValueChanged, true);
            app.nSpinner.Position = [357 482 100 22];
            app.nSpinner.Value = 1;

            % Create productstoichiometriccoeffficientLabel
            app.productstoichiometriccoeffficientLabel = uilabel(app.Batchreactor2UIFigure);
            app.productstoichiometriccoeffficientLabel.HorizontalAlignment = 'right';
            app.productstoichiometriccoeffficientLabel.Position = [163 447 182 22];
            app.productstoichiometriccoeffficientLabel.Text = 'product stoichiometric coeffficient';

            % Create nuSpinner
            app.nuSpinner = uispinner(app.Batchreactor2UIFigure);
            app.nuSpinner.Step = 0.5;
            app.nuSpinner.Limits = [0.5 4];
            app.nuSpinner.ValueChangedFcn = createCallbackFcn(app, @nuSpinnerValueChanged, true);
            app.nuSpinner.Position = [357 447 100 22];
            app.nuSpinner.Value = 1;

            % Show the figure after all components are created
            app.Batchreactor2UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = batch2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
```

```
            registerApp(app, app.Batchreactor2UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Batchreactor2UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Batch reactor 3

batch3.mlapp

```
classdef batch3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Batchreactor3UIFigure  matlab.ui.Figure
        k3Slider               matlab.ui.control.Slider
        k3SliderLabel          matlab.ui.control.Label
        k2Slider               matlab.ui.control.Slider
        k2SliderLabel          matlab.ui.control.Label
        k1Slider               matlab.ui.control.Slider
        k1SliderLabel          matlab.ui.control.Label
        rubrik                 matlab.ui.control.Label
        TabGroup               matlab.ui.container.TabGroup
        AllTab                 matlab.ui.container.Tab
        UIAxes                 matlab.ui.control.UIAxes
        ATab                   matlab.ui.container.Tab
        UIAxes_2               matlab.ui.control.UIAxes
        BTab                   matlab.ui.container.Tab
        UIAxes_3               matlab.ui.control.UIAxes
        CTab                   matlab.ui.container.Tab
        UIAxes_4               matlab.ui.control.UIAxes
        DTab                   matlab.ui.container.Tab
        UIAxes_5               matlab.ui.control.UIAxes
        ETab                   matlab.ui.container.Tab
        UIAxes_6               matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Batchreactor3UIFigure,'off');
                [t,y]=ode45(@diffekv,[0 5],[10 5 0 0 0]);
                plot(app.UIAxes,t,y);
                app.UIAxes.Title.Interpreter='latex';
                app.UIAxes.TitleFontSizeMultiplier=1.8;
                app.UIAxes.Title.String='$A + B \stackrel{k_1}{\longrightarrow} C\quad C \stackrel{k_2}{\longrightarrow} 2E\quad
                legend(app.UIAxes,'C_A','C_B','C_C','C_D','C_E',"Location","best");
                plot(app.UIAxes_2,t,y(:,1),'k');
                app.UIAxes_2.Title.Interpreter='latex';
                app.UIAxes_2.TitleFontSizeMultiplier=1.8;
                app.UIAxes_2.Title.String='$A + B \stackrel{k_1}{\longrightarrow} C\quad C \stackrel{k_2}{\longrightarrow} 2E\qu
                plot(app.UIAxes_3,t,y(:,2),'k');
                app.UIAxes_3.Title.Interpreter='latex';
                app.UIAxes_3.TitleFontSizeMultiplier=1.8;
                app.UIAxes_3.Title.String='$A + B \stackrel{k_1}{\longrightarrow} C\quad C \stackrel{k_2}{\longrightarrow} 2E\qu
                plot(app.UIAxes_4,t,y(:,3),'k');
                app.UIAxes_4.Title.Interpreter='latex';
                app.UIAxes_4.TitleFontSizeMultiplier=1.8;
                app.UIAxes_4.Title.String='$A + B \stackrel{k_1}{\longrightarrow} C\quad C \stackrel{k_2}{\longrightarrow} 2E\qu
                plot(app.UIAxes_5,t,y(:,4),'k');
                app.UIAxes_5.Title.Interpreter='latex';
                app.UIAxes_5.TitleFontSizeMultiplier=1.8;
                app.UIAxes_5.Title.String='$A + B \stackrel{k_1}{\longrightarrow} C\quad C \stackrel{k_2}{\longrightarrow} 2E\qu
                plot(app.UIAxes_6,t,y(:,5),'k');
                app.UIAxes_6.Title.Interpreter='latex';
                app.UIAxes_6.TitleFontSizeMultiplier=1.8;
                app.UIAxes_6.Title.String='$A + B \stackrel{k_1}{\longrightarrow} C\quad C \stackrel{k_2}{\longrightarrow} 2E\qu
                toogleUIC(app.Batchreactor3UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

            function dy = diffekv(~,y)
                dy=zeros(size(y));
                CA=y(1);
```

```
            CB=y(2);
            CC=y(3);
            r1=app.k1Slider.Value*CA*CB;
            r2=app.k2Slider.Value*CC;
            r3=app.k3Slider.Value*CA^2;
            dy(1)=-r1-2*r3;
            dy(2)=-r1;
            dy(3)=r1-r2;
            dy(4)=r3;
            dy(5)=2*r2;
        end

    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        simulera(app);
    end

    % Value changed function: k1Slider
    function k1SliderValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: k2Slider
    function k2SliderValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: k3Slider
    function k3SliderValueChanged(app, event)
        simulera(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Batchreactor3UIFigure and hide until all components are created
        app.Batchreactor3UIFigure = uifigure('Visible', 'off');
        app.Batchreactor3UIFigure.Color = [0.94 0.94 0.94];
        app.Batchreactor3UIFigure.Position = [100 100 738 583];
        app.Batchreactor3UIFigure.Name = 'Batch reactor 3';

        % Create TabGroup
        app.TabGroup = uitabgroup(app.Batchreactor3UIFigure);
        app.TabGroup.Position = [32 26 688 432];

        % Create AllTab
        app.AllTab = uitab(app.TabGroup);
        app.AllTab.Title = 'All';

        % Create UIAxes
        app.UIAxes = uiaxes(app.AllTab);
        title(app.UIAxes, 'Reactions')
        xlabel(app.UIAxes, 'Time')
        ylabel(app.UIAxes, 'Mole')
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.Position = [11 20 656 362];

        % Create ATab
```

```
app.ATab = uitab(app.TabGroup);
app.ATab.Title = 'A';

% Create UIAxes_2
app.UIAxes_2 = uiaxes(app.ATab);
title(app.UIAxes_2, 'Reactions')
xlabel(app.UIAxes_2, 'Time')
ylabel(app.UIAxes_2, 'Mole')
app.UIAxes_2.XTickLabelRotation = 0;
app.UIAxes_2.YTickLabelRotation = 0;
app.UIAxes_2.ZTickLabelRotation = 0;
app.UIAxes_2.Position = [11 20 656 362];

% Create BTab
app.BTab = uitab(app.TabGroup);
app.BTab.Title = 'B';

% Create UIAxes_3
app.UIAxes_3 = uiaxes(app.BTab);
title(app.UIAxes_3, 'Reactions')
xlabel(app.UIAxes_3, 'Time')
ylabel(app.UIAxes_3, 'Mole')
app.UIAxes_3.XTickLabelRotation = 0;
app.UIAxes_3.YTickLabelRotation = 0;
app.UIAxes_3.ZTickLabelRotation = 0;
app.UIAxes_3.Position = [11 20 656 362];

% Create CTab
app.CTab = uitab(app.TabGroup);
app.CTab.Title = 'C';

% Create UIAxes_4
app.UIAxes_4 = uiaxes(app.CTab);
title(app.UIAxes_4, 'Reactions')
xlabel(app.UIAxes_4, 'Time')
ylabel(app.UIAxes_4, 'Mole')
app.UIAxes_4.XTickLabelRotation = 0;
app.UIAxes_4.YTickLabelRotation = 0;
app.UIAxes_4.ZTickLabelRotation = 0;
app.UIAxes_4.Position = [11 20 656 362];

% Create DTab
app.DTab = uitab(app.TabGroup);
app.DTab.Title = 'D';

% Create UIAxes_5
app.UIAxes_5 = uiaxes(app.DTab);
title(app.UIAxes_5, 'Reactions')
xlabel(app.UIAxes_5, 'Time')
ylabel(app.UIAxes_5, 'Mole')
app.UIAxes_5.XTickLabelRotation = 0;
app.UIAxes_5.YTickLabelRotation = 0;
app.UIAxes_5.ZTickLabelRotation = 0;
app.UIAxes_5.Position = [11 20 656 362];

% Create ETab
app.ETab = uitab(app.TabGroup);
app.ETab.Title = 'E';

% Create UIAxes_6
app.UIAxes_6 = uiaxes(app.ETab);
title(app.UIAxes_6, 'Reactions')
xlabel(app.UIAxes_6, 'Time')
ylabel(app.UIAxes_6, 'Mole')
app.UIAxes_6.XTickLabelRotation = 0;
app.UIAxes_6.YTickLabelRotation = 0;
app.UIAxes_6.ZTickLabelRotation = 0;
app.UIAxes_6.Position = [11 20 656 362];

% Create rubrik
app.rubrik = uilabel(app.Batchreactor3UIFigure);
```

```
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [149 543 442 30];
            app.rubrik.Text = 'Batch reactor with Multiple Reactions';

            % Create k1SliderLabel
            app.k1SliderLabel = uilabel(app.Batchreactor3UIFigure);
            app.k1SliderLabel.HorizontalAlignment = 'right';
            app.k1SliderLabel.Position = [33 489 25 22];
            app.k1SliderLabel.Text = 'k1';

            % Create k1Slider
            app.k1Slider = uislider(app.Batchreactor3UIFigure);
            app.k1Slider.Limits = [0 1];
            app.k1Slider.ValueChangedFcn = createCallbackFcn(app, @k1SliderValueChanged, true);
            app.k1Slider.MinorTicks = [];
            app.k1Slider.Position = [79 498 150 3];
            app.k1Slider.Value = 0.4;

            % Create k2SliderLabel
            app.k2SliderLabel = uilabel(app.Batchreactor3UIFigure);
            app.k2SliderLabel.HorizontalAlignment = 'right';
            app.k2SliderLabel.Position = [275 489 25 22];
            app.k2SliderLabel.Text = 'k2';

            % Create k2Slider
            app.k2Slider = uislider(app.Batchreactor3UIFigure);
            app.k2Slider.Limits = [0 1];
            app.k2Slider.ValueChangedFcn = createCallbackFcn(app, @k2SliderValueChanged, true);
            app.k2Slider.MinorTicks = [];
            app.k2Slider.Position = [321 498 150 3];
            app.k2Slider.Value = 0.6;

            % Create k3SliderLabel
            app.k3SliderLabel = uilabel(app.Batchreactor3UIFigure);
            app.k3SliderLabel.HorizontalAlignment = 'right';
            app.k3SliderLabel.Position = [513 489 25 22];
            app.k3SliderLabel.Text = 'k3';

            % Create k3Slider
            app.k3Slider = uislider(app.Batchreactor3UIFigure);
            app.k3Slider.Limits = [0 1];
            app.k3Slider.ValueChangedFcn = createCallbackFcn(app, @k3SliderValueChanged, true);
            app.k3Slider.MinorTicks = [];
            app.k3Slider.Position = [559 498 150 3];
            app.k3Slider.Value = 0.3;

            % Show the figure after all components are created
            app.Batchreactor3UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = batch3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Batchreactor3UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end
```

```
        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Batchreactor3UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

CSTR 1

cstr1.mlapp

```
classdef cstr1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        CSTR1UIFigure               matlab.ui.Figure
        reaction                    matlab.ui.control.Label
        FA0                         matlab.ui.control.Label
        CBout                       matlab.ui.control.Label
        FB                          matlab.ui.control.Label
        CAout                       matlab.ui.control.Label
        FA                          matlab.ui.control.Label
        CB                          matlab.ui.control.Label
        CA                          matlab.ui.control.Label
        tau                         matlab.ui.control.Label
        alfa                        matlab.ui.control.DropDown
        reactionorderDropDownLabel  matlab.ui.control.Label
        kEdit                       matlab.ui.control.EditField
        k                           matlab.ui.control.Slider
        rateconstantSliderLabel     matlab.ui.control.Label
        ypsilonEdit                 matlab.ui.control.EditField
        ypsilon                     matlab.ui.control.Slider
        VolumetricflowrateLsLabel   matlab.ui.control.Label
        CA0Edit                     matlab.ui.control.EditField
        CA0                         matlab.ui.control.Slider
        CA0molLSliderLabel          matlab.ui.control.Label
        titel                       matlab.ui.control.Label
        Image                       matlab.ui.control.Image
    end


    methods (Access = private)

        function compute(app)
            try
                toogleUIC(app.CSTR1UIFigure,'off');
                theta=1000/app.ypsilon.Value;
                app.FA0.Text=sprintf('FA\x2080 = %5.2f mol/s',app.CA0.Value*app.ypsilon.Value);
                app.tau.Text=sprintf('\x03a4 = %4.0f s',theta);
                kTheta=theta*app.k.Value;
                if app.alfa.Value == '2'
                    cA=(sqrt(1+4*kTheta*app.CA0.Value)-1)/2/kTheta;
                else
                    cA=app.CA0.Value/(1+kTheta);
                end
                app.CA.Text=sprintf('CA = %4.2f mol/L',cA);
                app.CAout.Text=sprintf('CA = %4.2f mol/L',cA);
                app.CB.Text=sprintf('CB = %4.2f mol/L',app.CA0.Value-cA);
                app.CBout.Text=sprintf('CB = %4.2f mol/L',app.CA0.Value-cA);
                app.FA.Text=sprintf('FA = %5.2f mol/s',cA*app.ypsilon.Value);
                app.FB.Text=sprintf('FB = %5.2f mol/s',(app.CA0.Value-cA)*app.ypsilon.Value);
                toogleUIC(app.CSTR1UIFigure,'on');
           catch ME
                errordlg(ME.message);
            end

        end


        function texter(app)
            if app.alfa.Value == '2'
                app.reaction.Text=sprintf('A \x27f6 B\t\t r = %5.3fCA\x00b2 mol/(L\x00b7s)',app.k.Value);
            else
                app.reaction.Text=sprintf('A \x27f6 B\t\t r = %5.3fCA mol/(L\x00b7s)',app.k.Value);
            end
        end

        function results = between(~,value,item)
```

```
            results=0;
            if (str2double(value) >= min(item)) && (str2double(value) <= max(item))
                results=1;
            end
        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            app.CA0Edit.Value=num2str(app.CA0.Value);
            app.ypsilonEdit.Value=num2str(app.ypsilon.Value);
            app.kEdit.Value=num2str(app.k.Value);
            texter(app);
            compute(app);
        end

        % Value changed function: CA0
        function CA0ValueChanged(app, event)
            value = app.CA0.Value;
            app.CA0Edit.Value=sprintf('%4.2f',value);
            compute(app);
        end

        % Value changed function: CA0Edit
        function CA0EditValueChanged(app, event)
            value = app.CA0Edit.Value;
            if between(app,value,app.CA0.Limits)
                app.CA0.Value=str2double(value);
                compute(app);
            else
                app.CA0Edit.Value=num2str(app.CA0.Value);
            end
        end

        % Value changed function: ypsilon
        function ypsilonValueChanged(app, event)
            value = app.ypsilon.Value;
            app.ypsilonEdit.Value=sprintf('%4.1f',value);
            compute(app);
        end

        % Value changed function: ypsilonEdit
        function ypsilonEditValueChanged(app, event)
            value = app.ypsilonEdit.Value;
            if between(app,value,app.ypsilon.Limits)
                app.ypsilon.Value=str2double(value);
                compute(app);
            else
                app.ypsilonEdit.Value=num2str(app.ypsilon.Value);
            end
        end

        % Value changed function: k
        function kValueChanged(app, event)
            value = app.k.Value;
            app.kEdit.Value=sprintf('%4.3f',value);
            texter(app);
            compute(app);
        end

        % Value changed function: kEdit
        function kEditValueChanged(app, event)
            value = app.kEdit.Value;
            if between(app,value,app.k.Limits)
                app.k.Value=str2double(value);
                texter(app);
                compute(app);
```

```
            else
                app.kEdit.Value=num2str(app.k.Value);
            end
        end

        % Value changed function: alfa
        function alfaValueChanged(app, event)
            texter(app);
            compute(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create CSTR1UIFigure and hide until all components are created
            app.CSTR1UIFigure = uifigure('Visible', 'off');
            app.CSTR1UIFigure.Color = [0.94 0.94 0.94];
            app.CSTR1UIFigure.Position = [100 100 740 676];
            app.CSTR1UIFigure.Name = 'CSTR 1';

            % Create Image
            app.Image = uiimage(app.CSTR1UIFigure);
            app.Image.BackgroundColor = [0.9412 0.9412 0.9412];
            app.Image.Position = [134 61 474 404];
            app.Image.ImageSource = 'tankreaktor1_bakgrund.jpg';

            % Create titel
            app.titel = uilabel(app.CSTR1UIFigure);
            app.titel.FontSize = 20;
            app.titel.FontWeight = 'bold';
            app.titel.Position = [214 630 313 24];
            app.titel.Text = 'Reaction in an Isothermal CSTR';

            % Create CA0molLSliderLabel
            app.CA0molLSliderLabel = uilabel(app.CSTR1UIFigure);
            app.CA0molLSliderLabel.HorizontalAlignment = 'right';
            app.CA0molLSliderLabel.Position = [11 586 70 22];
            app.CA0molLSliderLabel.Text = 'CA0 (mol/L)';

            % Create CA0
            app.CA0 = uislider(app.CSTR1UIFigure);
            app.CA0.Limits = [0.1 1];
            app.CA0.ValueChangedFcn = createCallbackFcn(app, @CA0ValueChanged, true);
            app.CA0.MinorTicks = [];
            app.CA0.Position = [102 595 150 3];
            app.CA0.Value = 0.1;

            % Create CA0Edit
            app.CA0Edit = uieditfield(app.CSTR1UIFigure, 'text');
            app.CA0Edit.ValueChangedFcn = createCallbackFcn(app, @CA0EditValueChanged, true);
            app.CA0Edit.Position = [282 574 45 22];

            % Create VolumetricflowrateLsLabel
            app.VolumetricflowrateLsLabel = uilabel(app.CSTR1UIFigure);
            app.VolumetricflowrateLsLabel.HorizontalAlignment = 'right';
            app.VolumetricflowrateLsLabel.Position = [341 580 138 22];
            app.VolumetricflowrateLsLabel.Text = 'Volumetric flow rate (L/s)';

            % Create ypsilon
            app.ypsilon = uislider(app.CSTR1UIFigure);
            app.ypsilon.Limits = [10 50];
            app.ypsilon.MajorTicks = [10 20 30 40 50];
            app.ypsilon.ValueChangedFcn = createCallbackFcn(app, @ypsilonValueChanged, true);
            app.ypsilon.MinorTicks = [];
            app.ypsilon.Position = [500 589 150 3];
            app.ypsilon.Value = 10;

            % Create ypsilonEdit
```

```
app.ypsilonEdit = uieditfield(app.CSTR1UIFigure, 'text');
app.ypsilonEdit.ValueChangedFcn = createCallbackFcn(app, @ypsilonEditValueChanged, true);
app.ypsilonEdit.Position = [676 576 45 22];

% Create rateconstantSliderLabel
app.rateconstantSliderLabel = uilabel(app.CSTR1UIFigure);
app.rateconstantSliderLabel.HorizontalAlignment = 'right';
app.rateconstantSliderLabel.Position = [7 531 75 22];
app.rateconstantSliderLabel.Text = 'rate constant';

% Create k
app.k = uislider(app.CSTR1UIFigure);
app.k.Limits = [0.001 0.07];
app.k.MajorTicks = [0.001 0.02 0.04 0.07];
app.k.ValueChangedFcn = createCallbackFcn(app, @kValueChanged, true);
app.k.MinorTicks = [];
app.k.Position = [103 540 150 3];
app.k.Value = 0.001;

% Create kEdit
app.kEdit = uieditfield(app.CSTR1UIFigure, 'text');
app.kEdit.ValueChangedFcn = createCallbackFcn(app, @kEditValueChanged, true);
app.kEdit.Position = [282 521 45 22];

% Create reactionorderDropDownLabel
app.reactionorderDropDownLabel = uilabel(app.CSTR1UIFigure);
app.reactionorderDropDownLabel.HorizontalAlignment = 'right';
app.reactionorderDropDownLabel.Position = [461 519 80 22];
app.reactionorderDropDownLabel.Text = 'reaction order';

% Create alfa
app.alfa = uidropdown(app.CSTR1UIFigure);
app.alfa.Items = {'1', '2'};
app.alfa.ValueChangedFcn = createCallbackFcn(app, @alfaValueChanged, true);
app.alfa.Position = [556 519 100 22];
app.alfa.Value = '1';

% Create tau
app.tau = uilabel(app.CSTR1UIFigure);
app.tau.BackgroundColor = [0 1 0];
app.tau.HorizontalAlignment = 'right';
app.tau.FontSize = 14;
app.tau.FontWeight = 'bold';
app.tau.Position = [341 220 109 22];
app.tau.Text = 'tau';

% Create CA
app.CA = uilabel(app.CSTR1UIFigure);
app.CA.BackgroundColor = [0 1 0];
app.CA.HorizontalAlignment = 'right';
app.CA.FontSize = 14;
app.CA.FontWeight = 'bold';
app.CA.Position = [341 199 109 22];
app.CA.Text = 'CA';

% Create CB
app.CB = uilabel(app.CSTR1UIFigure);
app.CB.BackgroundColor = [0 1 0];
app.CB.HorizontalAlignment = 'right';
app.CB.FontSize = 14;
app.CB.FontWeight = 'bold';
app.CB.Position = [341 178 109 22];
app.CB.Text = 'CB';

% Create FA
app.FA = uilabel(app.CSTR1UIFigure);
app.FA.BackgroundColor = [1 1 0];
app.FA.FontSize = 14;
app.FA.FontWeight = 'bold';
app.FA.Position = [556 210 133 22];
app.FA.Text = 'FA';
```

```
            % Create CAout
            app.CAout = uilabel(app.CSTR1UIFigure);
            app.CAout.BackgroundColor = [1 1 0];
            app.CAout.FontSize = 14;
            app.CAout.FontWeight = 'bold';
            app.CAout.Position = [556 189 133 22];
            app.CAout.Text = 'CA';

            % Create FB
            app.FB = uilabel(app.CSTR1UIFigure);
            app.FB.BackgroundColor = [1 1 0];
            app.FB.FontSize = 14;
            app.FB.FontWeight = 'bold';
            app.FB.Position = [556 147 133 22];
            app.FB.Text = 'FB';

            % Create CBout
            app.CBout = uilabel(app.CSTR1UIFigure);
            app.CBout.BackgroundColor = [1 1 0];
            app.CBout.FontSize = 14;
            app.CBout.FontWeight = 'bold';
            app.CBout.Position = [556 126 134 22];
            app.CBout.Text = 'CB';

            % Create FA0
            app.FA0 = uilabel(app.CSTR1UIFigure);
            app.FA0.BackgroundColor = [1 1 0];
            app.FA0.HorizontalAlignment = 'right';
            app.FA0.FontSize = 14;
            app.FA0.FontWeight = 'bold';
            app.FA0.Position = [122 423 136 22];
            app.FA0.Text = 'FA0';

            % Create reaction
            app.reaction = uilabel(app.CSTR1UIFigure);
            app.reaction.HorizontalAlignment = 'center';
            app.reaction.FontSize = 16;
            app.reaction.FontWeight = 'bold';
            app.reaction.Position = [264 472 325 22];
            app.reaction.Text = 'A -> B rA = kCA';

            % Show the figure after all components are created
            app.CSTR1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = cstr1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.CSTR1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.CSTR1UIFigure)
```

```
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

CSTR 2

cstr2.mlapp

```
classdef cstr2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        CSTR2UIFigure                 matlab.ui.Figure
        mumaxSlider                   matlab.ui.control.Slider
        maximumgrowthrateSliderLabel  matlab.ui.control.Label
        rubrik                        matlab.ui.control.Label
        ksSlider                      matlab.ui.control.Slider
        MonodconstantSliderLabel      matlab.ui.control.Label
        CS0Slider                     matlab.ui.control.Slider
        feedsubstrateconcentrationSliderLabel  matlab.ui.control.Label
        celldeathrateconstantSliderLabel  matlab.ui.control.Label
        kdSlider                      matlab.ui.control.Slider
        TabGroup                      matlab.ui.container.TabGroup
        ConcentrationTab              matlab.ui.container.Tab
        UIAxes                        matlab.ui.control.UIAxes
        CellproductionTab             matlab.ui.container.Tab
        UIAxes2                       matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function Draw(app)
            try
                toogleUIC(app.CSTR2UIFigure,'off');
                mumax = app.mumaxSlider.Value;
                app. maximumgrowthrateSliderLabel.Text=sprintf('maximum growth rate, \x03bc\x2098\x2090\x2093');
                ks = app.ksSlider.Value;
                kd = app.kdSlider.Value;
                CS0 = app.CS0Slider.Value;
                Ycs=0.5;
                Dmax=-(CS0*kd+kd*ks-CS0*mumax)/(CS0+ks);
                D=linspace(0,Dmax,100);
                CS=-ks*(D+kd)./(D+kd-mumax);
                Dplot=[D ones(1,10)*1.0];CSplot=[CS ones(1,10)*CS0];
                my=mumax*CS./(ks+CS);
                CC=Ycs*D.*(CS0-CS)./my;
                CCplot=[CC ones(1,10)*0];
                Prod=D.*CC;
                ProdPlot=[Prod ones(1,10)*0];
                plot(app.UIAxes,Dplot,CSplot,'k-',Dplot,CCplot,'k:','LineWidth',1.5)
                app.UIAxes.YLabel.String='Concentration [gL^{-1}]';
                app.UIAxes.Title.String=['D_{max} = ',sprintf('%4.2f',Dmax),' h^{-1}'];
                app.UIAxes.XLabel.String='dilution rate [h^{-1}]';
                app.UIAxes.YLim=[0 CS0];
                legend(app.UIAxes,'C_{substrate}','C_{cell}','location','best')
                plot(app.UIAxes2,Dplot,ProdPlot,'k-','LineWidth',1.5)
                app.UIAxes2.YLabel.String='Cell production [gL^{-1}h^{-1}';
                app.UIAxes2.XLabel.String='dilution rate [h^{-1}]';
                [M,pos]=max(ProdPlot);
                app.UIAxes2.Title.String=['D_{opt} = ',sprintf('%4.2f',D(pos)),' h^{-1}'];
                app.UIAxes2.YLim=[0 M];
                toogleUIC(app.CSTR2UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
```

```
            Draw(app);
        end

        % Value changed function: CS0Slider, kdSlider, ksSlider,
        % ...and 1 other component
        function mumaxSliderValueChanged(app, event)
            Draw(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create CSTR2UIFigure and hide until all components are created
            app.CSTR2UIFigure = uifigure('Visible', 'off');
            app.CSTR2UIFigure.Color = [0.94 0.94 0.94];
            app.CSTR2UIFigure.Position = [100 100 740 754];
            app.CSTR2UIFigure.Name = 'CSTR 2';

            % Create TabGroup
            app.TabGroup = uitabgroup(app.CSTR2UIFigure);
            app.TabGroup.Position = [26 36 685 511];

            % Create ConcentrationTab
            app.ConcentrationTab = uitab(app.TabGroup);
            app.ConcentrationTab.Title = 'Concentration';

            % Create UIAxes
            app.UIAxes = uiaxes(app.ConcentrationTab);
            app.UIAxes.PlotBoxAspectRatio = [1.56847545219638 1 1];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [33 29 626 429];

            % Create CellproductionTab
            app.CellproductionTab = uitab(app.TabGroup);
            app.CellproductionTab.Title = 'Cell production';

            % Create UIAxes2
            app.UIAxes2 = uiaxes(app.CellproductionTab);
            app.UIAxes2.PlotBoxAspectRatio = [1.4937343358396 1 1];
            app.UIAxes2.XTickLabelRotation = 0;
            app.UIAxes2.YTickLabelRotation = 0;
            app.UIAxes2.ZTickLabelRotation = 0;
            app.UIAxes2.Position = [24 29 646 441];

            % Create kdSlider
            app.kdSlider = uislider(app.CSTR2UIFigure);
            app.kdSlider.Limits = [0 0.1];
            app.kdSlider.ValueChangedFcn = createCallbackFcn(app, @mumaxSliderValueChanged, true);
            app.kdSlider.MinorTicks = [];
            app.kdSlider.Position = [550 596 150 3];

            % Create celldeathrateconstantSliderLabel
            app.celldeathrateconstantSliderLabel = uilabel(app.CSTR2UIFigure);
            app.celldeathrateconstantSliderLabel.HorizontalAlignment = 'right';
            app.celldeathrateconstantSliderLabel.Position = [400 586 130 22];
            app.celldeathrateconstantSliderLabel.Text = 'cell death rate constant';

            % Create feedsubstrateconcentrationSliderLabel
            app.feedsubstrateconcentrationSliderLabel = uilabel(app.CSTR2UIFigure);
            app.feedsubstrateconcentrationSliderLabel.HorizontalAlignment = 'right';
            app.feedsubstrateconcentrationSliderLabel.Position = [26 596 157 22];
            app.feedsubstrateconcentrationSliderLabel.Text = 'feed substrate concentration';

            % Create CS0Slider
            app.CS0Slider = uislider(app.CSTR2UIFigure);
```

```
            app.CS0Slider.Limits = [0.5 2];
            app.CS0Slider.MajorTicks = [0.5 0.75 1 1.25 1.5 1.75 2 15];
            app.CS0Slider.ValueChangedFcn = createCallbackFcn(app, @mumaxSliderValueChanged, true);
            app.CS0Slider.MinorTicks = [];
            app.CS0Slider.Position = [204 605 150 3];
            app.CS0Slider.Value = 1;

            % Create MonodconstantSliderLabel
            app.MonodconstantSliderLabel = uilabel(app.CSTR2UIFigure);
            app.MonodconstantSliderLabel.HorizontalAlignment = 'right';
            app.MonodconstantSliderLabel.Position = [420 657 110 22];
            app.MonodconstantSliderLabel.Text = {'Monod constant'; ''};

            % Create ksSlider
            app.ksSlider = uislider(app.CSTR2UIFigure);
            app.ksSlider.Limits = [0.1 1];
            app.ksSlider.MajorTicks = [0.1 0.25 0.5 0.75 1];
            app.ksSlider.ValueChangedFcn = createCallbackFcn(app, @mumaxSliderValueChanged, true);
            app.ksSlider.MinorTicks = [];
            app.ksSlider.Position = [550 667 150 3];
            app.ksSlider.Value = 0.5;

            % Create rubrik
            app.rubrik = uilabel(app.CSTR2UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.Position = [295 708 256 30];
            app.rubrik.Text = 'Operating a Chemostat';

            % Create maximumgrowthrateSliderLabel
            app.maximumgrowthrateSliderLabel = uilabel(app.CSTR2UIFigure);
            app.maximumgrowthrateSliderLabel.HorizontalAlignment = 'right';
            app.maximumgrowthrateSliderLabel.Position = [38 660 165 22];
            app.maximumgrowthrateSliderLabel.Text = {'maximum growth rate'; ''};

            % Create mumaxSlider
            app.mumaxSlider = uislider(app.CSTR2UIFigure);
            app.mumaxSlider.Limits = [0.15 1];
            app.mumaxSlider.MajorTicks = [0.15 0.3 0.45 0.6 0.75 1];
            app.mumaxSlider.ValueChangedFcn = createCallbackFcn(app, @mumaxSliderValueChanged, true);
            app.mumaxSlider.MinorTicks = [];
            app.mumaxSlider.Position = [227 670 150 3];
            app.mumaxSlider.Value = 0.5;

            % Show the figure after all components are created
            app.CSTR2UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = cstr2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.CSTR2UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
```

```
            delete(app.CSTR2UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

CSTR 3

cstr3.mlapp

```
classdef cstr3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        CSTR3UIFigure               matlab.ui.Figure
        tauSlider                   matlab.ui.control.Slider
        residencetimesSliderLabel   matlab.ui.control.Label
        T0Slider                    matlab.ui.control.Slider
        feedtemperatureKSliderLabel  matlab.ui.control.Label
        krSlider                    matlab.ui.control.Slider
        reversereactionpreexponentialfactor1sSliderLabel  matlab.ui.control.Label
        USlider                     matlab.ui.control.Slider
        heattransfercoefficientcaldmx00b3KsLabel  matlab.ui.control.Label
        rubrik                      matlab.ui.control.Label
        UIAxes                      matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.CSTR3UIFigure,'off');
                U=app.USlider.Value;
                kr=app.krSlider.Value;
                Tf=app.T0Slider.Value;
                tau=app.tauSlider.Value;
                %V=50000;
                kf=100000000000;
                Ef=20000;
                Er=24000;
                R=1.987;
                rho=0.8;
                Cp=2;
                cA0=0.01;
                A=8.94;
                DeltaHrxn=-15000;
                Tc=310;
                v=100;
                T=linspace(250,400);
                eb=(-v*rho*Cp*(T-Tf)-U*A*(T-Tc))/v/DeltaHrxn*1000;
                k1=kf*exp(-Ef/R./T);
                k2=kr*exp(-Er/R./T);
                mb=tau*k1*cA0./(1+k1*tau+k2*tau)*1000;
                plot(app.UIAxes,T,mb,'g',T,eb,'b','LineWidth',1.5);
                app.UIAxes.YLim=[0 12];
                app.heattransfercoefficientcaldmx00b3KsLabel.Text=sprintf('heat transfer coefficient [cal/(dm\x00b3 K s]');
                toogleUIC(app.CSTR3UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            draw(app);
        end

        % Value changed function: T0Slider, USlider, krSlider, tauSlider
        function USliderValueChanged(app, event)
            draw(app);
        end
```

```
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create CSTR3UIFigure and hide until all components are created
        app.CSTR3UIFigure = uifigure('Visible', 'off');
        app.CSTR3UIFigure.Color = [0.94 0.94 0.94];
        app.CSTR3UIFigure.Position = [100 100 740 667];
        app.CSTR3UIFigure.Name = 'CSTR 3';
        app.CSTR3UIFigure.Scrollable = 'on';

        % Create UIAxes
        app.UIAxes = uiaxes(app.CSTR3UIFigure);
        xlabel(app.UIAxes, 'Temperature [K]')
        ylabel(app.UIAxes, {'Product concentration'; '[mmol/dm^3]'; ''})
        app.UIAxes.PlotBoxAspectRatio = [1.65568862275449 1 1];
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.Position = [27 22 635 376];

        % Create rubrik
        app.rubrik = uilabel(app.CSTR3UIFigure);
        app.rubrik.FontSize = 24;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [27 607 725 30];
        app.rubrik.Text = 'Multiple Steady States in a Continuously Stirred Tank Reactor';

        % Create heattransfercoefficientcaldmx00b3KsLabel
        app.heattransfercoefficientcaldmx00b3KsLabel = uilabel(app.CSTR3UIFigure);
        app.heattransfercoefficientcaldmx00b3KsLabel.HorizontalAlignment = 'right';
        app.heattransfercoefficientcaldmx00b3KsLabel.Position = [48 564 239 22];
        app.heattransfercoefficientcaldmx00b3KsLabel.Text = {'heat transfer coefficient [cal/(dm\x00b3 K s)'; ''};

        % Create USlider
        app.USlider = uislider(app.CSTR3UIFigure);
        app.USlider.Limits = [0 20];
        app.USlider.ValueChangedFcn = createCallbackFcn(app, @USliderValueChanged, true);
        app.USlider.MinorTicks = [];
        app.USlider.Position = [308 573 150 3];

        % Create reversereactionpreexponentialfactor1sSliderLabel
        app.reversereactionpreexponentialfactor1sSliderLabel = uilabel(app.CSTR3UIFigure);
        app.reversereactionpreexponentialfactor1sSliderLabel.HorizontalAlignment = 'right';
        app.reversereactionpreexponentialfactor1sSliderLabel.Position = [47 499 239 22];
        app.reversereactionpreexponentialfactor1sSliderLabel.Text = {'reverse reaction pre-exponential factor [1/s]'; ''};

        % Create krSlider
        app.krSlider = uislider(app.CSTR3UIFigure);
        app.krSlider.Limits = [0 30000000000000];
        app.krSlider.ValueChangedFcn = createCallbackFcn(app, @USliderValueChanged, true);
        app.krSlider.MinorTicks = [];
        app.krSlider.Position = [307 508 150 3];

        % Create feedtemperatureKSliderLabel
        app.feedtemperatureKSliderLabel = uilabel(app.CSTR3UIFigure);
        app.feedtemperatureKSliderLabel.HorizontalAlignment = 'right';
        app.feedtemperatureKSliderLabel.Position = [35 437 115 22];
        app.feedtemperatureKSliderLabel.Text = 'feed temperature [K]';

        % Create T0Slider
        app.T0Slider = uislider(app.CSTR3UIFigure);
        app.T0Slider.Limits = [250 325];
        app.T0Slider.MajorTicks = [250 265 280 295 310 325];
        app.T0Slider.ValueChangedFcn = createCallbackFcn(app, @USliderValueChanged, true);
        app.T0Slider.MinorTicks = [];
        app.T0Slider.Position = [181 447 150 3];
```

```
            app.T0Slider.Value = 250;

            % Create residencetimesSliderLabel
            app.residencetimesSliderLabel = uilabel(app.CSTR3UIFigure);
            app.residencetimesSliderLabel.HorizontalAlignment = 'right';
            app.residencetimesSliderLabel.Position = [370 437 100 22];
            app.residencetimesSliderLabel.Text = 'residence time [s]';

            % Create tauSlider
            app.tauSlider = uislider(app.CSTR3UIFigure);
            app.tauSlider.Limits = [100 1000];
            app.tauSlider.MajorTicks = [100 300 500 700 850 1000];
            app.tauSlider.ValueChangedFcn = createCallbackFcn(app, @USliderValueChanged, true);
            app.tauSlider.MinorTicks = [];
            app.tauSlider.Position = [489 447 150 3];
            app.tauSlider.Value = 100;

            % Show the figure after all components are created
            app.CSTR3UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = cstr3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.CSTR3UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.CSTR3UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

CSTR 4

cstr4.mlapp

```
classdef cstr4 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        CSTR4UIFigure           matlab.ui.Figure
        TabGroup                matlab.ui.container.TabGroup
        PhaseplaneTab           matlab.ui.container.Tab
        UIAxes_2                matlab.ui.control.UIAxes
        TemperatureTab          matlab.ui.container.Tab
        UIAxes_4                matlab.ui.control.UIAxes
        ConversionTab           matlab.ui.container.Tab
        UIAxes_5                matlab.ui.control.UIAxes
        PhaseplanefullTab       matlab.ui.container.Tab
        UIAxes_1                matlab.ui.control.UIAxes
        EnergyTab               matlab.ui.container.Tab
        UIAxes_3                matlab.ui.control.UIAxes
        C0Slider                matlab.ui.control.Slider
        Initialreactorconcentrationkmolm3Label  matlab.ui.control.Label
        T0Slider                matlab.ui.control.Slider
        IntitalreactortemperatureKLabel  matlab.ui.control.Label
        tauSlider               matlab.ui.control.Slider
        ResidencetimeminLabel  matlab.ui.control.Label
        Label                   matlab.ui.control.Label
    end


    methods (Access = public)

        function draw(app)
            try
                toogleUIC(app.CSTR4UIFigure,'off');
                app.Initialreactorconcentrationkmolm3Label.Text=sprintf('Initial reactor concentration [kmol/m\x00b3]');
                Cp=4000;
                UA=3400;
                V=10;
                DeltaH=-220000;
                tau=app.tauSlider.Value;
                T0=app.T0Slider.Value;
                C0=app.C0Slider.Value;
                [~,y1]=ode23s(@model,[0 200],[2,T0]);
                [~,y2]=ode23s(@model,[0 200],[1.5,T0]);
                [~,y3]=ode23s(@model,[0 200],[1.0,T0]);
                [~,y4]=ode23s(@model,[0 200],[0.5,T0]);
                plot(app.UIAxes_1,y1(:,2),y1(:,1),'k',y2(:,2),y2(:,1),'k',y3(:,2),y3(:,1),'k',y4(:,2),y4(:,1),'k','Linewidth',1.5
                app.UIAxes_1.XLim=[300 500];
                app.UIAxes_1.YLim=[0 2];
                app.UIAxes_1.XLabel.String='Temperature [K]';
                app.UIAxes_1.YLabel.String='Reactor concentration [kmol/m^3]';

                [~,y1]=ode23s(@model,[0 1000],[C0,T0]);
                oms=(C0-y1(:,1))./C0;
                plot(app.UIAxes_2,y1(:,2),oms,'k','Linewidth',1.5)
                app.UIAxes_2.XLim=[300 500];
                app.UIAxes_2.YLim=[0 1];
                app.UIAxes_2.XLabel.String='Temperature [K]';
                app.UIAxes_2.YLabel.String='Conversion';

                T=linspace(300,425,20);
                k=0.004*exp(-15000*(1./T-1/298));
                Qg=-k./(1+k*tau)*DeltaH*2;
                Qr=Cp/tau*(T-298)+UA/V*(T-300);
                plot(app.UIAxes_3,T,Qg/1000,'g',T,Qr/1000,'b','Linewidth',1.5)
                app.UIAxes_3.YLimMode='auto';
                app.UIAxes_3.XLim=[300 425];
                app.UIAxes_3.XLabel.String='Temperature [K]';
                app.UIAxes_3.YLabel.String='Energy rate [MJ/m^3 min]';
```

```
            C0=app.C0Slider.Value;
            [t1,y1]=ode23s(@mode1,[0 200],[C0,T0]);
            plot(app.UIAxes_4,t1,y1(:,2),'k','Linewidth',1.5)
            app.UIAxes_4.XLim=[0 200];
            app.UIAxes_4.YLim=[300 550];
            app.UIAxes_4.XLabel.String='Time [min]';
            app.UIAxes_4.YLabel.String='Temperature [K]';

            oms=(C0-y1(:,1))./C0;
            plot(app.UIAxes_5,t1,oms,'k','Linewidth',1.5)
            axis(app.UIAxes_5,[0 200 0 1])
            app.UIAxes_5.XLabel.String='Time [min]';
            app.UIAxes_5.YLabel.String='Conversion';
            toogleUIC(app.CSTR4UIFigure,'on');
        catch ME
            errordlg(ME.message);
        end
        function dy = mode1(~,y)
            dy=zeros(size(y));
            k=0.004*exp(-15000*(1/y(2)-1/298));
            dy(1)=1/tau*(2-y(1))-k*y(1);
            dy(2)=UA/V/Cp*(300-y(2))+1/tau*(298-y(2))-DeltaH/Cp*k*y(1);
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        draw(app);
    end

    % Value changed function: C0Slider, T0Slider, tauSlider
    function T0SliderValueChanged(app, event)
        draw(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create CSTR4UIFigure and hide until all components are created
        app.CSTR4UIFigure = uifigure('Visible', 'off');
        app.CSTR4UIFigure.Color = [0.94 0.94 0.94];
        app.CSTR4UIFigure.Position = [100 100 740 700];
        app.CSTR4UIFigure.Name = 'CSTR 4';
        app.CSTR4UIFigure.Scrollable = 'on';

        % Create Label
        app.Label = uilabel(app.CSTR4UIFigure);
        app.Label.FontSize = 24;
        app.Label.Position = [69 644 588 43];
        app.Label.Text = 'Multiple Steady States in a CSTR with Heat Exchange';

        % Create ResidencetimeminLabel
        app.ResidencetimeminLabel = uilabel(app.CSTR4UIFigure);
        app.ResidencetimeminLabel.HorizontalAlignment = 'right';
        app.ResidencetimeminLabel.Position = [10 596 118 22];
        app.ResidencetimeminLabel.Text = {'Residence time [min]'; ''};

        % Create tauSlider
        app.tauSlider = uislider(app.CSTR4UIFigure);
        app.tauSlider.Limits = [1 50];
        app.tauSlider.MajorTicks = [1 10 20 30 40 50];
        app.tauSlider.ValueChangedFcn = createCallbackFcn(app, @T0SliderValueChanged, true);
```

```
app.tauSlider.MinorTicks = [5 15 25 35 45];
app.tauSlider.Position = [149 605 190 3];
app.tauSlider.Value = 30;

% Create IntitalreactortemperatureKLabel
app.IntitalreactortemperatureKLabel = uilabel(app.CSTR4UIFigure);
app.IntitalreactortemperatureKLabel.HorizontalAlignment = 'right';
app.IntitalreactortemperatureKLabel.Position = [310 590 215 28];
app.IntitalreactortemperatureKLabel.Text = 'Intital reactor temperature [K]';

% Create T0Slider
app.T0Slider = uislider(app.CSTR4UIFigure);
app.T0Slider.Limits = [298 450];
app.T0Slider.MajorTicks = [298 325 350 375 400 425 450];
app.T0Slider.ValueChangedFcn = createCallbackFcn(app, @T0SliderValueChanged, true);
app.T0Slider.MinorTicks = [];
app.T0Slider.Position = [547 605 151 3];
app.T0Slider.Value = 340;

% Create Initialreactorconcentrationkmolm3Label
app.Initialreactorconcentrationkmolm3Label = uilabel(app.CSTR4UIFigure);
app.Initialreactorconcentrationkmolm3Label.HorizontalAlignment = 'right';
app.Initialreactorconcentrationkmolm3Label.Position = [296 501 240 43];
app.Initialreactorconcentrationkmolm3Label.Text = {'Initial reactor concentration [kmol/m3]'; ''};

% Create C0Slider
app.C0Slider = uislider(app.CSTR4UIFigure);
app.C0Slider.Limits = [0 2];
app.C0Slider.MajorTicks = [0 0.4 0.8 1.2 1.6 2];
app.C0Slider.ValueChangedFcn = createCallbackFcn(app, @T0SliderValueChanged, true);
app.C0Slider.MinorTicks = [];
app.C0Slider.Position = [558 531 150 3];
app.C0Slider.Value = 1;

% Create TabGroup
app.TabGroup = uitabgroup(app.CSTR4UIFigure);
app.TabGroup.Position = [25 18 690 418];

% Create PhaseplaneTab
app.PhaseplaneTab = uitab(app.TabGroup);
app.PhaseplaneTab.Title = 'Phase plane';

% Create UIAxes_2
app.UIAxes_2 = uiaxes(app.PhaseplaneTab);
app.UIAxes_2.PlotBoxAspectRatio = [2.04934210526316 1 1];
app.UIAxes_2.XTickLabelRotation = 0;
app.UIAxes_2.YTickLabelRotation = 0;
app.UIAxes_2.ZTickLabelRotation = 0;
app.UIAxes_2.Position = [22 24 651 346];

% Create TemperatureTab
app.TemperatureTab = uitab(app.TabGroup);
app.TemperatureTab.Title = 'Temperature';

% Create UIAxes_4
app.UIAxes_4 = uiaxes(app.TemperatureTab);
app.UIAxes_4.PlotBoxAspectRatio = [2.04934210526316 1 1];
app.UIAxes_4.XTickLabelRotation = 0;
app.UIAxes_4.YTickLabelRotation = 0;
app.UIAxes_4.ZTickLabelRotation = 0;
app.UIAxes_4.Position = [17 24 656 346];

% Create ConversionTab
app.ConversionTab = uitab(app.TabGroup);
app.ConversionTab.Title = 'Conversion';

% Create UIAxes_5
app.UIAxes_5 = uiaxes(app.ConversionTab);
app.UIAxes_5.PlotBoxAspectRatio = [2.04934210526316 1 1];
app.UIAxes_5.XTickLabelRotation = 0;
app.UIAxes_5.YTickLabelRotation = 0;
```

```
            app.UIAxes_5.ZTickLabelRotation = 0;
            app.UIAxes_5.Position = [22 24 647 346];

            % Create PhaseplanefullTab
            app.PhaseplanefullTab = uitab(app.TabGroup);
            app.PhaseplanefullTab.Title = 'Phase plane full';

            % Create UIAxes_1
            app.UIAxes_1 = uiaxes(app.PhaseplanefullTab);
            app.UIAxes_1.XTickLabelRotation = 0;
            app.UIAxes_1.YTickLabelRotation = 0;
            app.UIAxes_1.ZTickLabelRotation = 0;
            app.UIAxes_1.Position = [10 24 659 346];

            % Create EnergyTab
            app.EnergyTab = uitab(app.TabGroup);
            app.EnergyTab.Title = 'Energy';

            % Create UIAxes_3
            app.UIAxes_3 = uiaxes(app.EnergyTab);
            app.UIAxes_3.PlotBoxAspectRatio = [2.04934210526316 1 1];
            app.UIAxes_3.XTickLabelRotation = 0;
            app.UIAxes_3.YTickLabelRotation = 0;
            app.UIAxes_3.ZTickLabelRotation = 0;
            app.UIAxes_3.Position = [16 16 644 346];

            % Show the figure after all components are created
            app.CSTR4UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = cstr4

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.CSTR4UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.CSTR4UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
```

```
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

CSTR 5

cstr5.mlapp

```
classdef cstr5 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        CSTR5UIFigure      matlab.ui.Figure
        ypsilonField2      matlab.ui.control.EditField
        tField             matlab.ui.control.EditField
        TabGroup           matlab.ui.container.TabGroup
        TemperatureTab     matlab.ui.container.Tab
        UIAxes_1           matlab.ui.control.UIAxes
        ConcentrationTab   matlab.ui.container.Tab
        UIAxes_2           matlab.ui.control.UIAxes
        ypsilonSlider      matlab.ui.control.Slider
        newvolumetricflowratedmx00b3minLabel  matlab.ui.control.Label
        tSlider            matlab.ui.control.Slider
        timeatwhichflowrateischangedminSliderLabel  matlab.ui.control.Label
        rubrik             matlab.ui.control.Label
    end




    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.CSTR5UIFigure,'off');
                app.newvolumetricflowratedmx00b3minLabel.Text=sprintf('new volumetric flow rate [dm\x00b3/min]');
                tid=app.tSlider.Value;
                ypsilonNew=app.ypsilonSlider.Value;
                vo=2;
                DeltaH=-50000;
                CpA=150;
                CpB=95;
                V=10;
                Ta=325;
                UA=3800;
                To=350;
                vo=2;
                Cao=10;
                ta=6;
                tc=25;
                Cas=5;
                Ts=370;
                [y,~,exitflag]=fsolve(@ss,[Cas Ts],optimset('Display','off'));
                if exitflag==1
                    [t,y]=ode23s(@cstr,[0 tc],[y(1) y(2) y(1) y(2)]);
                    plot(app.UIAxes_1,t,y(:,2),'--',t,y(:,4),'-','LineWidth',1.5);
                    app.UIAxes_1.XLabel.String='Time [min]';
                    app.UIAxes_1.YLabel.String='Temperature [K]';
                    legend(app.UIAxes_1,'off');
                    plot(app.UIAxes_2,t,y(:,3),'b',t,Cao-y(:,3),'r','LineWidth',1.5);
                    app.UIAxes_2.XLabel.String='Time [min]';
                    app.UIAxes_2.YLabel.String='Concentration [mol/L]';
                    legend(app.UIAxes_2,'C_A','C_B','location','best')
                end
                toogleUIC(app.CSTR5UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function dy = cstr(t,y)
                dy=zeros(size(y));
                if t > tid
                    ypsilon1=vo;
                    ypsilon2=ypsilonNew;
                else
                    ypsilon1=vo;
```

```
                ypsilon2=vo;
            end
            k1=7000*exp(-36000/8.314./y(2));
            ra1=-k1*y(1);
            k2=7000*exp(-36000/8.314./y(4));
            ra2=-k2*y(3);
            dy(1)=ypsilon1/V*(Cao-y(1))+ra1;
            Cb1=(Cao-y(1));
            dy(3)=ypsilon2/V*(Cao-y(3))+ra2;
            Cb2=(Cao-y(3));
            if t > ta
                dy(2)=(-ypsilon1*Cao*CpA*(y(2)-To)+DeltaH*ra1*V)/V/(y(1)*CpA+Cb1*CpB);
                dy(4)=(-ypsilon2*Cao*CpA*(y(4)-To)+DeltaH*ra2*V)/V/(y(3)*CpA+Cb2*CpB);
            end
        end
        function res = ss(y)
            res=zeros(size(y));
            Ca=y(1);
            T=y(2);
            k=7000*exp(-36000/8.314./T);
            ra=-k*Ca;
            res(1)=Ca-vo*Cao/(vo+V*k);
            res(2)=T-(-vo*Cao*CpA*To-Ta*UA-ra*V*DeltaH)/(-vo*Cao*CpA-UA);
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.tField.Value=sprintf('%4.1f',app.tSlider.Value);
        app.ypsilonField2.Value=sprintf('%4.1f',app.ypsilonSlider.Value);
        draw(app);
    end

    % Value changed function: tField
    function tFieldValueChanged(app, event)
        value = app.tField.Value;
        app.tSlider.Value=str2double(value);
        draw(app);
    end

    % Value changed function: ypsilonField2
    function ypsilonField2ValueChanged(app, event)
        value = app.ypsilonField2.Value;
        app.ypsilonSlider.Value=str2double(value);
        draw(app);
    end

    % Value changed function: tSlider
    function tSliderValueChanged(app, event)
        value = app.tSlider.Value;
        app.tField.Value=sprintf('%4.1f',value);
        draw(app);
    end

    % Value changed function: ypsilonSlider
    function ypsilonSliderValueChanged(app, event)
        value = app.ypsilonSlider.Value;
        app.ypsilonField2.Value=sprintf('%4.1f',value);
        draw(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
```

```
function createComponents(app)

    % Create CSTR5UIFigure and hide until all components are created
    app.CSTR5UIFigure = uifigure('Visible', 'off');
    app.CSTR5UIFigure.Color = [0.94 0.94 0.94];
    app.CSTR5UIFigure.Position = [100 100 720 641];
    app.CSTR5UIFigure.Name = 'CSTR 5';
    app.CSTR5UIFigure.Scrollable = 'on';

    % Create rubrik
    app.rubrik = uilabel(app.CSTR5UIFigure);
    app.rubrik.FontSize = 24;
    app.rubrik.FontWeight = 'bold';
    app.rubrik.Position = [63 591 623 30];
    app.rubrik.Text = 'Continuous Stirred-Tank Reactor That Loses Cooling';

    % Create timeatwhichflowrateischangedminSliderLabel
    app.timeatwhichflowrateischangedminSliderLabel = uilabel(app.CSTR5UIFigure);
    app.timeatwhichflowrateischangedminSliderLabel.HorizontalAlignment = 'right';
    app.timeatwhichflowrateischangedminSliderLabel.Position = [135 554 216 22];
    app.timeatwhichflowrateischangedminSliderLabel.Text = {'time at which flow rate is changed [min]'; ''};

    % Create tSlider
    app.tSlider = uislider(app.CSTR5UIFigure);
    app.tSlider.Limits = [7 12];
    app.tSlider.ValueChangedFcn = createCallbackFcn(app, @tSliderValueChanged, true);
    app.tSlider.MinorTicks = [];
    app.tSlider.Position = [372 563 150 3];
    app.tSlider.Value = 9;

    % Create newvolumetricflowratedmx00b3minLabel
    app.newvolumetricflowratedmx00b3minLabel = uilabel(app.CSTR5UIFigure);
    app.newvolumetricflowratedmx00b3minLabel.HorizontalAlignment = 'right';
    app.newvolumetricflowratedmx00b3minLabel.Position = [141 496 217 22];
    app.newvolumetricflowratedmx00b3minLabel.Text = 'new volumetric flow rate [dm\x00b3/min';

    % Create ypsilonSlider
    app.ypsilonSlider = uislider(app.CSTR5UIFigure);
    app.ypsilonSlider.Limits = [0 25];
    app.ypsilonSlider.ValueChangedFcn = createCallbackFcn(app, @ypsilonSliderValueChanged, true);
    app.ypsilonSlider.MinorTicks = [];
    app.ypsilonSlider.Position = [379 505 150 3];
    app.ypsilonSlider.Value = 4;

    % Create TabGroup
    app.TabGroup = uitabgroup(app.CSTR5UIFigure);
    app.TabGroup.Position = [17 10 686 455];

    % Create TemperatureTab
    app.TemperatureTab = uitab(app.TabGroup);
    app.TemperatureTab.Title = 'Temperature';

    % Create UIAxes_1
    app.UIAxes_1 = uiaxes(app.TemperatureTab);
    xlabel(app.UIAxes_1, 'Time [min]')
    app.UIAxes_1.PlotBoxAspectRatio = [1.50857142857143 1 1];
    app.UIAxes_1.XTickLabelRotation = 0;
    app.UIAxes_1.YTickLabelRotation = 0;
    app.UIAxes_1.ZTickLabelRotation = 0;
    app.UIAxes_1.NextPlot = 'replace';
    app.UIAxes_1.Position = [55 24 578 392];

    % Create ConcentrationTab
    app.ConcentrationTab = uitab(app.TabGroup);
    app.ConcentrationTab.Title = 'Concentration';

    % Create UIAxes_2
    app.UIAxes_2 = uiaxes(app.ConcentrationTab);
    xlabel(app.UIAxes_2, 'Time [min]')
    app.UIAxes_2.PlotBoxAspectRatio = [1.50857142857143 1 1];
    app.UIAxes_2.XTickLabelRotation = 0;
```

```
            app.UIAxes_2.YTickLabelRotation = 0;
            app.UIAxes_2.ZTickLabelRotation = 0;
            app.UIAxes_2.NextPlot = 'replace';
            app.UIAxes_2.Position = [55 24 578 392];

            % Create tField
            app.tField = uieditfield(app.CSTR5UIFigure, 'text');
            app.tField.ValueChangedFcn = createCallbackFcn(app, @tFieldValueChanged, true);
            app.tField.Position = [547 554 37 22];

            % Create ypsilonField2
            app.ypsilonField2 = uieditfield(app.CSTR5UIFigure, 'text');
            app.ypsilonField2.ValueChangedFcn = createCallbackFcn(app, @ypsilonField2ValueChanged, true);
            app.ypsilonField2.Position = [546 495 38 22];

            % Show the figure after all components are created
            app.CSTR5UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = cstr5

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.CSTR5UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.CSTR5UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

CSTR 6

cstr6.mlapp

```
classdef cstr6 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        CSTR6UIFigure                matlab.ui.Figure
        TField                       matlab.ui.control.EditField
        EaEbEdit                     matlab.ui.control.EditField
        tauEdit                      matlab.ui.control.EditField
        FindToptCheckBox             matlab.ui.control.CheckBox
        TSlider                      matlab.ui.control.Slider
        TemperatureKSliderLabel      matlab.ui.control.Label
        EaEbSlider                   matlab.ui.control.Slider
        EaEbSliderLabel              matlab.ui.control.Label
        tauSlider                    matlab.ui.control.Slider
        ResidencetimeminSliderLabel  matlab.ui.control.Label
        rubrik                       matlab.ui.control.Label
        UIAxes                       matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.CSTR6UIFigure,'off');
                TK=linspace(600,1200);
                Ea=15000;
                Eb=Ea*app.EaEbSlider.Value;
                k1=@(T) 1e8*exp(-Ea./T);
                k2=@(T) 1e15*exp(-Eb./T);
                theta=app.tauSlider.Value;
                ca0=1;
                ca=@(T) ca0./(1+k1(T)*theta);
                cb=@(T) k1(T).*ca0*theta./(1+k1(T)*theta)./(1+k2(T)*theta);
                p1=plot(app.UIAxes,TK,ca(TK),'k--');
                app.UIAxes.NextPlot='add';
                p2=plot(app.UIAxes,TK,cb(TK),'k-');
                if app.FindToptCheckBox.Value == 1
                    Tspan=linspace(600,1200,500);
                    [~,J]=max(cb(Tspan));
                    plot(app.UIAxes,Tspan(J),cb(Tspan(J)),'ro',"LineWidth",2);
                    app.UIAxes.Title.String=sprintf('Topt=%4.0f K Conversion %4.1f %% Selectivity %4.1f %%',...
                        Tspan(J),...
                        (1-ca(Tspan(J)))*100,...
                        cb(Tspan(J))/(1-ca(Tspan(J)))*100);
                else
                    Topt=app.TSlider.Value;
                    plot(app.UIAxes,Topt,cb(Topt),'go',"LineWidth",2);
                    app.UIAxes.Title.String=sprintf('T=%4.0f K Conversion %4.1f %% Selectivity %4.1f %%',...
                        Topt,...
                        (1-ca(Topt))*100,...
                        cb(Topt)/(1-ca(Topt))*100);
                end
                legend(app.UIAxes,[p1,p2],{'c_A','c_B'},'Location','best','FontSize',12);
                app.UIAxes.NextPlot='replace';
                toogleUIC(app.CSTR6UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end
    end


    % Callbacks that handle component events
    methods (Access = private)
```

```
% Code that executes after component creation
function startupFcn(app)
    app.tauEdit.Value=sprintf('%5.2f',app.tauSlider.Value);
    app.EaEbEdit.Value=sprintf('%4.2f',app.EaEbSlider.Value);
    app.TField.Value=sprintf('%3.0f',app.TSlider.Value);
    simulera(app);
end

% Value changed function: FindToptCheckBox
function FindToptCheckBoxValueChanged(app, event)
    value = app.FindToptCheckBox.Value;
    if value == 1
        app.TemperatureKSliderLabel.Visible=0;
        app.TSlider.Visible=0;
        app.TField.Visible=0;
    else
        app.TemperatureKSliderLabel.Visible=1;
        app.TSlider.Visible=1;
        app.TField.Visible=1;
    end
    simulera(app);
end

% Value changed function: tauEdit
function tauEditValueChanged(app, event)
    value = app.tauEdit.Value;
    if between(app,value,app.tauSlider.Limits)
        app.tauSlider.Value=str2double(value);
        simulera(app);
    else
        app.tauEdit.Value=num2str(app.tauSlider.Value);
    end

end

% Value changed function: EaEbEdit
function EaEbEditValueChanged(app, event)
    value = app.EaEbEdit.Value;
    if between(app,value,app.EaEbSlider.Limits)
        app.EaEbSlider.Value=str2double(value);
        simulera(app);
    else
        app.EaEbEdit.Value=num2str(app.EaEbSlider.Value);
    end

end

% Value changed function: TField
function TFieldValueChanged(app, event)
    value = app.TField.Value;
    if between(app,value,app.TSlider.Limits)
        app.TSlider.Value=str2double(value);
        simulera(app);
    else
        app.TField.Value=num2str(app.TSlider.Value);
    end

end

% Value changed function: tauSlider
function tauSliderValueChanged(app, event)
    value = app.tauSlider.Value;
    app.tauEdit.Value=sprintf('%5.2f',value);
    simulera(app);
end

% Value changed function: EaEbSlider
function EaEbSliderValueChanged(app, event)
    value = app.EaEbSlider.Value;
    app.EaEbEdit.Value=sprintf('%4.2f',value);
    simulera(app);
```

```
        end

        % Value changed function: TSlider
        function TSliderValueChanged(app, event)
            value = app.TSlider.Value;
            app.TField.Value=sprintf('%3.0f',value);
            simulera(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create CSTR6UIFigure and hide until all components are created
            app.CSTR6UIFigure = uifigure('Visible', 'off');
            app.CSTR6UIFigure.Color = [0.94 0.94 0.94];
            app.CSTR6UIFigure.Position = [100 100 740 527];
            app.CSTR6UIFigure.Name = 'CSTR 6';

            % Create UIAxes
            app.UIAxes = uiaxes(app.CSTR6UIFigure);
            title(app.UIAxes, 'Title')
            xlabel(app.UIAxes, 'Temperature [K]')
            ylabel(app.UIAxes, 'Concentration [mol/L]')
            app.UIAxes.XLim = [600 1200];
            app.UIAxes.YLim = [0 1];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [87 21 563 318];

            % Create rubrik
            app.rubrik = uilabel(app.CSTR6UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [92 479 558 30];
            app.rubrik.Text = 'Maximizing an Intermediate Reactant in a CSTR';

            % Create ResidencetimeminSliderLabel
            app.ResidencetimeminSliderLabel = uilabel(app.CSTR6UIFigure);
            app.ResidencetimeminSliderLabel.HorizontalAlignment = 'right';
            app.ResidencetimeminSliderLabel.Position = [3 436 132 22];
            app.ResidencetimeminSliderLabel.Text = 'Residence time (min)';

            % Create tauSlider
            app.tauSlider = uislider(app.CSTR6UIFigure);
            app.tauSlider.Limits = [0.1 10];
            app.tauSlider.MajorTicks = [0.1 2 4 6 8 10];
            app.tauSlider.ValueChangedFcn = createCallbackFcn(app, @tauSliderValueChanged, true);
            app.tauSlider.MinorTicks = [];
            app.tauSlider.Position = [154 445 150 3];
            app.tauSlider.Value = 0.1;

            % Create EaEbSliderLabel
            app.EaEbSliderLabel = uilabel(app.CSTR6UIFigure);
            app.EaEbSliderLabel.HorizontalAlignment = 'right';
            app.EaEbSliderLabel.Position = [407 436 38 22];
            app.EaEbSliderLabel.Text = 'Ea/Eb';

            % Create EaEbSlider
            app.EaEbSlider = uislider(app.CSTR6UIFigure);
            app.EaEbSlider.Limits = [1.6 2];
            app.EaEbSlider.MajorTicks = [1.6 1.7 1.8 1.9 2 11.6];
            app.EaEbSlider.ValueChangedFcn = createCallbackFcn(app, @EaEbSliderValueChanged, true);
            app.EaEbSlider.MinorTicks = [];
            app.EaEbSlider.Position = [466 445 150 3];
            app.EaEbSlider.Value = 1.8;

            % Create TemperatureKSliderLabel
```

```
            app.TemperatureKSliderLabel = uilabel(app.CSTR6UIFigure);
            app.TemperatureKSliderLabel.HorizontalAlignment = 'right';
            app.TemperatureKSliderLabel.Position = [353 377 92 22];
            app.TemperatureKSliderLabel.Text = 'Temperature (K)';

            % Create TSlider
            app.TSlider = uislider(app.CSTR6UIFigure);
            app.TSlider.Limits = [600 1200];
            app.TSlider.ValueChangedFcn = createCallbackFcn(app, @TSliderValueChanged, true);
            app.TSlider.MinorTicks = [];
            app.TSlider.Position = [466 386 150 3];
            app.TSlider.Value = 800;

            % Create FindToptCheckBox
            app.FindToptCheckBox = uicheckbox(app.CSTR6UIFigure);
            app.FindToptCheckBox.ValueChangedFcn = createCallbackFcn(app, @FindToptCheckBoxValueChanged, true);
            app.FindToptCheckBox.Text = 'Find Topt';
            app.FindToptCheckBox.Position = [143 352 71 22];

            % Create tauEdit
            app.tauEdit = uieditfield(app.CSTR6UIFigure, 'text');
            app.tauEdit.ValueChangedFcn = createCallbackFcn(app, @tauEditValueChanged, true);
            app.tauEdit.Position = [315 435 39 22];

            % Create EaEbEdit
            app.EaEbEdit = uieditfield(app.CSTR6UIFigure, 'text');
            app.EaEbEdit.ValueChangedFcn = createCallbackFcn(app, @EaEbEditValueChanged, true);
            app.EaEbEdit.Position = [629 436 39 22];

            % Create TField
            app.TField = uieditfield(app.CSTR6UIFigure, 'text');
            app.TField.ValueChangedFcn = createCallbackFcn(app, @TFieldValueChanged, true);
            app.TField.Position = [629 373 39 22];

            % Show the figure after all components are created
            app.CSTR6UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = cstr6

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.CSTR6UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.CSTR6UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
```

```
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Chemistry 1

chemistry1.mlapp

```
classdef chemistry1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Chemistry1UIFigure  matlab.ui.Figure
        EaSlider            matlab.ui.control.Slider
        EakJmolLabel        matlab.ui.control.Label
        rubrik              matlab.ui.control.Label
        UIAxes_2            matlab.ui.control.UIAxes
        UIAxes              matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Chemistry1UIFigure,'off');
                R=8.314e-3;
                f=@(T1,T2) exp(-app.EaSlider.Value/R*(1./T2-1./T1));
                T=linspace(0,500);
                TK=T+273;
                plot(app.UIAxes,T,f(TK,TK+10),'k');
                app.UIAxes.NextPlot='add';
                T18=find(f(TK,TK+10) >1.8,1,'last');
                T22=find(f(TK,TK+10) >2.2,1,'last');
                if T18 >= 1
                    plot(app.UIAxes,[T(T18) T(T18)],app.UIAxes.YLim,'r--');
                end
                if T22 >= 1
                    plot(app.UIAxes,[T(T22) T(T22)],app.UIAxes.YLim,'r--');
                end
                app.UIAxes.XGrid=1;
                app.UIAxes.YGrid=1;
                app.UIAxes.NextPlot='replace';
                plot(app.UIAxes_2,T,log10(f(TK,TK+100)),'k');
                toogleUIC(app.Chemistry1UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            simulera(app);
        end

        % Value changed function: EaSlider
        function EaSliderValueChanged(app, event)
            simulera(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Chemistry1UIFigure and hide until all components are created
            app.Chemistry1UIFigure = uifigure('Visible', 'off');
```

```
            app.Chemistry1UIFigure.AutoResizeChildren = 'off';
            app.Chemistry1UIFigure.Color = [0.94 0.94 0.94];
            app.Chemistry1UIFigure.Position = [0 0 676 632];
            app.Chemistry1UIFigure.Name = 'Chemistry 1';
            app.Chemistry1UIFigure.Resize = 'off';

            % Create UIAxes
            app.UIAxes = uiaxes(app.Chemistry1UIFigure);
            title(app.UIAxes, 'Increase 10 K')
            xlabel(app.UIAxes, 'Temperature [^oC]')
            ylabel(app.UIAxes, {'k_2/k_1'; ''})
            app.UIAxes.FontName = 'Helvetica';
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.XGrid = 'on';
            app.UIAxes.YGrid = 'on';
            app.UIAxes.Position = [103 275 455 218];

            % Create UIAxes_2
            app.UIAxes_2 = uiaxes(app.Chemistry1UIFigure);
            title(app.UIAxes_2, 'Increase 100 K')
            xlabel(app.UIAxes_2, 'Temperature [^oC]')
            ylabel(app.UIAxes_2, {'log(k_2/k_1)'; ''})
            app.UIAxes_2.FontName = 'Helvetica';
            app.UIAxes_2.XTickLabelRotation = 0;
            app.UIAxes_2.YTickLabelRotation = 0;
            app.UIAxes_2.ZTickLabelRotation = 0;
            app.UIAxes_2.XGrid = 'on';
            app.UIAxes_2.YGrid = 'on';
            app.UIAxes_2.Position = [103 35 455 223];

            % Create rubrik
            app.rubrik = uilabel(app.Chemistry1UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [60 573 557 30];
            app.rubrik.Text = 'The effect of Temperature on the Rate Constant';

            % Create EakJmolLabel
            app.EakJmolLabel = uilabel(app.Chemistry1UIFigure);
            app.EakJmolLabel.HorizontalAlignment = 'right';
            app.EakJmolLabel.Position = [172 539 66 22];
            app.EakJmolLabel.Text = 'Ea (kJ/mol)';

            % Create EaSlider
            app.EaSlider = uislider(app.Chemistry1UIFigure);
            app.EaSlider.Limits = [40 400];
            app.EaSlider.MajorTicks = [40 100 150 200 250 300 350 400];
            app.EaSlider.ValueChangedFcn = createCallbackFcn(app, @EaSliderValueChanged, true);
            app.EaSlider.MinorTicks = [];
            app.EaSlider.Position = [259 548 204 3];
            app.EaSlider.Value = 40;

            % Show the figure after all components are created
            app.Chemistry1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = chemistry1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Chemistry1UIFigure)

            % Execute the startup function
```

```
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Chemistry1UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Chemistry 2

chemistry2.mlapp

```
classdef chemistry2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Chemistry2UIFigure  matlab.ui.Figure
        TabGroup            matlab.ui.container.TabGroup
        ExohermicTab        matlab.ui.container.Tab
        EarSlider           matlab.ui.control.Slider
        EarSliderLabel      matlab.ui.control.Label
        EafSlider           matlab.ui.control.Slider
        EafSliderLabel      matlab.ui.control.Label
        UIAxes              matlab.ui.control.UIAxes
        EndothermicTab      matlab.ui.container.Tab
        ErSlider_2          matlab.ui.control.Slider
        ErSlider_2Label     matlab.ui.control.Label
        EfSlider_2          matlab.ui.control.Slider
        EfSlider_2Label     matlab.ui.control.Label
        UIAxes_2            matlab.ui.control.UIAxes
        rubtik              matlab.ui.control.Label
    end


    methods (Access = private)

        function simulera1(app)
            try
                toogleUIC(app.Chemistry2UIFigure,'off');
                R=8.314e-3;
                k=@(Ea,TK) exp(-Ea/R*(1./TK-1/298));
                T=linspace(260,340);
                K1=@(TK) k(app.EafSlider.Value,TK)./k(app.EarSlider.Value,TK);
                x=@(TK) K1(TK)./(1+K1(TK));
                Trate=linspace(260,340,500);
                xrate=linspace(0,1,500);
                [T1,X1]=meshgrid(Trate,xrate);
                rate=k(app.EafSlider.Value,T1).*(1-X1)-k(app.EarSlider.Value,T1).*X1;
                [I,J]=find(rate<0);
                for i=1:length(I)
                    rate(I(i),J(i))=NaN;
                end
                contourf(app.UIAxes,T1,X1,log10(rate),30);
                c=colorbar(app.UIAxes);
                c.Label.String='log_{10}(r_{net})';
                app.UIAxes.NextPlot='add';
                plot(app.UIAxes,T,x(T),'r','LineWidth',2);
                [~,J]=max(rate,[],2);
                plot(app.UIAxes,Trate(J),xrate,'b','LineWidth',2);
                app.UIAxes.NextPlot='replace';
                toogleUIC(app.Chemistry2UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end

        function simulera2(app)
            try
                toogleUIC(app.Chemistry2UIFigure,'off');
                R=8.314e-3;
                k=@(Ea,TK) exp(-Ea/R*(1./TK-1/298));
                T=linspace(260,320);
                K1=@(TK) k(app.EfSlider_2.Value,TK)./k(app.ErSlider_2.Value,TK);
                x=@(TK) K1(TK)./(1+K1(TK));
                Trate=linspace(260,320,500);
                xrate=linspace(0,1,500);
                [T1,X1]=meshgrid(Trate,xrate);
                rate=k(app.EfSlider_2.Value,T1).*(1-X1)-k(app.ErSlider_2.Value,T1).*X1;
```

```
            [I,J]=find(rate<0);
            for i=1:length(I)
                rate(I(i),J(i))=NaN;
            end
            contourf(app.UIAxes_2,T1,X1,log10(rate),30);
            c=colorbar(app.UIAxes_2);
            c.Label.String='log_{10}(r_{net})';
            app.UIAxes_2.NextPlot='add';
            plot(app.UIAxes_2,T,x(T),'r','LineWidth',2);
            app.UIAxes_2.NextPlot='replace';
            toogleUIC(app.Chemistry2UIFigure,'on');
        catch ME
            errordlg(ME.message);
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        simulera1(app);
        simulera2(app);
    end

    % Value changed function: EafSlider
    function EafSliderValueChanged(app, event)
        simulera1(app);
    end

    % Value changed function: EarSlider
    function EarSliderValueChanged(app, event)
        simulera1(app);
    end

    % Value changed function: EfSlider_2
    function EfSlider_2ValueChanged(app, event)
        simulera2(app);
    end

    % Value changed function: ErSlider_2
    function ErSlider_2ValueChanged(app, event)
        simulera2(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Chemistry2UIFigure and hide until all components are created
        app.Chemistry2UIFigure = uifigure('Visible', 'off');
        app.Chemistry2UIFigure.Color = [0.94 0.94 0.94];
        app.Chemistry2UIFigure.Position = [100 100 740 569];
        app.Chemistry2UIFigure.Name = 'Chemistry 2';

        % Create rubtik
        app.rubtik = uilabel(app.Chemistry2UIFigure);
        app.rubtik.FontSize = 24;
        app.rubtik.FontWeight = 'bold';
        app.rubtik.Position = [185 515 359 30];
        app.rubtik.Text = 'Equilibrium and Reaction Rate';

        % Create TabGroup
        app.TabGroup = uitabgroup(app.Chemistry2UIFigure);
        app.TabGroup.Position = [32 42 678 425];

        % Create ExohermicTab
```

```
app.ExohermicTab = uitab(app.TabGroup);
app.ExohermicTab.Title = 'Exohermic';

% Create UIAxes
app.UIAxes = uiaxes(app.ExohermicTab);
xlabel(app.UIAxes, 'Temperature [K]')
ylabel(app.UIAxes, 'Conversion')
app.UIAxes.XTickLabelRotation = 0;
app.UIAxes.YTickLabelRotation = 0;
app.UIAxes.ZTickLabelRotation = 0;
app.UIAxes.Position = [14 1 637 317];

% Create EafSliderLabel
app.EafSliderLabel = uilabel(app.ExohermicTab);
app.EafSliderLabel.HorizontalAlignment = 'right';
app.EafSliderLabel.Position = [34 361 25 22];
app.EafSliderLabel.Text = 'Eaf';

% Create EafSlider
app.EafSlider = uislider(app.ExohermicTab);
app.EafSlider.Limits = [80 100];
app.EafSlider.ValueChangedFcn = createCallbackFcn(app, @EafSliderValueChanged, true);
app.EafSlider.MinorTicks = [];
app.EafSlider.Position = [80 370 150 3];
app.EafSlider.Value = 80;

% Create EarSliderLabel
app.EarSliderLabel = uilabel(app.ExohermicTab);
app.EarSliderLabel.HorizontalAlignment = 'right';
app.EarSliderLabel.Position = [405 361 25 22];
app.EarSliderLabel.Text = 'Ear';

% Create EarSlider
app.EarSlider = uislider(app.ExohermicTab);
app.EarSlider.Limits = [200 280];
app.EarSlider.MajorTicks = [200 220 240 260 280];
app.EarSlider.ValueChangedFcn = createCallbackFcn(app, @EarSliderValueChanged, true);
app.EarSlider.MinorTicks = [];
app.EarSlider.Position = [451 370 150 3];
app.EarSlider.Value = 200;

% Create EndothermicTab
app.EndothermicTab = uitab(app.TabGroup);
app.EndothermicTab.Title = 'Endothermic';

% Create UIAxes_2
app.UIAxes_2 = uiaxes(app.EndothermicTab);
xlabel(app.UIAxes_2, 'Temperature [K]')
ylabel(app.UIAxes_2, 'Conversion')
app.UIAxes_2.XTickLabelRotation = 0;
app.UIAxes_2.YTickLabelRotation = 0;
app.UIAxes_2.ZTickLabelRotation = 0;
app.UIAxes_2.Position = [14 1 637 317];

% Create EfSlider_2Label
app.EfSlider_2Label = uilabel(app.EndothermicTab);
app.EfSlider_2Label.HorizontalAlignment = 'right';
app.EfSlider_2Label.Position = [34 361 25 22];
app.EfSlider_2Label.Text = 'Ef';

% Create EfSlider_2
app.EfSlider_2 = uislider(app.EndothermicTab);
app.EfSlider_2.Limits = [235 295];
app.EfSlider_2.ValueChangedFcn = createCallbackFcn(app, @EfSlider_2ValueChanged, true);
app.EfSlider_2.MinorTicks = [];
app.EfSlider_2.Position = [80 370 150 3];
app.EfSlider_2.Value = 240;

% Create ErSlider_2Label
app.ErSlider_2Label = uilabel(app.EndothermicTab);
app.ErSlider_2Label.HorizontalAlignment = 'right';
```

```
            app.ErSlider_2Label.Position = [405 361 25 22];
            app.ErSlider_2Label.Text = 'Er';

            % Create ErSlider_2
            app.ErSlider_2 = uislider(app.EndothermicTab);
            app.ErSlider_2.Limits = [80 160];
            app.ErSlider_2.MajorTicks = [80 100 120 140 160];
            app.ErSlider_2.ValueChangedFcn = createCallbackFcn(app, @ErSlider_2ValueChanged, true);
            app.ErSlider_2.MinorTicks = [];
            app.ErSlider_2.Position = [451 370 150 3];
            app.ErSlider_2.Value = 80;

            % Show the figure after all components are created
            app.Chemistry2UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = chemistry2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Chemistry2UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Chemistry2UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Chemistry 3

chemistry3.mlapp

```
classdef chemistry3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Chemistry3UIFigure          matlab.ui.Figure
        alfaDropDown                matlab.ui.control.DropDown
        ReactionorderDropDownLabel  matlab.ui.control.Label
        rubrik                      matlab.ui.control.Label
        UIAxes                      matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Chemistry3UIFigure,'off');
                order=app.alfaDropDown.Value;
                ca=linspace(0,1);
                k=10;
                switch order
                    case '1'
                        rate=ca/k;
                        app.UIAxes.Title.String='$r=c_A/k$';
                    case '2'
                        rate=k*ones(size(ca));
                        app.UIAxes.Title.String='$r=k$';
                    case '3'
                        rate=k*ca.^0.5;
                        app.UIAxes.Title.String='$r=kc_A^{0.5}$';
                    case '4'
                        rate=k*ca;
                        app.UIAxes.Title.String='$r=kc_A$';
                    case '5'
                        rate=k*ca.^2;
                        app.UIAxes.Title.String='$r=kc_A^2$';
                    case '6'
                        K=10;
                        rate=k*ca./(1+K*ca).^2;
                        app.UIAxes.Title.String='$r=\frac{kc_A}{(1+Kc_A)^2}$';
                    case '7'
                        mu=10;
                        Km=0.05;
                        rate=mu*ca./(Km+ca);
                        app.UIAxes.Title.String='$r=\frac{\mu_{max}c_A}{K_m+C_A}$';
                    case '8'
                        rate=k*ca.*(ca(end)-ca);
                        app.UIAxes.Title.String='$r=kc_Ac_R$';
                end
                plot(app.UIAxes,ca,rate,'k')
                app.UIAxes.Title.Interpreter='latex';
                %app.UIAxes.TitleFontSizeMultiplier=1.1;
                toogleUIC(app.Chemistry3UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            app.UIAxes.InnerPosition
```

```
            simulera(app);
        end

        % Value changed function: alfaDropDown
        function alfaDropDownValueChanged(app, event)
            simulera(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Chemistry3UIFigure and hide until all components are created
            app.Chemistry3UIFigure = uifigure('Visible', 'off');
            app.Chemistry3UIFigure.Color = [0.94 0.94 0.94];
            app.Chemistry3UIFigure.Position = [100 100 640 475];
            app.Chemistry3UIFigure.Name = 'Chemistry 3';

            % Create UIAxes
            app.UIAxes = uiaxes(app.Chemistry3UIFigure);
            title(app.UIAxes, 'rate equation')
            xlabel(app.UIAxes, 'Concentration')
            ylabel(app.UIAxes, 'rate')
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.BoxStyle = 'full';
            app.UIAxes.TitleFontWeight = 'normal';
            app.UIAxes.TitleFontSizeMultiplier = 1.5;
            app.UIAxes.Position = [27 15 587 338];

            % Create rubrik
            app.rubrik = uilabel(app.Chemistry3UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [153 425 402 30];
            app.rubrik.Text = {'Reaction Rate and Reaction Order'; ''};

            % Create ReactionorderDropDownLabel
            app.ReactionorderDropDownLabel = uilabel(app.Chemistry3UIFigure);
            app.ReactionorderDropDownLabel.HorizontalAlignment = 'right';
            app.ReactionorderDropDownLabel.Position = [191 381 84 22];
            app.ReactionorderDropDownLabel.Text = 'Reaction order';

            % Create alfaDropDown
            app.alfaDropDown = uidropdown(app.Chemistry3UIFigure);
            app.alfaDropDown.Items = {'-1', '0', '1/2', '1', '2', 'Langmuir-Hinshelwood', 'Michaelis-Menten', 'Autocatalytic'};
            app.alfaDropDown.ItemsData = {'1', '2', '3', '4', '5', '6', '7', '8'};
            app.alfaDropDown.ValueChangedFcn = createCallbackFcn(app, @alfaDropDownValueChanged, true);
            app.alfaDropDown.Position = [290 381 201 22];
            app.alfaDropDown.Value = '4';

            % Show the figure after all components are created
            app.Chemistry3UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = chemistry3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Chemistry3UIFigure)
```

```
            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Chemistry3UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Heterogeneous 1

heterogenous1.mlapp

```
classdef heterogenous1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Heterogeneous1UIFigure      matlab.ui.Figure
        Label_2                     matlab.ui.control.Label
        Label                       matlab.ui.control.Label
        kSlider                     matlab.ui.control.Slider
        rateconstant1sSliderLabel   matlab.ui.control.Label
        DSlider                     matlab.ui.control.Slider
        diffusivitycm2sSliderLabel  matlab.ui.control.Label
        rSlider                     matlab.ui.control.Slider
        pelletradiuscmSliderLabel   matlab.ui.control.Label
        DiffusionandReactioninaCatalystPelletLabel  matlab.ui.control.Label
        UIAxes                      matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.Heterogeneous1UIFigure,'off');
                phi=0.4;sigma=0.8;tau=3;
                app.diffusivitycm2sSliderLabel.Text=sprintf('diffusivity [cm\x00b2/s]');
                R=app.rSlider.Value;
                D=app.DSlider.Value*phi*sigma/tau;
                k=app.kSlider.Value;
                Thiele=R*sqrt(k/D);
                app.Label.Text=sprintf('Thiele modules=%4.2f',Thiele);
                eta=3./Thiele.^2.*(Thiele.*coth(Thiele)-1);
                app.Label_2.Text=sprintf('effectiveness factor=%4.2f',eta);
                cAs=0.04;
                r=linspace(0,R,100);
                cA=cAs*R./r.*sinh(Thiele*r./R)/sinh(Thiele);
                plot(app.UIAxes,r,cA,'k-',[0 R],[cAs cAs],'k--','LineWidth',1.5)
                toogleUIC(app.Heterogeneous1UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end

    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            draw(app);
        end

        % Value changed function: DSlider, kSlider, rSlider
        function rSliderValueChanged(app, event)
            draw(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Heterogeneous1UIFigure and hide until all components are created
```

```
app.Heterogeneous1UIFigure = uifigure('Visible', 'off');
app.Heterogeneous1UIFigure.Color = [0.94 0.94 0.94];
app.Heterogeneous1UIFigure.Position = [100 100 740 542];
app.Heterogeneous1UIFigure.Name = 'Heterogeneous 1';
app.Heterogeneous1UIFigure.Scrollable = 'on';

% Create UIAxes
app.UIAxes = uiaxes(app.Heterogeneous1UIFigure);
xlabel(app.UIAxes, 'catalyst pellet raduis [cm]')
ylabel(app.UIAxes, {'concentration [mmol/cm^3]'; ''})
app.UIAxes.YLim = [0 0.05];
app.UIAxes.XTickLabelRotation = 0;
app.UIAxes.YTickLabelRotation = 0;
app.UIAxes.ZTickLabelRotation = 0;
app.UIAxes.Position = [23 42 477 344];

% Create DiffusionandReactioninaCatalystPelletLabel
app.DiffusionandReactioninaCatalystPelletLabel = uilabel(app.Heterogeneous1UIFigure);
app.DiffusionandReactioninaCatalystPelletLabel.FontSize = 24;
app.DiffusionandReactioninaCatalystPelletLabel.FontWeight = 'bold';
app.DiffusionandReactioninaCatalystPelletLabel.Position = [137 492 495 30];
app.DiffusionandReactioninaCatalystPelletLabel.Text = 'Diffusion and Reaction in a Catalyst Pellet';

% Create pelletradiuscmSliderLabel
app.pelletradiuscmSliderLabel = uilabel(app.Heterogeneous1UIFigure);
app.pelletradiuscmSliderLabel.HorizontalAlignment = 'right';
app.pelletradiuscmSliderLabel.Position = [15 462 96 22];
app.pelletradiuscmSliderLabel.Text = 'pellet radius [cm]';

% Create rSlider
app.rSlider = uislider(app.Heterogeneous1UIFigure);
app.rSlider.Limits = [0.1 1];
app.rSlider.MajorTicks = [0.1 0.3 0.5 0.7 0.9 1];
app.rSlider.ValueChangedFcn = createCallbackFcn(app, @rSliderValueChanged, true);
app.rSlider.MinorTicks = [0 0.2 0.4 0.6 0.8 0.9 4];
app.rSlider.Position = [132 471 150 3];
app.rSlider.Value = 0.5;

% Create diffusivitycm2sSliderLabel
app.diffusivitycm2sSliderLabel = uilabel(app.Heterogeneous1UIFigure);
app.diffusivitycm2sSliderLabel.HorizontalAlignment = 'right';
app.diffusivitycm2sSliderLabel.Position = [413 466 102 22];
app.diffusivitycm2sSliderLabel.Text = {'diffusivity [cm^2/s]'; ''};

% Create DSlider
app.DSlider = uislider(app.Heterogeneous1UIFigure);
app.DSlider.Limits = [1 100];
app.DSlider.MajorTicks = [1 20 40 60 80 100];
app.DSlider.ValueChangedFcn = createCallbackFcn(app, @rSliderValueChanged, true);
app.DSlider.MinorTicks = [];
app.DSlider.Position = [536 475 150 3];
app.DSlider.Value = 50;

% Create rateconstant1sSliderLabel
app.rateconstant1sSliderLabel = uilabel(app.Heterogeneous1UIFigure);
app.rateconstant1sSliderLabel.HorizontalAlignment = 'right';
app.rateconstant1sSliderLabel.Position = [413 420 101 22];
app.rateconstant1sSliderLabel.Text = {'rate constant [1/s]'; ''};

% Create kSlider
app.kSlider = uislider(app.Heterogeneous1UIFigure);
app.kSlider.Limits = [5 45];
app.kSlider.ValueChangedFcn = createCallbackFcn(app, @rSliderValueChanged, true);
app.kSlider.MinorTicks = [];
app.kSlider.Position = [535 429 150 3];
app.kSlider.Value = 20;

% Create Label
app.Label = uilabel(app.Heterogeneous1UIFigure);
app.Label.FontSize = 14;
app.Label.FontWeight = 'bold';
```

```
            app.Label.Position = [508 353 229 22];
            app.Label.Text = '';

            % Create Label_2
            app.Label_2 = uilabel(app.Heterogeneous1UIFigure);
            app.Label_2.FontSize = 14;
            app.Label_2.FontWeight = 'bold';
            app.Label_2.Position = [508 323 229 22];
            app.Label_2.Text = '';

            % Show the figure after all components are created
            app.Heterogeneous1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = heterogenous1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Heterogeneous1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Heterogeneous1UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Multiple reactors 1

multiple1.mlapp

```
classdef multiple1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Multiplereactors1UIFigure      matlab.ui.Figure
        ExitconversionPanel            matlab.ui.container.Panel
        x2Field                        matlab.ui.control.EditField
        x1Field                        matlab.ui.control.EditField
        x2Slider                       matlab.ui.control.Slider
        Reactor2Label                  matlab.ui.control.Label
        x1Slider                       matlab.ui.control.Slider
        Reactor1Label                  matlab.ui.control.Label
        Label_2                        matlab.ui.control.Label
        Label_3                        matlab.ui.control.Label
        Label                          matlab.ui.control.Label
        kineticsDropDown               matlab.ui.control.DropDown
        reactionkineticsDropDownLabel  matlab.ui.control.Label
        reactorDropDown                matlab.ui.control.DropDown
        reactor1reactor2DropDownLabel  matlab.ui.control.Label
        rubrik                         matlab.ui.control.Label
        UIAxes                         matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.Multiplereactors1UIFigure,'off');
                Ca0=2;
                x1=app.x1Slider.Value;
                x2=app.x2Slider.Value;
                app.x2Slider.Limits=[x1 1];
                app.x1Slider.Limits=[0 x2];
                conv=linspace(0,1);
                ra=rate(conv);
                app.UIAxes.NextPlot='replaceall';
                plot(app.UIAxes,conv,ra,'k','LineWidth',1.5)
                app.UIAxes.XLabel.String='conversion';
                app.UIAxes.YLabel.String='-1/r_A';
                app.UIAxes.YLim=[0 max(rate(linspace(0,x2,50)))];
                app.UIAxes.NextPlot='add';
                if app.reactorDropDown.Value == '1' || app.reactorDropDown.Value == '3'
                    V1=Ca0*x1*rate(x1);
                    x=[0 x1 x1 0 0];
                    y=[0 0 rate(x1) rate(x1) 0];
                    fill(app.UIAxes,x,y,'y');
                elseif app.reactorDropDown.Value == '2' || app.reactorDropDown.Value == '4'
                    V1=Ca0*integral(@rate,0,x1);
                    x=[0 x1 linspace(x1,0,50) 0];
                    y=[0 0 rate(linspace(x1,0,50)) 0];
                    fill(app.UIAxes,x,y,'r');
                else
                    V1=0;
                end
                if app.reactorDropDown.Value == '2' || app.reactorDropDown.Value == '3'
                    V2=Ca0*(x2-x1)*rate(x2);
                    x=[x1 x2 x2 x1 x1];
                    y=[0 0 rate(x2) rate(x2) 0];
                    fill(app.UIAxes,x,y,'g');
                elseif app.reactorDropDown.Value == '1' || app.reactorDropDown.Value == '4'
                    V2=Ca0*integral(@rate,x1,x2);
                    x=[x1 x2 linspace(x2,x1,50) x1];
                    y=[0 0 rate(linspace(x2,x1,50)) 0];
                    fill(app.UIAxes,x,y,'r');
                else
                    V2=0;
```

```
            end
            app.Label.Text=sprintf('Reactor 1=%5.2f L',V1);
            app.Label_3.Text=sprintf('Reactor 2=%5.2f L',V2);
            app.Label_2.Text=sprintf('        Total=%5.2f L',V1+V2);
            plot(app.UIAxes,conv,ra,'k','LineWidth',1.5)
            toogleUIC(app.Multiplereactors1UIFigure,'on');
        catch ME
            errordlg(ME.message);
        end
        function y = rate(oms)
            k=1;
            if app.kineticsDropDown.Value == '1'
                y=k*Ca0*(1-oms);
            elseif app.kineticsDropDown.Value == '2'
                y=k*Ca0*(1-oms).^2;
            elseif app.kineticsDropDown.Value == '3'
                y=(0.5+1.2*Ca0)^2*k*Ca0*(1-oms)./(0.5+1.2*Ca0*(1-oms)).^2;
            else
                y=ones(size(oms));
            end
            y=1./y;
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.x1Field.Value=sprintf('%4.2f',app.x1Slider.Value);
        app.x2Field.Value=sprintf('%4.2f',app.x2Slider.Value);
        draw(app);
    end

    % Value changed function: x1Slider
    function x1SliderValueChanged(app, event)
        value = app.x1Slider.Value;
        app.x1Field.Value=sprintf('%4.2f',value);
        draw(app);
    end

    % Callback function
    function x1FieldValueChanged(app, event)
        value = app.x1Field.Value;
        if between(app,value,app.x1Slider.Limits)
            app.x1Slider.Value=str2double(value);
            draw(app);
        else
            app.x1Field.Value=num2str(app.x1Slider.Value);
        end
    end

    % Callback function
    function x2FieldValueChanged(app, event)
        value = app.x2Field.Value;
        if between(app,value,app.x2Slider.Limits)
            app.x2Slider.Value=str2double(value);
            draw(app);
        else
            app.x2Field.Value=sprintf('%4.2f',app.x2Slider.Value);
        end
    end

    % Value changed function: x2Slider
    function x2SliderValueChanged(app, event)
        value = app.x2Slider.Value;
        app.x2Field.Value=sprintf('%4.2f',value);
        draw(app);
    end
```

```
        % Value changed function: kineticsDropDown
        function kineticsDropDownValueChanged(app, event)
            draw(app);
        end

        % Value changed function: reactorDropDown
        function reactorDropDownValueChanged(app, event)
            draw(app);
        end

        % Value changed function: x1Field
        function x1FieldValueChanged2(app, event)
            value = app.x1Field.Value;
            if between(app,value,app.x1Slider.Limits)
                app.x1Slider.Value=str2double(value);
                draw(app);
            else
                app.x1Field.Value=num2str(app.x1Slider.Value);
            end
        end

        % Value changed function: x2Field
        function x2FieldValueChanged2(app, event)
            value = app.x2Field.Value;
            if between(app,value,app.x2Slider.Limits)
                app.x2Slider.Value=str2double(value);
                draw(app);
            else
                app.x2Field.Value=sprintf('%4.2f',app.x2Slider.Value);
            end
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Multiplereactors1UIFigure and hide until all components are created
            app.Multiplereactors1UIFigure = uifigure('Visible', 'off');
            app.Multiplereactors1UIFigure.Color = [0.94 0.94 0.94];
            app.Multiplereactors1UIFigure.Position = [100 100 681 599];
            app.Multiplereactors1UIFigure.Name = 'Multiple reactors 1';
            app.Multiplereactors1UIFigure.Scrollable = 'on';

            % Create UIAxes
            app.UIAxes = uiaxes(app.Multiplereactors1UIFigure);
            xlabel(app.UIAxes, 'conversion')
            ylabel(app.UIAxes, '-1/r_A')
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.NextPlot = 'add';
            app.UIAxes.Position = [70 28 427 274];

            % Create rubrik
            app.rubrik = uilabel(app.Multiplereactors1UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [109 561 477 30];
            app.rubrik.Text = 'Minimized Volume for Reactors in Series';

            % Create reactor1reactor2DropDownLabel
            app.reactor1reactor2DropDownLabel = uilabel(app.Multiplereactors1UIFigure);
            app.reactor1reactor2DropDownLabel.HorizontalAlignment = 'right';
            app.reactor1reactor2DropDownLabel.Position = [15 524 141 22];
            app.reactor1reactor2DropDownLabel.Text = 'reactor 1, reactor 2';

            % Create reactorDropDown
            app.reactorDropDown = uidropdown(app.Multiplereactors1UIFigure);
```

```
app.reactorDropDown.Items = {'CSTR,PFR', 'PFR,CSTR', 'CSTR,CSTR', 'PFR,PFR', ''};
app.reactorDropDown.ItemsData = {'1', '2', '3', '4'};
app.reactorDropDown.ValueChangedFcn = createCallbackFcn(app, @reactorDropDownValueChanged, true);
app.reactorDropDown.Position = [174 524 100 22];
app.reactorDropDown.Value = '1';

% Create reactionkineticsDropDownLabel
app.reactionkineticsDropDownLabel = uilabel(app.Multiplereactors1UIFigure);
app.reactionkineticsDropDownLabel.HorizontalAlignment = 'right';
app.reactionkineticsDropDownLabel.Position = [342 524 92 22];
app.reactionkineticsDropDownLabel.Text = 'reaction kinetics';

% Create kineticsDropDown
app.kineticsDropDown = uidropdown(app.Multiplereactors1UIFigure);
app.kineticsDropDown.Items = {'first order', 'second order', 'Langmuir-Hinshelwood'};
app.kineticsDropDown.ItemsData = {'1', '2', '3', ''};
app.kineticsDropDown.ValueChangedFcn = createCallbackFcn(app, @kineticsDropDownValueChanged, true);
app.kineticsDropDown.Position = [449 524 137 22];
app.kineticsDropDown.Value = '1';

% Create Label
app.Label = uilabel(app.Multiplereactors1UIFigure);
app.Label.FontSize = 14;
app.Label.FontWeight = 'bold';
app.Label.Position = [495 322 162 22];
app.Label.Text = '';

% Create Label_3
app.Label_3 = uilabel(app.Multiplereactors1UIFigure);
app.Label_3.FontSize = 14;
app.Label_3.FontWeight = 'bold';
app.Label_3.Position = [495 301 162 22];
app.Label_3.Text = '';

% Create Label_2
app.Label_2 = uilabel(app.Multiplereactors1UIFigure);
app.Label_2.FontSize = 14;
app.Label_2.FontWeight = 'bold';
app.Label_2.Position = [495 270 162 22];
app.Label_2.Text = '';

% Create ExitconversionPanel
app.ExitconversionPanel = uipanel(app.Multiplereactors1UIFigure);
app.ExitconversionPanel.Title = 'Exit conversion';
app.ExitconversionPanel.FontWeight = 'bold';
app.ExitconversionPanel.FontSize = 14;
app.ExitconversionPanel.Position = [71 351 515 154];

% Create Reactor1Label
app.Reactor1Label = uilabel(app.ExitconversionPanel);
app.Reactor1Label.HorizontalAlignment = 'right';
app.Reactor1Label.Position = [68 102 58 22];
app.Reactor1Label.Text = 'Reactor 1';

% Create x1Slider
app.x1Slider = uislider(app.ExitconversionPanel);
app.x1Slider.Limits = [0 1];
app.x1Slider.ValueChangedFcn = createCallbackFcn(app, @x1SliderValueChanged, true);
app.x1Slider.MinorTicks = [];
app.x1Slider.Position = [147 111 150 3];
app.x1Slider.Value = 0.4;

% Create Reactor2Label
app.Reactor2Label = uilabel(app.ExitconversionPanel);
app.Reactor2Label.HorizontalAlignment = 'right';
app.Reactor2Label.Position = [23 27 103 43];
app.Reactor2Label.Text = 'Reactor 2';

% Create x2Slider
app.x2Slider = uislider(app.ExitconversionPanel);
app.x2Slider.Limits = [0 1];
```

```
            app.x2Slider.ValueChangedFcn = createCallbackFcn(app, @x2SliderValueChanged, true);
            app.x2Slider.MinorTicks = [];
            app.x2Slider.Position = [147 47 150 3];
            app.x2Slider.Value = 0.6;

            % Create x1Field
            app.x1Field = uieditfield(app.ExitconversionPanel, 'text');
            app.x1Field.ValueChangedFcn = createCallbackFcn(app, @x1FieldValueChanged2, true);
            app.x1Field.Position = [314 101 49 22];

            % Create x2Field
            app.x2Field = uieditfield(app.ExitconversionPanel, 'text');
            app.x2Field.ValueChangedFcn = createCallbackFcn(app, @x2FieldValueChanged2, true);
            app.x2Field.Position = [314 37 49 22];

            % Show the figure after all components are created
            app.Multiplereactors1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = multiple1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Multiplereactors1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Multiplereactors1UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
```

```
        uic.Enable = state;
    end
end
drawnow; % flush the ui events queue
end
```

Multiple reactors 2

multiple2.mlapp

```
classdef multiple2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Multiplereactors2UIFigure      matlab.ui.Figure
        x1Field                        matlab.ui.control.EditField
        ProductionLabel                matlab.ui.control.Label
        xLabel                         matlab.ui.control.Label
        Fa0Label                       matlab.ui.control.Label
        VarningLabel                   matlab.ui.control.Label
        reactor2DropDown               matlab.ui.control.DropDown
        reactor2Label                  matlab.ui.control.Label
        x1Slider                       matlab.ui.control.Slider
        conversionexitreactor1Label    matlab.ui.control.Label
        kineticsDropDown               matlab.ui.control.DropDown
        reactionkineticsDropDownLabel  matlab.ui.control.Label
        reactor1DropDown               matlab.ui.control.DropDown
        reactor1Label                  matlab.ui.control.Label
        rubrik                         matlab.ui.control.Label
        UIAxes                         matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.Multiplereactors2UIFigure,'off');
                Ca0=2;
                x1=app.x1Slider.Value;
                kinetics=app.kineticsDropDown.Value;
                reactor1=app.reactor1DropDown.Value;
                reactor2=app.reactor2DropDown.Value;
                if reactor1 == '1' || reactor1 == '3'
                    V1=1;
                elseif reactor1 =='2' || reactor1 =='4'
                    V1=2;
                end
                if reactor2 == '1' || reactor2 == '3'
                    V2=1;
                elseif reactor2 =='2' || reactor2 =='4'
                    V2=2;
                end
                conv=linspace(0,1);
                ra=rate(conv);
                app.UIAxes.NextPlot='replaceall';
                plot(app.UIAxes,conv,ra,'k','LineWidth',1.5)
                app.UIAxes.XLabel.String='conversion';
                app.UIAxes.YLabel.String='-1/r_A';
                app.UIAxes.NextPlot='add';
                if reactor1 == '1' || reactor1 == '2'
                    Fa0=V1/rate(x1)/x1;
                    x=[0 x1 x1 0 0];
                    y=[0 0 rate(x1) rate(x1) 0];
                    fill(app.UIAxes,x,y,'y');
                elseif reactor1 == '3' || reactor1 == '4'
                    Fa0=V1/integral(@rate,0,x1);
                    x=[0 x1 linspace(x1,0,50) 0];
                    y=[0 0 rate(linspace(x1,0,50)) 0];
                    fill(app.UIAxes,x,y,'r');
                else
                    Fa0=0;
                end
                if reactor(x1)*reactor(0.9999) > 0
                    app.VarningLabel.Text='Reactor to large';
                    app.VarningLabel.Visible=1;
                    x2=0.9999;
```

```
            else
                x2=fzero(@reactor,[x1 0.9999],optimset('Display','off'));
                app.VarningLabel.Visible=0;
            end
            x2plot=min(x2,0.99);
            app.xLabel.Text=sprintf('Total conversion %4.1f %%',x2*100);
            if reactor2 == '1' || reactor2 == '2'
                x=[x1 x2plot x2plot x1 x1];
                y=[0 0 rate(x2plot) rate(x2plot) 0];
                fill(app.UIAxes,x,y,'g');
            elseif reactor2 == '3' || reactor2 == '4'
                x=[x1 x2plot linspace(x2plot,x1,50) x1];
                y=[0 0 rate(linspace(x2plot,x1,50)) 0];
                fill(app.UIAxes,x,y,'r');
            else
                Fa0=0;
            end
            app.UIAxes.YLim=[0 max(rate(linspace(0,x2plot,50)))];
            app.Fa0Label.Text=sprintf('Molar flow feed %5.2f mol/s',Fa0);
            app.ProductionLabel.Text=sprintf('Total production %5.2f mol/s',Fa0*x2);
            plot(app.UIAxes,conv,ra,'k','LineWidth',1.5)
            toogleUIC(app.Multiplereactors2UIFigure,'on');
        catch ME
            errordlg(ME.message);
        end
        function y = rate(oms)
            k=1;
            if kinetics == '1'
                y=k*Ca0*(1-oms);
            elseif kinetics == '2'
                y=k*Ca0*(1-oms).^2;
            elseif kinetics == '3'
                y=(0.5+1.2*Ca0)^2*k*Ca0*(1-oms)./(0.5+1.2*Ca0*(1-oms)).^2;
            else
                y=ones(size(oms));
            end
            y=1./y;
        end
        function res = reactor(x)
            if reactor2 == '1' || reactor2 == '2'
                res=V2/Fa0-(x-x1)*rate(x);
            elseif reactor2 == '3' || reactor2 == '4'
                res=V2/Fa0-integral(@rate,x1,x);
            else
                res=0;
            end
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.x1Field.Value=sprintf('%4.2f',app.x1Slider.Value);
        draw(app);
    end

    % Value changed function: x1Field
    function x1EditFieldValueChanged(app, event)
        value = app.x1Field.Value;
        if between(app,value,app.x1Slider.Limits)
            app.x1Slider.Value=str2double(value);
            draw(app);
        else
            app.x1Field.Value=num2str(app.x1Slider.Value);
        end
    end

    % Value changed function: reactor1DropDown
```

```
    function reactor1DropDownValueChanged(app, event)
        draw(app);
    end

    % Value changed function: reactor2DropDown
    function reactor2DropDownValueChanged(app, event)
        draw(app);
    end

    % Value changed function: x1Slider
    function x1SliderValueChanged(app, event)
        value = app.x1Slider.Value;
        app.x1Field.Value=sprintf('%4.2f',value);
        draw(app);
    end

    % Value changed function: kineticsDropDown
    function kineticsDropDownValueChanged(app, event)
        draw(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Multiplereactors2UIFigure and hide until all components are created
        app.Multiplereactors2UIFigure = uifigure('Visible', 'off');
        app.Multiplereactors2UIFigure.Color = [0.94 0.94 0.94];
        app.Multiplereactors2UIFigure.Position = [100 100 589 675];
        app.Multiplereactors2UIFigure.Name = 'Multiple reactors 2';
        app.Multiplereactors2UIFigure.Scrollable = 'on';

        % Create UIAxes
        app.UIAxes = uiaxes(app.Multiplereactors2UIFigure);
        xlabel(app.UIAxes, 'conversion')
        ylabel(app.UIAxes, '-1/r_A')
        app.UIAxes.PlotBoxAspectRatio = [1.35174418604651 1 1];
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.NextPlot = 'add';
        app.UIAxes.Position = [36 11 516 386];

        % Create rubrik
        app.rubrik = uilabel(app.Multiplereactors2UIFigure);
        app.rubrik.FontSize = 24;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [225 637 200 30];
        app.rubrik.Text = 'Reactors in serie';

        % Create reactor1Label
        app.reactor1Label = uilabel(app.Multiplereactors2UIFigure);
        app.reactor1Label.HorizontalAlignment = 'right';
        app.reactor1Label.Position = [50 602 61 22];
        app.reactor1Label.Text = 'reactor 1';

        % Create reactor1DropDown
        app.reactor1DropDown = uidropdown(app.Multiplereactors2UIFigure);
        app.reactor1DropDown.Items = {'CSTR 1000 L', 'CSTR 2000 L', 'PFR 1000 L', 'PFR 2000 L', '', ''};
        app.reactor1DropDown.ItemsData = {'1', '2', '3', '4'};
        app.reactor1DropDown.ValueChangedFcn = createCallbackFcn(app, @reactor1DropDownValueChanged, true);
        app.reactor1DropDown.Position = [129 602 156 22];
        app.reactor1DropDown.Value = '1';

        % Create reactionkineticsDropDownLabel
        app.reactionkineticsDropDownLabel = uilabel(app.Multiplereactors2UIFigure);
        app.reactionkineticsDropDownLabel.HorizontalAlignment = 'right';
        app.reactionkineticsDropDownLabel.Position = [303 602 92 22];
```

```
app.reactionkineticsDropDownLabel.Text = 'reaction kinetics';

% Create kineticsDropDown
app.kineticsDropDown = uidropdown(app.Multiplereactors2UIFigure);
app.kineticsDropDown.Items = {'first order', 'second order', 'Langmuir-Hinshelwood'};
app.kineticsDropDown.ItemsData = {'1', '2', '3', ''};
app.kineticsDropDown.ValueChangedFcn = createCallbackFcn(app, @kineticsDropDownValueChanged, true);
app.kineticsDropDown.Position = [410 602 157 22];
app.kineticsDropDown.Value = '1';

% Create conversionexitreactor1Label
app.conversionexitreactor1Label = uilabel(app.Multiplereactors2UIFigure);
app.conversionexitreactor1Label.HorizontalAlignment = 'right';
app.conversionexitreactor1Label.Position = [62 503 136 22];
app.conversionexitreactor1Label.Text = 'conversion exit reactor 1';

% Create x1Slider
app.x1Slider = uislider(app.Multiplereactors2UIFigure);
app.x1Slider.Limits = [0 1];
app.x1Slider.ValueChangedFcn = createCallbackFcn(app, @x1SliderValueChanged, true);
app.x1Slider.MinorTicks = [];
app.x1Slider.Position = [219 512 150 3];
app.x1Slider.Value = 0.5;

% Create reactor2Label
app.reactor2Label = uilabel(app.Multiplereactors2UIFigure);
app.reactor2Label.HorizontalAlignment = 'right';
app.reactor2Label.Position = [44 553 71 22];
app.reactor2Label.Text = 'reactor 2';

% Create reactor2DropDown
app.reactor2DropDown = uidropdown(app.Multiplereactors2UIFigure);
app.reactor2DropDown.Items = {'CSTR 1000 L', 'CSTR 2000 L', 'PFR 1000 L', 'PFR 2000 L', ''};
app.reactor2DropDown.ItemsData = {'1', '2', '3', '4'};
app.reactor2DropDown.ValueChangedFcn = createCallbackFcn(app, @reactor2DropDownValueChanged, true);
app.reactor2DropDown.Position = [133 553 152 22];
app.reactor2DropDown.Value = '1';

% Create VarningLabel
app.VarningLabel = uilabel(app.Multiplereactors2UIFigure);
app.VarningLabel.FontSize = 16;
app.VarningLabel.FontWeight = 'bold';
app.VarningLabel.FontColor = [1 0 0];
app.VarningLabel.Position = [70 437 215 22];
app.VarningLabel.Text = 'Varning';

% Create Fa0Label
app.Fa0Label = uilabel(app.Multiplereactors2UIFigure);
app.Fa0Label.FontWeight = 'bold';
app.Fa0Label.Position = [70 415 171 22];
app.Fa0Label.Text = 'Fa0';

% Create xLabel
app.xLabel = uilabel(app.Multiplereactors2UIFigure);
app.xLabel.FontWeight = 'bold';
app.xLabel.Position = [225 416 170 22];
app.xLabel.Text = 'x';

% Create ProductionLabel
app.ProductionLabel = uilabel(app.Multiplereactors2UIFigure);
app.ProductionLabel.FontWeight = 'bold';
app.ProductionLabel.Position = [368 416 170 21];
app.ProductionLabel.Text = 'Production';

% Create x1Field
app.x1Field = uieditfield(app.Multiplereactors2UIFigure, 'text');
app.x1Field.ValueChangedFcn = createCallbackFcn(app, @x1EditFieldValueChanged, true);
app.x1Field.Position = [390 499 50 22];

% Show the figure after all components are created
app.Multiplereactors2UIFigure.Visible = 'on';
```

```
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = multiple2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Multiplereactors2UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Multiplereactors2UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Multiple reactors 3

multiple3.mlapp

```
classdef multiple3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Multiplereactors3UIFigure  matlab.ui.Figure
        Compare                    matlab.ui.control.Label
        N                          matlab.ui.control.Spinner
        NoofCSTRSpinnerLabel       matlab.ui.control.Label
        tau                        matlab.ui.control.Slider
        totalresidencetimeLabel    matlab.ui.control.Label
        Rubrik                     matlab.ui.control.Label
        UIAxes                     matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Multiplereactors3UIFigure,'off');
                k=1;
                Tau=app.tau.Value;
                nReactors=app.N.Value;
                Ca0=1;
                tau1=linspace(0,Tau);
                C1=Ca0*exp(-k.*tau1);
                tau2=0:Tau/nReactors:Tau;
                C2=[1 Ca0./(1+k*Tau/nReactors).^(1:nReactors)];
                tau3=zeros(2*nReactors+1,1);C3=zeros(2*nReactors+1,1);
                for i=1:nReactors
                    tau3(2*i-1)=tau2(i);tau3(2*i)=tau2(i);
                    C3(2*i-1)=C2(i);C3(2*i)=C2(i+1);
                end
                tau3(end)=tau2(end);C3(end)=C2(end);
                app.Compare.Text=sprintf('PFR have %4.1f%% higher conversion than CSTR',(C2(end)-C1(end))/(1-C1(end))*100);
                plot(app.UIAxes,tau1,C1,'k',...
                    tau3,C3,'r');
                toogleUIC(app.Multiplereactors3UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            simulera(app);
        end

        % Value changed function: tau
        function tauValueChanged(app, event)
            simulera(app);
        end

        % Value changed function: N
        function NValueChanged(app, event)
            simulera(app);
        end
    end

    % Component initialization
    methods (Access = private)
```

```
    % Create UIFigure and components
    function createComponents(app)

        % Create Multiplereactors3UIFigure and hide until all components are created
        app.Multiplereactors3UIFigure = uifigure('Visible', 'off');
        app.Multiplereactors3UIFigure.Color = [0.94 0.94 0.94];
        app.Multiplereactors3UIFigure.Position = [100 100 717 548];
        app.Multiplereactors3UIFigure.Name = 'Multiple reactors 3';

        % Create UIAxes
        app.UIAxes = uiaxes(app.Multiplereactors3UIFigure);
        xlabel(app.UIAxes, 'Residence time')
        ylabel(app.UIAxes, 'Concentration')
        app.UIAxes.PlotBoxAspectRatio = [1.83946488294314 1 1];
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTick = [0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1];
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.Position = [40 26 597 341];

        % Create Rubrik
        app.Rubrik = uilabel(app.Multiplereactors3UIFigure);
        app.Rubrik.FontSize = 18;
        app.Rubrik.FontWeight = 'bold';
        app.Rubrik.Position = [115 511 490 22];
        app.Rubrik.Text = 'Equally Sized CSTR in Series versus Plug Flow Reactor';

        % Create totalresidencetimeLabel
        app.totalresidencetimeLabel = uilabel(app.Multiplereactors3UIFigure);
        app.totalresidencetimeLabel.HorizontalAlignment = 'right';
        app.totalresidencetimeLabel.Position = [40 457 110 22];
        app.totalresidencetimeLabel.Text = 'total residence time';

        % Create tau
        app.tau = uislider(app.Multiplereactors3UIFigure);
        app.tau.Limits = [1 5];
        app.tau.MajorTicks = [1 2 3 4 5];
        app.tau.ValueChangedFcn = createCallbackFcn(app, @tauValueChanged, true);
        app.tau.MinorTicks = [];
        app.tau.Position = [171 466 150 3];
        app.tau.Value = 2;

        % Create NoofCSTRSpinnerLabel
        app.NoofCSTRSpinnerLabel = uilabel(app.Multiplereactors3UIFigure);
        app.NoofCSTRSpinnerLabel.HorizontalAlignment = 'right';
        app.NoofCSTRSpinnerLabel.Position = [440 447 70 22];
        app.NoofCSTRSpinnerLabel.Text = 'No of CSTR';

        % Create N
        app.N = uispinner(app.Multiplereactors3UIFigure);
        app.N.Limits = [1 30];
        app.N.ValueChangedFcn = createCallbackFcn(app, @NValueChanged, true);
        app.N.Position = [525 447 100 22];
        app.N.Value = 1;

        % Create Compare
        app.Compare = uilabel(app.Multiplereactors3UIFigure);
        app.Compare.FontSize = 14;
        app.Compare.FontWeight = 'bold';
        app.Compare.Position = [83 389 332 22];

        % Show the figure after all components are created
        app.Multiplereactors3UIFigure.Visible = 'on';
    end
end

% App creation and deletion
methods (Access = public)

    % Construct app
    function app = multiple3
```

```
            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Multiplereactors3UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Multiplereactors3UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Nonideal 1

nonideal1.mlapp

```
classdef nonideal1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Nonideal1UIFigure          matlab.ui.Figure
        TabGroup                   matlab.ui.container.TabGroup
        XTab                       matlab.ui.container.Tab
        UIAxes_1                   matlab.ui.control.UIAxes
        EtTab                      matlab.ui.container.Tab
        UIAxes_2                   matlab.ui.control.UIAxes
        FtTab                      matlab.ui.container.Tab
        UIAxes_3                   matlab.ui.control.UIAxes
        order                      matlab.ui.control.Slider
        ReactionorderSliderLabel   matlab.ui.control.Label
        ca0                        matlab.ui.control.Slider
        IntitalconcentrationSliderLabel  matlab.ui.control.Label
        k                          matlab.ui.control.Slider
        RateconstantSliderLabel    matlab.ui.control.Label
        rubrik                     matlab.ui.control.Label
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Nonideal1UIFigure,'off');
                tspan=[0 14];
                tplot=linspace(tspan(1),tspan(2));
                area=integral(@meas,tspan(1),tspan(2));
                Et=meas(tplot)/area;
                plot(app.UIAxes_2,tplot,Et);
                [t,y]=ode45(@ODEfun,tspan,zeros(4,1));
                plot(app.UIAxes_1,t,y(:,1:2))
                legend(app.UIAxes_1,'Ideal','Non-ideal','location','best')
                plot(app.UIAxes_3,t,y(:,4))
                toogleUIC(app.Nonideal1UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function Ct = meas(tplot)
                Ct=zeros(size(tplot));
                for i=1:length(tplot)
                    t=tplot(i);
                    if t <= 4 && t>=0
                        Ct(i)= 0.0038746 + 0.2739782*t + 1.574621*t.^2 - 0.2550041*t.^3;
                    elseif t > 4 && t<=14
                        Ct(i)= -33.43818 + 37.18972*t - 11.58838*t.^2 + 1.695303*t.^3 - 0.1298667*t.^4 + 0.005028*t.^5 - 7.743*1
                    else
                        Ct(i)=0;
                    end
                end
            end
            function dy = ODEfun(t,y)
                dy=zeros(size(y));
                X= y(1);
                E=meas(t)/area;
                dy(1)=app.k.Value*(app.ca0.Value^app.order.Value)*((1-X)^app.order.Value)/app.ca0.Value;
                dy(2) = X*E;
                dy(3) = t*E;
                dy(4) = E;
            end

        end
    end

    % Callbacks that handle component events
```

```
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        simulera(app);
    end

    % Value changed function: k
    function kValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: ca0
    function ca0ValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: order
    function orderValueChanged(app, event)
        value = app.order.Value;
        app.order.Value=round(2*value)/2;
        simulera(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Nonideal1UIFigure and hide until all components are created
        app.Nonideal1UIFigure = uifigure('Visible', 'off');
        app.Nonideal1UIFigure.Color = [0.94 0.94 0.94];
        app.Nonideal1UIFigure.Position = [100 100 740 686];
        app.Nonideal1UIFigure.Name = 'Nonideal 1';

        % Create rubrik
        app.rubrik = uilabel(app.Nonideal1UIFigure);
        app.rubrik.FontSize = 18;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [226 642 416 22];
        app.rubrik.Text = 'Mean Conversion Calculation in a Real Reactor';

        % Create RateconstantSliderLabel
        app.RateconstantSliderLabel = uilabel(app.Nonideal1UIFigure);
        app.RateconstantSliderLabel.HorizontalAlignment = 'right';
        app.RateconstantSliderLabel.Position = [77 590 80 22];
        app.RateconstantSliderLabel.Text = 'Rate constant';

        % Create k
        app.k = uislider(app.Nonideal1UIFigure);
        app.k.Limits = [0.01 1];
        app.k.MajorTicks = [0.01 0.2 0.4 0.6 0.8 1];
        app.k.ValueChangedFcn = createCallbackFcn(app, @kValueChanged, true);
        app.k.MinorTicks = [];
        app.k.Position = [178 599 150 3];
        app.k.Value = 0.1;

        % Create IntitalconcentrationSliderLabel
        app.IntitalconcentrationSliderLabel = uilabel(app.Nonideal1UIFigure);
        app.IntitalconcentrationSliderLabel.HorizontalAlignment = 'right';
        app.IntitalconcentrationSliderLabel.Position = [36 521 110 22];
        app.IntitalconcentrationSliderLabel.Text = 'Intital concentration';

        % Create ca0
        app.ca0 = uislider(app.Nonideal1UIFigure);
        app.ca0.Limits = [0.1 20];
        app.ca0.MajorTicks = [0.1 4 8 12 16 20];
        app.ca0.ValueChangedFcn = createCallbackFcn(app, @ca0ValueChanged, true);
        app.ca0.MinorTicks = [];
```

```
        app.ca0.Position = [178 531 150 3];
        app.ca0.Value = 4;

        % Create ReactionorderSliderLabel
        app.ReactionorderSliderLabel = uilabel(app.Nonideal1UIFigure);
        app.ReactionorderSliderLabel.HorizontalAlignment = 'right';
        app.ReactionorderSliderLabel.Position = [447 584 84 22];
        app.ReactionorderSliderLabel.Text = 'Reaction order';

        % Create order
        app.order = uislider(app.Nonideal1UIFigure);
        app.order.Limits = [0.5 2];
        app.order.MajorTicks = [0.5 1 1.5 2];
        app.order.ValueChangedFcn = createCallbackFcn(app, @orderValueChanged, true);
        app.order.MinorTicks = [];
        app.order.Position = [552 593 150 3];
        app.order.Value = 1;

        % Create TabGroup
        app.TabGroup = uitabgroup(app.Nonideal1UIFigure);
        app.TabGroup.Position = [55 52 654 413];

        % Create XTab
        app.XTab = uitab(app.TabGroup);
        app.XTab.Title = 'X';

        % Create UIAxes_1
        app.UIAxes_1 = uiaxes(app.XTab);
        xlabel(app.UIAxes_1, 'Time')
        ylabel(app.UIAxes_1, 'Conversion')
        app.UIAxes_1.PlotBoxAspectRatio = [1.94909090909091 1 1];
        app.UIAxes_1.XTickLabelRotation = 0;
        app.UIAxes_1.YTickLabelRotation = 0;
        app.UIAxes_1.ZTickLabelRotation = 0;
        app.UIAxes_1.Position = [37 23 584 330];

        % Create EtTab
        app.EtTab = uitab(app.TabGroup);
        app.EtTab.Title = 'E(t)';

        % Create UIAxes_2
        app.UIAxes_2 = uiaxes(app.EtTab);
        xlabel(app.UIAxes_2, 'Time')
        app.UIAxes_2.PlotBoxAspectRatio = [1.94909090909091 1 1];
        app.UIAxes_2.XTickLabelRotation = 0;
        app.UIAxes_2.YTickLabelRotation = 0;
        app.UIAxes_2.ZTickLabelRotation = 0;
        app.UIAxes_2.Position = [37 23 584 330];

        % Create FtTab
        app.FtTab = uitab(app.TabGroup);
        app.FtTab.Title = 'F(t)';

        % Create UIAxes_3
        app.UIAxes_3 = uiaxes(app.FtTab);
        xlabel(app.UIAxes_3, 'Time')
        app.UIAxes_3.PlotBoxAspectRatio = [1.94909090909091 1 1];
        app.UIAxes_3.XTickLabelRotation = 0;
        app.UIAxes_3.YTickLabelRotation = 0;
        app.UIAxes_3.ZTickLabelRotation = 0;
        app.UIAxes_3.Position = [37 23 584 330];

        % Show the figure after all components are created
        app.Nonideal1UIFigure.Visible = 'on';
    end
end

% App creation and deletion
methods (Access = public)

    % Construct app
```

```
        function app = nonideal1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Nonideal1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Nonideal1UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Nonideal 2

nonideal2.mlapp

```
classdef nonideal2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Nonideal2UIFigure      matlab.ui.Figure
        CIS                    matlab.ui.control.Label
        sigma                  matlab.ui.control.Label
        tmText                 matlab.ui.control.Label
        MakeexperimentButton   matlab.ui.control.StateButton
        NoofCSTRSpinner        matlab.ui.control.Spinner
        TracerDropDown         matlab.ui.control.DropDown
        TracerDropDownLabel    matlab.ui.control.Label
        ReactorsDropDown       matlab.ui.control.DropDown
        ReactorsDropDownLabel  matlab.ui.control.Label
        rubrik                 matlab.ui.control.Label
        UIAxes_2               matlab.ui.control.UIAxes
        UIAxes                 matlab.ui.control.UIAxes
        UIAxes3                matlab.ui.control.UIAxes
        UIAxes2                matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function toggleUIC(app, state, label)
            % toggle the state of the uicontrols in the app
            uics = findall(app.Nonideal2UIFigure);
            for idx = 1:length(uics)
                uic = uics(idx);
                if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')
                    uic.Enable = state;
                end
            end
            % update the simulate button text
            app.MakeexperimentButton.Text = label;
            drawnow; % flush the ui events queue
        end % toggleUIC
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Value changed function: ReactorsDropDown
        function ReactorsDropDownValueChanged(app, event)
            value = app.ReactorsDropDown.Value;
            if value == '5'
                app.NoofCSTRSpinner.Visible=1;
                %app.NoofCSTRSpinnerLabel.Visible=1;
            else
                app.NoofCSTRSpinner.Visible=0;
                %app.NoofCSTRSpinnerLabel.Visible=0;
            end
            app.UIAxes.cla;
            app.UIAxes_2.cla;
            app.UIAxes2.cla;
            app.UIAxes3.cla;
            app.tmText.Text='';
            app.sigma.Text='';
            app.CIS.Text='';
        end

        % Value changed function: TracerDropDown
        function TracerDropDownValueChanged(app, event)
            %value = app.TracerDropDown.Value;
            app.UIAxes.cla;
            app.UIAxes_2.cla;
```

```
    app.UIAxes2.cla;
    app.UIAxes3.cla;
    app.tmText.Text='';
    app.sigma.Text='';
    app.CIS.Text='';
end

% Value changed function: NoofCSTRSpinner
function NoofCSTRSpinnerValueChanged(app, event)
    %value = app.NoofCSTRSpinner.Value;
    app.UIAxes.cla;
    app.UIAxes_2.cla;
    app.UIAxes2.cla;
    app.UIAxes3.cla;
    app.tmText.Text='';
    app.sigma.Text='';
    app.CIS.Text='';
end

% Value changed function: MakeexperimentButton
function MakeexperimentButtonValueChanged(app, event)
    try
        % create the simulation input
        app.toggleUIC('off', 'Running ...');
        t=0:0.01:25;t=t';
        tracer=str2double(app.TracerDropDown.Value);
        switch app.TracerDropDown.Value
            case '1'
                c=t>=1 & t<1.4;
            case '2'
                c=t>=1;
        end
        plot(app.UIAxes,t,c,'k');
        switch app.ReactorsDropDown.Value
            case '1'
                t=t+5;
                plot(app.UIAxes_2,t,c,'k');
            case '2'
                addtime=1;
                theta=5;
                if tracer == 1
                    [t,c]=ode23s(@diffekv,[addtime,25],0);
                else
                    [t,c]=ode23s(@diffekv,[addtime,25],0);
                end
                plot(app.UIAxes_2,t,c,'k');
            case '3'
                addtime=3.5;
                theta=2.5;
                if tracer == 1
                    [t,c]=ode23s(@diffekv,[addtime,25],0);
                else
                    [t,c]=ode23s(@diffekv,[addtime,25],0);
                end
                plot(app.UIAxes_2,t,c,'k');
            case '4'
                addtime=1;
                theta=2.5;
                if tracer == 1
                    [t,c]=ode23s(@diffekv,[addtime,25],0);
                else
                    [t,c]=ode23s(@diffekv,[addtime,25],0);
                end
                t=t+2.5;
                plot(app.UIAxes_2,t,c,'k');
            case '5'
                N=app.NoofCSTRSpinner.Value;
                addtime=1;
                theta=5/N;
                start=zeros(N,1);
                if tracer == 1
```

```
                start(1,1)=0;
                [t,c]=ode23s(@diffekv,[addtime,25],start);
            else
                [t,c]=ode23s(@diffekv,[addtime,25],start);
            end
            plot(app.UIAxes_2,t,c(:,end),'k');
    end
    if app.ReactorsDropDown.Value == '5'
        app.UIAxes_2.YLim=[0 max(c(:,end))];
        calcRTD(t-1,c(:,end));
    else
        app.UIAxes_2.YLim=[0 max(c)];
        calcRTD(t-1,c);
    end
catch ME
    errordlg(ME.message);
end
app.toggleUIC('on', 'Make experiment');
function dy = diffekv(t,C)
    dy=zeros(size(C));
    N=length(C);
    switch app.TracerDropDown.Value
        case '1'
            if t>= addtime && t < addtime + 0.1
                ca0=1;
            else
                ca0=0;
            end
        case '2'
            if t>= addtime
                ca0=1;
            else
                ca0=0;
            end
    end
    dy(1)=1/theta*(ca0-C(1));
    for i=2:N
        dy(i)=1/theta*(C(i-1)-C(i));
    end
end
function calcRTD(time,conc)
    switch app.TracerDropDown.Value
        case '1'
            Ctot=trapz(time,conc);
            Et=conc./Ctot;
            plot(app.UIAxes2,time,Et,'k');
            app.UIAxes2.XLabel.String='t';
            app.UIAxes2.Title.String='E(t)';
            plot(app.UIAxes3,time,time.*Et,'k')
            app.UIAxes3.XLabel.String='t';
            app.UIAxes3.YLabel.String='t*E(t)';
            app.UIAxes3.Title.String='Red area = Average residence time ';
            app.UIAxes3.NextPlot='add';
            x=[time;time(end);0];
            y=[time.*Et;0;0];
            fill(app.UIAxes3,x,y,'r')
            app.UIAxes3.NextPlot='replace';
            tm=trapz(time,time.*Et);
            app.tmText.Text=sprintf('Average residence time %3.1f',tm);
            moment2=trapz(time,(time-tm).^2.*Et);
            app.sigma.Text=sprintf('Variance %4.2f',moment2);
            app.CIS.Text=sprintf('CIS Model N=%4.2f',tm^2/moment2);
        case '2'
            Ft=conc./conc(end);
            plot(app.UIAxes2,time,Ft,'k');
            app.UIAxes2.XLabel.String='t';
            app.UIAxes2.Title.String='F(t)';
            plot(app.UIAxes3,time,Ft,'k')
            app.UIAxes3.YLabel.String='F(t)';
            app.UIAxes3.XLabel.String='t';
            app.UIAxes3.Title.String='Red area = Average residence time ';
```

```
                    tm=trapz(Ft,time);
                    app.UIAxes3.NextPlot='add';
                    x=[time;time(end);0;0];
                    y=[Ft;1;1;0];
                    fill(app.UIAxes3,x,y,'r')
                    app.UIAxes3.NextPlot='replace';
                    app.tmText.Text=sprintf('Average residence time %3.1f',tm);
                    app.sigma.Text='';
                    app.CIS.Text='';
                end
            end

        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Nonideal2UIFigure and hide until all components are created
            app.Nonideal2UIFigure = uifigure('Visible', 'off');
            app.Nonideal2UIFigure.Color = [0.94 0.94 0.94];
            app.Nonideal2UIFigure.Position = [100 100 726 629];
            app.Nonideal2UIFigure.Name = 'Nonideal 2';

            % Create UIAxes2
            app.UIAxes2 = uiaxes(app.Nonideal2UIFigure);
            title(app.UIAxes2, 'Residence time distribution')
            xlabel(app.UIAxes2, 't')
            app.UIAxes2.FontWeight = 'bold';
            app.UIAxes2.XLim = [0 25];
            app.UIAxes2.XTickLabelRotation = 0;
            app.UIAxes2.YTickLabelRotation = 0;
            app.UIAxes2.ZTickLabelRotation = 0;
            app.UIAxes2.Position = [351 315 300 185];

            % Create UIAxes3
            app.UIAxes3 = uiaxes(app.Nonideal2UIFigure);
            title(app.UIAxes3, {'Average residence time'; ''})
            xlabel(app.UIAxes3, {'t'; ''; ''})
            app.UIAxes3.FontWeight = 'bold';
            app.UIAxes3.XLim = [0 25];
            app.UIAxes3.XTick = [0 5 10 15 20 25];
            app.UIAxes3.XTickLabelRotation = 0;
            app.UIAxes3.YTickLabelRotation = 0;
            app.UIAxes3.ZTickLabelRotation = 0;
            app.UIAxes3.Position = [351 115 300 185];

            % Create UIAxes
            app.UIAxes = uiaxes(app.Nonideal2UIFigure);
            title(app.UIAxes, 'Injected')
            xlabel(app.UIAxes, 't')
            ylabel(app.UIAxes, 'Concentration')
            app.UIAxes.FontWeight = 'bold';
            app.UIAxes.XLim = [0 25];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [34 315 300 185];

            % Create UIAxes_2
            app.UIAxes_2 = uiaxes(app.Nonideal2UIFigure);
            title(app.UIAxes_2, 'c(t)')
            xlabel(app.UIAxes_2, 't')
            ylabel(app.UIAxes_2, 'Concentration')
            app.UIAxes_2.FontWeight = 'bold';
            app.UIAxes_2.XLim = [0 25];
            app.UIAxes_2.XTickLabelRotation = 0;
            app.UIAxes_2.YTickLabelRotation = 0;
```

```
app.UIAxes_2.ZTickLabelRotation = 0;
app.UIAxes_2.Position = [34 115 300 185];

% Create rubrik
app.rubrik = uilabel(app.Nonideal2UIFigure);
app.rubrik.FontSize = 24;
app.rubrik.FontWeight = 'bold';
app.rubrik.Position = [120 567 512 30];
app.rubrik.Text = 'Tracer Experiment with Ideal Flow Reactors';

% Create ReactorsDropDownLabel
app.ReactorsDropDownLabel = uilabel(app.Nonideal2UIFigure);
app.ReactorsDropDownLabel.HorizontalAlignment = 'right';
app.ReactorsDropDownLabel.Position = [120 530 54 22];
app.ReactorsDropDownLabel.Text = 'Reactors';

% Create ReactorsDropDown
app.ReactorsDropDown = uidropdown(app.Nonideal2UIFigure);
app.ReactorsDropDown.Items = {'PFR', 'CSTR', 'PFR,CSTR', 'CSTR,PFR', 'CSTR in serie'};
app.ReactorsDropDown.ItemsData = {'1', '2', '3', '4', '5'};
app.ReactorsDropDown.ValueChangedFcn = createCallbackFcn(app, @ReactorsDropDownValueChanged, true);
app.ReactorsDropDown.Position = [189 530 129 22];
app.ReactorsDropDown.Value = '1';

% Create TracerDropDownLabel
app.TracerDropDownLabel = uilabel(app.Nonideal2UIFigure);
app.TracerDropDownLabel.HorizontalAlignment = 'right';
app.TracerDropDownLabel.Position = [452 530 40 22];
app.TracerDropDownLabel.Text = 'Tracer';

% Create TracerDropDown
app.TracerDropDown = uidropdown(app.Nonideal2UIFigure);
app.TracerDropDown.Items = {'Pulse', 'Step'};
app.TracerDropDown.ItemsData = {'1', '2', '3', '4'};
app.TracerDropDown.ValueChangedFcn = createCallbackFcn(app, @TracerDropDownValueChanged, true);
app.TracerDropDown.Position = [507 530 100 22];
app.TracerDropDown.Value = '1';

% Create NoofCSTRSpinner
app.NoofCSTRSpinner = uispinner(app.Nonideal2UIFigure);
app.NoofCSTRSpinner.Limits = [2 50];
app.NoofCSTRSpinner.ValueChangedFcn = createCallbackFcn(app, @NoofCSTRSpinnerValueChanged, true);
app.NoofCSTRSpinner.Visible = 'off';
app.NoofCSTRSpinner.Position = [324 530 75 22];
app.NoofCSTRSpinner.Value = 2;

% Create MakeexperimentButton
app.MakeexperimentButton = uibutton(app.Nonideal2UIFigure, 'state');
app.MakeexperimentButton.ValueChangedFcn = createCallbackFcn(app, @MakeexperimentButtonValueChanged, true);
app.MakeexperimentButton.Text = 'Make experiment';
app.MakeexperimentButton.BackgroundColor = [0.3922 0.8314 0.0745];
app.MakeexperimentButton.FontWeight = 'bold';
app.MakeexperimentButton.Position = [116 22 136 43];

% Create tmText
app.tmText = uilabel(app.Nonideal2UIFigure);
app.tmText.FontWeight = 'bold';
app.tmText.Position = [393 94 310 22];
app.tmText.Text = {''; ''};

% Create sigma
app.sigma = uilabel(app.Nonideal2UIFigure);
app.sigma.FontWeight = 'bold';
app.sigma.Position = [393 64 300 28];
app.sigma.Text = {''; ''};

% Create CIS
app.CIS = uilabel(app.Nonideal2UIFigure);
app.CIS.FontWeight = 'bold';
app.CIS.Position = [393 32 300 22];
app.CIS.Text = {''; ''};
```

```
            % Show the figure after all components are created
            app.Nonideal2UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = nonideal2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Nonideal2UIFigure)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Nonideal2UIFigure)
        end
    end
end
```

Nonideal 3

nonideal3.mlapp

```
classdef nonideal3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Nonideal3UIFigure      matlab.ui.Figure
        MakeexperimentButton   matlab.ui.control.StateButton
        NoofCSTRSpinner        matlab.ui.control.Spinner
        NoofCSTRSpinnerLabel   matlab.ui.control.Label
        ReactorsDropDown       matlab.ui.control.DropDown
        ReactorsDropDownLabel  matlab.ui.control.Label
        TracerDropDown         matlab.ui.control.DropDown
        TracerDropDownLabel    matlab.ui.control.Label
        rubrik                 matlab.ui.control.Label
        UIAxes_2               matlab.ui.control.UIAxes
        UIAxes                 matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function extInp = externalInput(app)
            Tstop = 25;
            T0 = 1;
            F = 1;

            switch (app.TracerDropDown.Value)
                case 'Pulse'
                    tv = [0 T0 T0 1.5*T0 1.5*T0 Tstop]';
                    uv = [0  0  F    F    0     0]';
                case 'Step'
                    tv = [0 T0 T0 Tstop]';
                    uv = [0  0  F     F]';
                otherwise
                    error('invalid input signal type');
            end
            plot(app.UIAxes, tv, uv, 'Marker','.');
            umin = min(uv); umax = max(uv); urng = 0.1*(umax-umin);
            app.UIAxes.YLim = [umin-urng umax+urng];
            extInp = [tv uv];
        end % externalInput

        function toggleUIC(app, state, label)
            % toggle the state of the uicontrols in the app
            uics = findall(app.Nonideal3UIFigure);
            for idx = 1:length(uics)
                uic = uics(idx);
                if isprop(uic,'Enable')
                    uic.Enable = state;
                end
            end
            % update the simulate button text
            app.MakeexperimentButton.Text = label;
            drawnow; % flush the ui events queue
        end % toggleUIC

    end


    % Callbacks that handle component events
    methods (Access = private)

        % Value changed function: ReactorsDropDown
        function ReactorsDropDownValueChanged(app, event)
            value = app.ReactorsDropDown.Value;
            if value == '5'
                app.NoofCSTRSpinner.Visible=1;
                app.NoofCSTRSpinnerLabel.Visible=1;
```

```
            else
                app.NoofCSTRSpinner.Visible=0;
                app.NoofCSTRSpinnerLabel.Visible=0;
            end
            app.UIAxes.cla;
            app.UIAxes_2.cla;
        end

        % Value changed function: NoofCSTRSpinner
        function NoofCSTRSpinnerValueChanged(app, event)
            %value = app.NoofCSTRSpinner.Value;
            app.UIAxes.cla;
            app.UIAxes_2.cla;
        end

        % Value changed function: MakeexperimentButton
        function MakeexperimentButtonValueChanged(app, event)
            try
                % create the simulation input
                app.toggleUIC('off', 'Simulating ...');
                simInp = Simulink.SimulationInput('RTD');
                simInp.ExternalInput = app.externalInput();
                %simInp = simulink.compiler.configureForDeployment(simInp);
                simOut = sim(simInp);
                simOut.yout{1}.Values.Data
                yp = simOut.yout{1}.Values.Data;
                plot(app.UIAxes_2,simOut.tout,yp);
            catch ME
                errordlg(ME.message);
            end
            app.toggleUIC('on', 'Simulate');
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Nonideal3UIFigure and hide until all components are created
            app.Nonideal3UIFigure = uifigure('Visible', 'off');
            app.Nonideal3UIFigure.Color = [0.94 0.94 0.94];
            app.Nonideal3UIFigure.Position = [100 100 819 505];
            app.Nonideal3UIFigure.Name = 'Nonideal 3';

            % Create UIAxes
            app.UIAxes = uiaxes(app.Nonideal3UIFigure);
            title(app.UIAxes, 'Injected')
            xlabel(app.UIAxes, 'Time')
            ylabel(app.UIAxes, 'Concentration')
            app.UIAxes.PlotBoxAspectRatio = [1.20849420849421 1 1];
            app.UIAxes.XLim = [0 25];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [46 48 359 315];

            % Create UIAxes_2
            app.UIAxes_2 = uiaxes(app.Nonideal3UIFigure);
            title(app.UIAxes_2, 'Response')
            xlabel(app.UIAxes_2, 'Time')
            ylabel(app.UIAxes_2, 'Concentration')
            app.UIAxes_2.PlotBoxAspectRatio = [1.20849420849421 1 1];
            app.UIAxes_2.XLim = [0 25];
            app.UIAxes_2.XTickLabelRotation = 0;
            app.UIAxes_2.YTickLabelRotation = 0;
            app.UIAxes_2.ZTickLabelRotation = 0;
            app.UIAxes_2.Position = [413 48 359 315];

            % Create rubrik
```

```
            app.rubrik = uilabel(app.Nonideal3UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [181 467 512 30];
            app.rubrik.Text = 'Tracer Experiment with Ideal Flow Reactors';

            % Create TracerDropDownLabel
            app.TracerDropDownLabel = uilabel(app.Nonideal3UIFigure);
            app.TracerDropDownLabel.HorizontalAlignment = 'right';
            app.TracerDropDownLabel.Position = [107 388 40 22];
            app.TracerDropDownLabel.Text = 'Tracer';

            % Create TracerDropDown
            app.TracerDropDown = uidropdown(app.Nonideal3UIFigure);
            app.TracerDropDown.Items = {'Pulse', 'Step'};
            app.TracerDropDown.Position = [162 388 100 22];
            app.TracerDropDown.Value = 'Pulse';

            % Create ReactorsDropDownLabel
            app.ReactorsDropDownLabel = uilabel(app.Nonideal3UIFigure);
            app.ReactorsDropDownLabel.HorizontalAlignment = 'right';
            app.ReactorsDropDownLabel.Position = [94 428 54 22];
            app.ReactorsDropDownLabel.Text = 'Reactors';

            % Create ReactorsDropDown
            app.ReactorsDropDown = uidropdown(app.Nonideal3UIFigure);
            app.ReactorsDropDown.Items = {'PFR', 'CSTR', 'PFR,CSTR', 'CSTR,PFR', 'CSTR in serie'};
            app.ReactorsDropDown.ItemsData = {'1', '2', '3', '4', '5'};
            app.ReactorsDropDown.ValueChangedFcn = createCallbackFcn(app, @ReactorsDropDownValueChanged, true);
            app.ReactorsDropDown.Position = [163 428 129 22];
            app.ReactorsDropDown.Value = '1';

            % Create NoofCSTRSpinnerLabel
            app.NoofCSTRSpinnerLabel = uilabel(app.Nonideal3UIFigure);
            app.NoofCSTRSpinnerLabel.HorizontalAlignment = 'right';
            app.NoofCSTRSpinnerLabel.Visible = 'off';
            app.NoofCSTRSpinnerLabel.Position = [390 428 70 22];
            app.NoofCSTRSpinnerLabel.Text = 'No of CSTR';

            % Create NoofCSTRSpinner
            app.NoofCSTRSpinner = uispinner(app.Nonideal3UIFigure);
            app.NoofCSTRSpinner.Limits = [2 50];
            app.NoofCSTRSpinner.ValueChangedFcn = createCallbackFcn(app, @NoofCSTRSpinnerValueChanged, true);
            app.NoofCSTRSpinner.Visible = 'off';
            app.NoofCSTRSpinner.Position = [475 428 100 22];
            app.NoofCSTRSpinner.Value = 2;

            % Create MakeexperimentButton
            app.MakeexperimentButton = uibutton(app.Nonideal3UIFigure, 'state');
            app.MakeexperimentButton.ValueChangedFcn = createCallbackFcn(app, @MakeexperimentButtonValueChanged, true);
            app.MakeexperimentButton.Text = 'Make experiment';
            app.MakeexperimentButton.BackgroundColor = [0.4667 0.6745 0.1882];
            app.MakeexperimentButton.Position = [654 18 108 22];

            % Show the figure after all components are created
            app.Nonideal3UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = nonideal3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Nonideal3UIFigure)

            if nargout == 0
```

```
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Nonideal3UIFigure)
        end
    end
end
```

PFR 1

pfr1.mlapp

```
classdef pfr1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR1UIFigure                   matlab.ui.Figure
        OutflowmolarflowPanel          matlab.ui.container.Panel
        Label_3                        matlab.ui.control.Label
        Label_2                        matlab.ui.control.Label
        Label                          matlab.ui.control.Label
        orderSpinner                   matlab.ui.control.Spinner
        ReactionorderBSpinnerLabel     matlab.ui.control.Label
        Editk                          matlab.ui.control.EditField
        EditxA                         matlab.ui.control.EditField
        k                              matlab.ui.control.Slider
        rateconstantLabel              matlab.ui.control.Label
        xA                             matlab.ui.control.Slider
        InletmolefractionASliderLabel  matlab.ui.control.Label
        rubrik                         matlab.ui.control.Label
        UIAxes                         matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function displayOrder(app)
            global rateConstant
            try
                CB0=1-app.xA.Value;
                Order=app.orderSpinner.Value;
                if Order == 2
                    rateConstant=app.k.Value/CB0;
                else
                    rateConstant=app.k.Value;
                end
                switch Order
                    case 0
                       app.UIAxes.Title.String=sprintf('A+2B->C\t\tr=%4.2fC_A [mol/(s L)]',rateConstant);
                    case 1
                       app.UIAxes.Title.String=sprintf('A+2B->C\t\tr=%4.2fC_AC_B [mol/(s L)]',rateConstant);
                    case 2
                        app.UIAxes.Title.String=sprintf('A+2B->C\t\tr=%4.2fC_AC_B^%1i [mol/(s L)]',rateConstant,Order);
                end
             catch ME
                errordlg(ME.message);
            end
        end

        function simulera(app)
            try
                toogleUIC(app.PFR1UIFigure,'off');
                Vspan=[0 5];
                F0=[app.xA.Value,(1-app.xA.Value),0]; %mol/s
                [V,F]=ode45(@diffekv,Vspan,F0);
                plot(app.UIAxes,V,F,"LineWidth",2);
                legend(app.UIAxes,'F_A','F_B','F_C','location','best')
                app.Label.Text=sprintf('A: %4.2f mol/s',F(end,1));
                app.Label_2.Text=sprintf('B: %4.2f mol/s',F(end,2));
                app.Label_3.Text=sprintf('C: %4.2f mol/s',F(end,3));
                toogleUIC(app.PFR1UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function dy = diffekv(~,F)
                global rateConstant
                dy=zeros(size(F));
                ypsilon=1; %L/S
                C=F./ypsilon;
```

```
            r=rateConstant*C(1)*C(2)^(app.orderSpinner.Value);
            dy(1)=-r;
            dy(2)=-2*r;
            dy(3)=r;
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.UIAxes.TitleFontWeight='B';
        app.UIAxes.TitleFontSizeMultiplier=1.5;
        displayOrder(app);
        app.EditxA.Value=sprintf('%4.2f',app.xA.Value);
        app.Editk.Value=sprintf('%4.2f',app.k.Value);
        simulera(app);
    end

    % Callback function
    function orderValueChanged(app, event)
        Border=round(app.order.Value);
        app.order.Value=Border;
        displayOrder(app);
        simulera(app);
    end

    % Value changed function: k
    function kValueChanged(app, event)
        app.Editk.Value=sprintf('%4.2f',app.k.Value);
        displayOrder(app);
        simulera(app);
    end

    % Value changed function: xA
    function xAValueChanged(app, event)
        app.EditxA.Value=sprintf('%4.2f',app.xA.Value);
        displayOrder(app);
        simulera(app);
    end

    % Value changed function: EditxA
    function EditxAValueChanged(app, event)
        value = app.EditxA.Value;
         if between(app,value,app.xA.Limits)
            app.xA.Value=str2double(value);
            displayOrder(app);
            simulera(app);
        else
            app.EditxA.Value=num2str(app.xA.Value);
         end
    end

    % Value changed function: Editk
    function EditkValueChanged(app, event)
        value = app.Editk.Value;
        if between(app,value,app.k.Limits)
            app.k.Value=str2double(value);
            displayOrder(app);
            simulera(app);
        else
            app.Editk.Value=num2str(app.k.Value);
        end
    end

    % Value changed function: orderSpinner
    function orderSpinnerValueChanged(app, event)
        displayOrder(app);
```

```
            simulera(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create PFR1UIFigure and hide until all components are created
            app.PFR1UIFigure = uifigure('Visible', 'off');
            app.PFR1UIFigure.Color = [0.94 0.94 0.94];
            app.PFR1UIFigure.Position = [100 100 745 742];
            app.PFR1UIFigure.Name = 'PFR 1';

            % Create UIAxes
            app.UIAxes = uiaxes(app.PFR1UIFigure);
            title(app.UIAxes, {'Rate equation'; ''})
            xlabel(app.UIAxes, {'Reactor volume [L]'; ''})
            ylabel(app.UIAxes, 'Molar Flow [mol/s]')
            app.UIAxes.PlotBoxAspectRatio = [1.84703196347032 1 1];
            app.UIAxes.FontWeight = 'bold';
            app.UIAxes.XLim = [0 5];
            app.UIAxes.YLim = [0 1];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.XGrid = 'on';
            app.UIAxes.YGrid = 'on';
            app.UIAxes.FontSize = 16;
            app.UIAxes.Position = [24 27 694 457];

            % Create rubrik
            app.rubrik = uilabel(app.PFR1UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [278 686 348 30];
            app.rubrik.Text = 'Isothermal Plug Flow Reactor';

            % Create InletmolefractionASliderLabel
            app.InletmolefractionASliderLabel = uilabel(app.PFR1UIFigure);
            app.InletmolefractionASliderLabel.HorizontalAlignment = 'right';
            app.InletmolefractionASliderLabel.Position = [47 582 112 22];
            app.InletmolefractionASliderLabel.Text = 'Inlet mole fraction A';

            % Create xA
            app.xA = uislider(app.PFR1UIFigure);
            app.xA.Limits = [0.1 0.9];
            app.xA.MajorTicks = [0.1 0.3 0.5 0.7 0.9];
            app.xA.ValueChangedFcn = createCallbackFcn(app, @xAValueChanged, true);
            app.xA.MinorTicks = [];
            app.xA.Position = [180 591 150 3];
            app.xA.Value = 0.5;

            % Create rateconstantLabel
            app.rateconstantLabel = uilabel(app.PFR1UIFigure);
            app.rateconstantLabel.HorizontalAlignment = 'right';
            app.rateconstantLabel.Position = [87 644 75 22];
            app.rateconstantLabel.Text = 'rate constant';

            % Create k
            app.k = uislider(app.PFR1UIFigure);
            app.k.Limits = [0 1];
            app.k.ValueChangedFcn = createCallbackFcn(app, @kValueChanged, true);
            app.k.MinorTicks = [];
            app.k.Position = [183 653 150 3];
            app.k.Value = 0.56;

            % Create EditxA
            app.EditxA = uieditfield(app.PFR1UIFigure, 'text');
```

```
        app.EditxA.ValueChangedFcn = createCallbackFcn(app, @EditxAValueChanged, true);
        app.EditxA.Position = [351 581 40 22];

        % Create Editk
        app.Editk = uieditfield(app.PFR1UIFigure, 'text');
        app.Editk.ValueChangedFcn = createCallbackFcn(app, @EditkValueChanged, true);
        app.Editk.Position = [351 644 40 22];

        % Create ReactionorderBSpinnerLabel
        app.ReactionorderBSpinnerLabel = uilabel(app.PFR1UIFigure);
        app.ReactionorderBSpinnerLabel.HorizontalAlignment = 'right';
        app.ReactionorderBSpinnerLabel.Position = [434 643 96 22];
        app.ReactionorderBSpinnerLabel.Text = 'Reaction order B';

        % Create orderSpinner
        app.orderSpinner = uispinner(app.PFR1UIFigure);
        app.orderSpinner.Limits = [0 2];
        app.orderSpinner.ValueChangedFcn = createCallbackFcn(app, @orderSpinnerValueChanged, true);
        app.orderSpinner.Position = [545 643 100 22];
        app.orderSpinner.Value = 1;

        % Create OutflowmolarflowPanel
        app.OutflowmolarflowPanel = uipanel(app.PFR1UIFigure);
        app.OutflowmolarflowPanel.Title = 'Outflow molar flow';
        app.OutflowmolarflowPanel.FontWeight = 'bold';
        app.OutflowmolarflowPanel.FontSize = 14;
        app.OutflowmolarflowPanel.Position = [562 501 142 102];

        % Create Label
        app.Label = uilabel(app.OutflowmolarflowPanel);
        app.Label.FontWeight = 'bold';
        app.Label.Position = [10 47 146 22];
        app.Label.Text = '';

        % Create Label_2
        app.Label_2 = uilabel(app.OutflowmolarflowPanel);
        app.Label_2.FontWeight = 'bold';
        app.Label_2.Position = [10 26 167 22];
        app.Label_2.Text = '';

        % Create Label_3
        app.Label_3 = uilabel(app.OutflowmolarflowPanel);
        app.Label_3.FontWeight = 'bold';
        app.Label_3.Position = [10 5 167 22];
        app.Label_3.Text = '';

        % Show the figure after all components are created
        app.PFR1UIFigure.Visible = 'on';
    end
end

% App creation and deletion
methods (Access = public)

    % Construct app
    function app = pfr1

        % Create UIFigure and components
        createComponents(app)

        % Register the app with App Designer
        registerApp(app, app.PFR1UIFigure)

        % Execute the startup function
        runStartupFcn(app, @startupFcn)

        if nargout == 0
            clear app
        end
    end

    % Code that executes before app deletion
```

```
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR1UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 2

pfr2.mlapp

```
classdef pfr2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR2UIFigure                matlab.ui.Figure
        cA0                         matlab.ui.control.Slider
        InitialconcentrationofASliderLabel  matlab.ui.control.Label
        k                           matlab.ui.control.Slider
        rateconstantSliderLabel     matlab.ui.control.Label
        order                       matlab.ui.control.DropDown
        reactionorderDropDownLabel  matlab.ui.control.Label
        rubrik                      matlab.ui.control.Label
        UIAxes                      matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.PFR2UIFigure,'off');
                SV=linspace(0.1,10);
                n=app.order.Value;
                rateconstant=app.k.Value;
                cA=app.cA0.Value;
                switch n
                    case '-1'
                        oms=1-sqrt(-(min(0,2*rateconstant-SV*cA^2))./SV)/cA;
                    case '-1/2'
                        oms=1-(1-min(1,3*rateconstant/2./SV/cA^(3/2))).^(3/2);
                    case '0'
                        oms=rateconstant./SV./cA;
                    case '1/2'
                        oms=1-min(0,rateconstant/2./SV/sqrt(cA)-1).^2;
                    case '1'
                        oms=1-exp(-rateconstant./SV);
                    case '2'
                        oms=rateconstant*cA./(SV+rateconstant*cA);
                end
                yyaxis(app.UIAxes,'left');
                plot(app.UIAxes,SV,oms);
                ylabel(app.UIAxes,'Conversion')
                ylim(app.UIAxes,[0 1]);
                yyaxis(app.UIAxes,'right');
                plot(app.UIAxes,SV,oms*cA.*SV);
                ylabel(app.UIAxes,'overall rate [mol/s]')
                ylim(app.UIAxes,[0 6]);
                toogleUIC(app.PFR2UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            simulera(app);
        end

        % Value changed function: cA0, k, order
        function orderValueChanged(app, event)
            simulera(app);
```

```
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create PFR2UIFigure and hide until all components are created
        app.PFR2UIFigure = uifigure('Visible', 'off');
        app.PFR2UIFigure.Color = [0.94 0.94 0.94];
        app.PFR2UIFigure.Position = [100 100 740 650];
        app.PFR2UIFigure.Name = 'PFR 2';

        % Create UIAxes
        app.UIAxes = uiaxes(app.PFR2UIFigure);
        xlabel(app.UIAxes, 'Space velocity [1/s]')
        ylabel(app.UIAxes, 'Y')
        app.UIAxes.PlotBoxAspectRatio = [1.61613691907215 1 1];
        app.UIAxes.FontWeight = 'bold';
        app.UIAxes.XLim = [0.1 10];
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.FontSize = 16;
        app.UIAxes.Position = [36 38 685 424];

        % Create rubrik
        app.rubrik = uilabel(app.PFR2UIFigure);
        app.rubrik.FontSize = 18;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [221 609 470 22];
        app.rubrik.Text = 'Reaction Rate and Conversion versus Space Velocity';

        % Create reactionorderDropDownLabel
        app.reactionorderDropDownLabel = uilabel(app.PFR2UIFigure);
        app.reactionorderDropDownLabel.HorizontalAlignment = 'right';
        app.reactionorderDropDownLabel.Position = [449 568 80 22];
        app.reactionorderDropDownLabel.Text = 'reaction order';

        % Create order
        app.order = uidropdown(app.PFR2UIFigure);
        app.order.Items = {'-1', '-1/2', '0', '1/2', '1', '2', ''};
        app.order.ValueChangedFcn = createCallbackFcn(app, @orderValueChanged, true);
        app.order.Position = [544 568 100 22];
        app.order.Value = '1';

        % Create rateconstantSliderLabel
        app.rateconstantSliderLabel = uilabel(app.PFR2UIFigure);
        app.rateconstantSliderLabel.HorizontalAlignment = 'right';
        app.rateconstantSliderLabel.Position = [141 577 75 22];
        app.rateconstantSliderLabel.Text = 'rate constant';

        % Create k
        app.k = uislider(app.PFR2UIFigure);
        app.k.Limits = [0.5 2];
        app.k.MajorTicks = [0.5 1 1.5 2];
        app.k.ValueChangedFcn = createCallbackFcn(app, @orderValueChanged, true);
        app.k.MinorTicks = [];
        app.k.Position = [236 587 150 3];
        app.k.Value = 1;

        % Create InitialconcentrationofASliderLabel
        app.InitialconcentrationofASliderLabel = uilabel(app.PFR2UIFigure);
        app.InitialconcentrationofASliderLabel.HorizontalAlignment = 'right';
        app.InitialconcentrationofASliderLabel.Position = [83 511 134 22];
        app.InitialconcentrationofASliderLabel.Text = 'Initial concentration of A';

        % Create cA0
        app.cA0 = uislider(app.PFR2UIFigure);
```

```
            app.cA0.Limits = [1 2];
            app.cA0.ValueChangedFcn = createCallbackFcn(app, @orderValueChanged, true);
            app.cA0.MinorTicks = [];
            app.cA0.Position = [236 521 150 3];
            app.cA0.Value = 1.5;

            % Show the figure after all components are created
            app.PFR2UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.PFR2UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR2UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 3

pfr3.mlapp

```
classdef pfr3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR3UIFigure                    matlab.ui.Figure
        WithdensitychangePanel          matlab.ui.container.Panel
        V2Slider                        matlab.ui.control.Slider
        VolumeLLabel_2                  matlab.ui.control.Label
        X2EditField                     matlab.ui.control.EditField
        ConversionEditField_2Label      matlab.ui.control.Label
        V2Edit                          matlab.ui.control.EditField
        IgnoredensitychangePanel        matlab.ui.container.Panel
        V1Slider                        matlab.ui.control.Slider
        VolumeLLabel                    matlab.ui.control.Label
        V1Edit                          matlab.ui.control.EditField
        X1EditField                     matlab.ui.control.EditField
        ConversionEditFieldLabel        matlab.ui.control.Label
        ypsilonSlider                   matlab.ui.control.Slider
        feedvolumetricflowLsLabel       matlab.ui.control.Label
        CA0Slider                       matlab.ui.control.Slider
        feedreactantconcentrationmolLLabel  matlab.ui.control.Label
        TSlider                         matlab.ui.control.Slider
        ReactortemperatureKSliderLabel  matlab.ui.control.Label
        TitleLabel                      matlab.ui.control.Label
        UIAxes                          matlab.ui.control.UIAxes
    end


    methods (Access = private)
        function draw(app)
            try
                toogleUIC(app.PFR3UIFigure,'off');
                T=app.TSlider.Value;
                CA0=app.CA0Slider.Value;
                ypsilon0=app.ypsilonSlider.Value;
                V1end=app.V1Slider.Value;
                V2end=app.V2Slider.Value;
                k=50*exp(-(10000/8.314/T));
                [V1,F1]=ode45(@model1,[0 V1end],[CA0*ypsilon0 0 0]);
                oms1=(F1(1,1)-F1(:,1))./F1(1,1);
                [V2,F2]=ode45(@model2,[0 V2end],[CA0*ypsilon0 0 0]);
                oms2=(F2(1,1)-F2(:,1))./F2(1,1);
                plot(app.UIAxes,V1,oms1,'g',V2,oms2,'b','LineWidth',1.5);
                app.UIAxes.XLim=[0 max([V1(end) V2(end)])];
                legend(app.UIAxes,'Without density change','With density change','location','best')
                x1=(F1(1,1)-F1(end,1))./F1(1,1);
                x2=(F2(1,1)-F2(end,1))./F2(1,1);
                app.X1EditField.Value=sprintf('%4.2f',x1);
                app.X2EditField.Value=sprintf('%4.2f',x2);
                app.V1Edit.Value=sprintf('%5.1f',V1(end));
                app.V2Edit.Value=sprintf('%5.1f',V2(end));
                toogleUIC(app.PFR3UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function dy = model1(~,F)
                dy=zeros(size(F));
                CA=F(1)/ypsilon0;
                r=k*CA.^2;
                dy(1)=-r;dy(2)=2*r;dy(3)=r;
            end
            function dy = model2(~,F)
                dy=zeros(size(F));
                ypsilon=sum(F)./CA0;
                CA=F(1)/ypsilon;
                r=k*CA.^2;
                dy(1)=-r;dy(2)=2*r;dy(3)=r;
```

```
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        draw(app);
    end

    % Value changed function: CA0Slider, TSlider, V1Slider, V2Slider,
    % ...and 1 other component
    function TSliderValueChanged(app, event)
        draw(app);
    end

    % Value changed function: X2EditField
    function X2EditFieldValueChanged(app, event)
        target = str2double(app.X1EditField.Value);
        T=app.TSlider.Value;
        CA0=app.CA0Slider.Value;
        ypsilon0=app.ypsilonSlider.Value;
        V1end=app.V1Slider.Value;
        k=50*exp(-(10000/8.314/T));
        FA0=CA0*ypsilon0;
        [V1,F1]=ode45(@model1,[0 V1end],[FA0 0 0]);
        oms1=(F1(1,1)-F1(:,1))./F1(1,1);
        [V2,F2]=ode45(@model2,[0 100],[FA0 0 0],odeset('Events',@VEvent));
        oms2=(F2(1,1)-F2(:,1))./F2(1,1);
        plot(app.UIAxes,V1,oms1,'g',V2,oms2,'b','LineWidth',1.5);
        app.UIAxes.XLim=[0 max([V1(end) V2(end)])];
        legend(app.UIAxes,'Without density change','With density change','location','best')
        x1=(F1(1,1)-F1(end,1))./F1(1,1);
        x2=(F2(1,1)-F2(end,1))./F2(1,1);
        app.X1EditField.Value=sprintf('%4.2f',x1);
        app.X2EditField.Value=sprintf('%4.2f',x2);
        app.V1Edit.Value=sprintf('%5.1f',V1(end));
        app.V2Edit.Value=sprintf('%5.1f',V2(end));
        app.V2Slider.Value=V2(end);
        function dy = model1(~,F)
            dy=zeros(size(F));
            CA=F(1)/ypsilon0;
            r=k*CA.^2;
            dy(1)=-r;dy(2)=2*r;dy(3)=r;
        end
        function dy = model2(~,F)
            dy=zeros(size(F));
            ypsilon=sum(F)./CA0;
            CA=F(1)/ypsilon;
            r=k*CA.^2;
            dy(1)=-r;dy(2)=2*r;dy(3)=r;
        end
        function[value,isterminal,direction] = VEvent(~,F)
            oms=(FA0-F(1))/FA0;
            value=oms-target;
            isterminal=1;
            direction=0;
        end
    end

    % Value changed function: X1EditField
    function X1EditFieldValueChanged(app, event)
        target = str2double(app.X2EditField.Value);
        T=app.TSlider.Value;
        CA0=app.CA0Slider.Value;
        ypsilon0=app.ypsilonSlider.Value;
        V2end=app.V2Slider.Value;
        k=50*exp(-(10000/8.314/T));
```

```
            FA0=CA0*ypsilon0;
            [V1,F1]=ode45(@model1,[0 100],[FA0 0 0],odeset('Events',@VEvent));
            oms1=(F1(1,1)-F1(:,1))./F1(1,1);
            [V2,F2]=ode45(@model2,[0 V2end],[FA0 0 0]);
            oms2=(F2(1,1)-F2(:,1))./F2(1,1);
            plot(app.UIAxes,V1,oms1,'g',V2,oms2,'b','LineWidth',1.5);
            app.UIAxes.XLim=[0 max([V1(end) V2(end)])];
            legend(app.UIAxes,'Without density change','With density change','location','best')
            x1=(F1(1,1)-F1(end,1))./F1(1,1);
            x2=(F2(1,1)-F2(end,1))./F2(1,1);
            app.X1EditField.Value=sprintf('%4.2f',x1);
            app.X2EditField.Value=sprintf('%4.2f',x2);
            app.V1Edit.Value=sprintf('%5.1f',V1(end));
            app.V2Edit.Value=sprintf('%5.1f',V2(end));
            app.V1Slider.Value=V1(end);
            function dy = model1(~,F)
                dy=zeros(size(F));
                CA=F(1)/ypsilon0;
                r=k*CA.^2;
                dy(1)=-r;dy(2)=2*r;dy(3)=r;
            end
            function dy = model2(~,F)
                dy=zeros(size(F));
                ypsilon=sum(F)./CA0;
                CA=F(1)/ypsilon;
                r=k*CA.^2;
                dy(1)=-r;dy(2)=2*r;dy(3)=r;
            end
            function[value,isterminal,direction] = VEvent(~,F)
                oms=(FA0-F(1))/FA0;
                value=oms-target;
                isterminal=1;
                direction=0;
            end

        end

        % Value changed function: V1Edit
        function V1EditValueChanged(app, event)
            value=app.V1Edit.Value;
            if between(app,value,app.V1Slider.Limits)
                app.V1Slider.Value=str2double(value);
                draw(app);
            else
                app.V1Edit.Value=num2str(app.V1Slider.Value);
            end
        end

        % Value changed function: V2Edit
        function V2EditValueChanged(app, event)
            value=app.V2Edit.Value;
            if between(app,value,app.V2Slider.Limits)
                app.V2Slider.Value=str2double(value);
                draw(app);
            else
                app.V2Edit.Value=num2str(app.V2Slider.Value);
            end
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create PFR3UIFigure and hide until all components are created
            app.PFR3UIFigure = uifigure('Visible', 'off');
            app.PFR3UIFigure.Color = [0.94 0.94 0.94];
            app.PFR3UIFigure.Position = [100 100 740 722];
            app.PFR3UIFigure.Name = 'PFR 3';
```

```
app.PFR3UIFigure.Scrollable = 'on';

% Create UIAxes
app.UIAxes = uiaxes(app.PFR3UIFigure);
xlabel(app.UIAxes, 'Reactor volume [L]')
ylabel(app.UIAxes, 'Conversion')
app.UIAxes.PlotBoxAspectRatio = [1.91558441558442 1 1];
app.UIAxes.XTickLabelRotation = 0;
app.UIAxes.YTickLabelRotation = 0;
app.UIAxes.ZTickLabelRotation = 0;
app.UIAxes.XGrid = 'on';
app.UIAxes.YGrid = 'on';
app.UIAxes.Interruptible = 'off';
app.UIAxes.Position = [51 31 640 350];

% Create TitleLabel
app.TitleLabel = uilabel(app.PFR3UIFigure);
app.TitleLabel.FontSize = 24;
app.TitleLabel.FontWeight = 'bold';
app.TitleLabel.Position = [71 671 580 30];
app.TitleLabel.Text = 'Why Density Change Cannot Be Ignored in a PFR';

% Create ReactortemperatureKSliderLabel
app.ReactortemperatureKSliderLabel = uilabel(app.PFR3UIFigure);
app.ReactortemperatureKSliderLabel.HorizontalAlignment = 'right';
app.ReactortemperatureKSliderLabel.Position = [71 631 134 22];
app.ReactortemperatureKSliderLabel.Text = 'Reactor temperature [K]';

% Create TSlider
app.TSlider = uislider(app.PFR3UIFigure);
app.TSlider.Limits = [298 450];
app.TSlider.MajorTicks = [300 350 400 450];
app.TSlider.ValueChangedFcn = createCallbackFcn(app, @TSliderValueChanged, true);
app.TSlider.MinorTicks = [];
app.TSlider.Interruptible = 'off';
app.TSlider.Position = [226 640 150 3];
app.TSlider.Value = 300;

% Create feedreactantconcentrationmolLLabel
app.feedreactantconcentrationmolLLabel = uilabel(app.PFR3UIFigure);
app.feedreactantconcentrationmolLLabel.HorizontalAlignment = 'right';
app.feedreactantconcentrationmolLLabel.Position = [16 571 191 22];
app.feedreactantconcentrationmolLLabel.Text = 'feed reactant concentration [mol/L]';

% Create CA0Slider
app.CA0Slider = uislider(app.PFR3UIFigure);
app.CA0Slider.Limits = [0.02 0.1];
app.CA0Slider.ValueChangedFcn = createCallbackFcn(app, @TSliderValueChanged, true);
app.CA0Slider.MinorTicks = [];
app.CA0Slider.Position = [228 580 150 3];
app.CA0Slider.Value = 0.05;

% Create feedvolumetricflowLsLabel
app.feedvolumetricflowLsLabel = uilabel(app.PFR3UIFigure);
app.feedvolumetricflowLsLabel.HorizontalAlignment = 'right';
app.feedvolumetricflowLsLabel.Position = [406 571 139 22];
app.feedvolumetricflowLsLabel.Text = {'feed volumetric flow (L/s)'; ''};

% Create ypsilonSlider
app.ypsilonSlider = uislider(app.PFR3UIFigure);
app.ypsilonSlider.Limits = [0.5 5];
app.ypsilonSlider.MajorTicks = [0.5 1 2 3 4 5];
app.ypsilonSlider.ValueChangedFcn = createCallbackFcn(app, @TSliderValueChanged, true);
app.ypsilonSlider.MinorTicks = [];
app.ypsilonSlider.Position = [566 580 150 3];
app.ypsilonSlider.Value = 2;

% Create IgnoredensitychangePanel
app.IgnoredensitychangePanel = uipanel(app.PFR3UIFigure);
app.IgnoredensitychangePanel.Title = 'Ignore density change';
app.IgnoredensitychangePanel.FontWeight = 'bold';
```

```
app.IgnoredensitychangePanel.FontSize = 14;
app.IgnoredensitychangePanel.Position = [71 394 307 141];

% Create ConversionEditFieldLabel
app.ConversionEditFieldLabel = uilabel(app.IgnoredensitychangePanel);
app.ConversionEditFieldLabel.HorizontalAlignment = 'right';
app.ConversionEditFieldLabel.Position = [12 28 66 22];
app.ConversionEditFieldLabel.Text = 'Conversion';

% Create X1EditField
app.X1EditField = uieditfield(app.IgnoredensitychangePanel, 'text');
app.X1EditField.ValueChangedFcn = createCallbackFcn(app, @X1EditFieldValueChanged, true);
app.X1EditField.Position = [93 28 100 22];

% Create V1Edit
app.V1Edit = uieditfield(app.IgnoredensitychangePanel, 'text');
app.V1Edit.ValueChangedFcn = createCallbackFcn(app, @V1EditValueChanged, true);
app.V1Edit.Position = [250 87 47 22];

% Create VolumeLLabel
app.VolumeLLabel = uilabel(app.IgnoredensitychangePanel);
app.VolumeLLabel.HorizontalAlignment = 'right';
app.VolumeLLabel.Position = [12 88 66 22];
app.VolumeLLabel.Text = 'Volume [L] ';

% Create V1Slider
app.V1Slider = uislider(app.IgnoredensitychangePanel);
app.V1Slider.Limits = [10 100];
app.V1Slider.ValueChangedFcn = createCallbackFcn(app, @TSliderValueChanged, true);
app.V1Slider.MinorTicks = [];
app.V1Slider.Position = [91 98 150 3];
app.V1Slider.Value = 40;

% Create WithdensitychangePanel
app.WithdensitychangePanel = uipanel(app.PFR3UIFigure);
app.WithdensitychangePanel.Title = 'With density change';
app.WithdensitychangePanel.FontWeight = 'bold';
app.WithdensitychangePanel.FontSize = 14;
app.WithdensitychangePanel.Position = [410 394 307 141];

% Create V2Edit
app.V2Edit = uieditfield(app.WithdensitychangePanel, 'text');
app.V2Edit.ValueChangedFcn = createCallbackFcn(app, @V2EditValueChanged, true);
app.V2Edit.Position = [240 86 49 22];

% Create ConversionEditField_2Label
app.ConversionEditField_2Label = uilabel(app.WithdensitychangePanel);
app.ConversionEditField_2Label.HorizontalAlignment = 'right';
app.ConversionEditField_2Label.Position = [1 28 66 22];
app.ConversionEditField_2Label.Text = 'Conversion';

% Create X2EditField
app.X2EditField = uieditfield(app.WithdensitychangePanel, 'text');
app.X2EditField.ValueChangedFcn = createCallbackFcn(app, @X2EditFieldValueChanged, true);
app.X2EditField.Position = [82 28 100 22];

% Create VolumeLLabel_2
app.VolumeLLabel_2 = uilabel(app.WithdensitychangePanel);
app.VolumeLLabel_2.HorizontalAlignment = 'right';
app.VolumeLLabel_2.Position = [8 87 62 22];
app.VolumeLLabel_2.Text = 'Volume [L]';

% Create V2Slider
app.V2Slider = uislider(app.WithdensitychangePanel);
app.V2Slider.Limits = [10 100];
app.V2Slider.ValueChangedFcn = createCallbackFcn(app, @TSliderValueChanged, true);
app.V2Slider.MinorTicks = [];
app.V2Slider.Position = [81 96 150 3];
app.V2Slider.Value = 40;

% Show the figure after all components are created
```

```
            app.PFR3UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.PFR3UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR3UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 4

pfr4.mlapp

```
classdef pfr4 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR4UIFigure                 matlab.ui.Figure
        alfaEdit                     matlab.ui.control.EditField
        T0Edit                       matlab.ui.control.EditField
        xeLabel                      matlab.ui.control.Label
        TeLabel                      matlab.ui.control.Label
        alfaSlider                   matlab.ui.control.Slider
        molarratioofinerttoreactantinfeedSliderLabel  matlab.ui.control.Label
        T0Slider                     matlab.ui.control.Slider
        FeedtemperatureKSliderLabel  matlab.ui.control.Label
        DropDownor                   matlab.ui.control.DropDown
        titleLabel                   matlab.ui.control.Label
        UIAxes                       matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.PFR4UIFigure,'off');
                if app.DropDownor.Value == '1'
                    DeltaH=-83680;
                    Cp=209;
                    Km=100000;
                    Tm=298;
                    Tmin=300;Tmax=550;
                    T1=300;T2=400;
                elseif app.DropDownor.Value == '2'
                    DeltaH=65000;
                    Cp=175;
                    Km=10;
                    Tm=398;
                    Tmin=250;
                    Tmax=500;
                    T1=400;T2=500;
                else
                    % Hit skall man inte komma
                end
                app.UIAxes.Title.String=sprintf('A \x21cc B');
                app.UIAxes.TitleFontSizeMultiplier=1.4;
                app.T0Slider.Limits=[T1 T2];
                alfa=app.alfaSlider.Value;
                Tf=app.T0Slider.Value;
                T=linspace(Tmin,Tmax);
                R=8.314;
                Ke=Km*exp(DeltaH/R*(1/Tm-1./T));
                xe=Ke./(1+Ke);
                xEB=(1+alfa)*Cp*(T-Tf)/(-DeltaH);
                plot(app.UIAxes,T,xe,'k',T,xEB,'b');
                app.UIAxes.XLim=[Tmin Tmax];
                app.UIAxes.YLim=[0 1.0];
                Te=fzero(@fun,(Tmin+Tmax)/2);
                Xe=(1+alfa)*Cp*(Te-Tf)/(-DeltaH);
                app.TeLabel.Text=sprintf('T\x2091=%3.0f K',Te);
                app.xeLabel.Text=sprintf('X\x2091=%3.2f',Xe);
                toogleUIC(app.PFR4UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function res=fun(T)
                Ke=Km*exp(DeltaH/R*(1/Tm-1./T));
                xe=Ke./(1+Ke);
                xEB=(1+alfa)*Cp*(T-Tf)/(-DeltaH);
```

```
            res=xe-xEB;
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.alfaEdit.Value=sprintf('%4.2f',app.alfaSlider.Value);
        app.T0Edit.Value=sprintf('%4.0f',app.T0Slider.Value);
        draw(app);
    end

    % Value changed function: DropDownor
    function DropDownorValueChanged(app, event)
        draw(app);
    end

    % Value changed function: alfaSlider
    function alfaSliderValueChanged(app, event)
        value = app.alfaSlider.Value;
        app.alfaEdit.Value=sprintf('%4.2f',value);
        draw(app);
    end

    % Value changed function: T0Slider
    function T0SliderValueChanged(app, event)
        value = app.T0Slider.Value;
        app.T0Edit.Value=sprintf('%4.0f',value);
        draw(app);
    end

    % Value changed function: T0Edit
    function T0EditValueChanged(app, event)
        value = app.T0Edit.Value;
        if between(app,value,app.T0Slider.Limits)
            app.T0Slider.Value=str2double(value);
            draw(app);
        else
            app.T0Edit.Value=num2str(app.T0Slider.Value);
        end
    end

    % Value changed function: alfaEdit
    function alfaEditValueChanged(app, event)
        value = app.alfaEdit.Value;
        if between(app,value,app.alfaSlider.Limits)
            app.alfaSlider.Value=str2double(value);
            draw(app);
        else
            app.alfaEdit.Value=num2str(app.alfaSlider.Value);
        end
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create PFR4UIFigure and hide until all components are created
        app.PFR4UIFigure = uifigure('Visible', 'off');
        app.PFR4UIFigure.Color = [0.94 0.94 0.94];
        app.PFR4UIFigure.Position = [100 100 733 597];
        app.PFR4UIFigure.Name = 'PFR 4';

        % Create UIAxes
```

```
app.UIAxes = uiaxes(app.PFR4UIFigure);
title(app.UIAxes, 'A <-> B')
xlabel(app.UIAxes, 'Temperature [K]')
ylabel(app.UIAxes, 'Conversion')
app.UIAxes.PlotBoxAspectRatio = [1.69508196721311 1 1];
app.UIAxes.XLim = [300 500];
app.UIAxes.XTickLabelRotation = 0;
app.UIAxes.YTickLabelRotation = 0;
app.UIAxes.ZTickLabelRotation = 0;
app.UIAxes.Position = [72 40 515 384];

% Create titleLabel
app.titleLabel = uilabel(app.PFR4UIFigure);
app.titleLabel.FontSize = 24;
app.titleLabel.FontWeight = 'bold';
app.titleLabel.Position = [37 544 644 30];
app.titleLabel.Text = 'Reversible Reaction in an Adiabatic Plug-Flow Reactor';

% Create DropDownor
app.DropDownor = uidropdown(app.PFR4UIFigure);
app.DropDownor.Items = {'Exothermic', 'Endothermic', ''};
app.DropDownor.ItemsData = {'1', '2'};
app.DropDownor.ValueChangedFcn = createCallbackFcn(app, @DropDownorValueChanged, true);
app.DropDownor.Position = [61 506 100 22];
app.DropDownor.Value = '1';

% Create FeedtemperatureKSliderLabel
app.FeedtemperatureKSliderLabel = uilabel(app.PFR4UIFigure);
app.FeedtemperatureKSliderLabel.HorizontalAlignment = 'right';
app.FeedtemperatureKSliderLabel.Position = [61 464 119 22];
app.FeedtemperatureKSliderLabel.Text = 'Feed temperature [K]';

% Create T0Slider
app.T0Slider = uislider(app.PFR4UIFigure);
app.T0Slider.Limits = [300 400];
app.T0Slider.ValueChangedFcn = createCallbackFcn(app, @T0SliderValueChanged, true);
app.T0Slider.MinorTicks = [];
app.T0Slider.Position = [201 473 150 3];
app.T0Slider.Value = 350;

% Create molarratioofinerttoreactantinfeedSliderLabel
app.molarratioofinerttoreactantinfeedSliderLabel = uilabel(app.PFR4UIFigure);
app.molarratioofinerttoreactantinfeedSliderLabel.HorizontalAlignment = 'right';
app.molarratioofinerttoreactantinfeedSliderLabel.Position = [268 506 203 22];
app.molarratioofinerttoreactantinfeedSliderLabel.Text = 'molar ratio of inert to reactant in feed';

% Create alfaSlider
app.alfaSlider = uislider(app.PFR4UIFigure);
app.alfaSlider.Limits = [0 1];
app.alfaSlider.ValueChangedFcn = createCallbackFcn(app, @alfaSliderValueChanged, true);
app.alfaSlider.MinorTicks = [];
app.alfaSlider.Position = [492 515 150 3];
app.alfaSlider.Value = 0.5;

% Create TeLabel
app.TeLabel = uilabel(app.PFR4UIFigure);
app.TeLabel.FontSize = 14;
app.TeLabel.FontWeight = 'bold';
app.TeLabel.Position = [586 346 80 22];
app.TeLabel.Text = 'Te';

% Create xeLabel
app.xeLabel = uilabel(app.PFR4UIFigure);
app.xeLabel.FontSize = 14;
app.xeLabel.FontWeight = 'bold';
app.xeLabel.Position = [586 315 95 22];
app.xeLabel.Text = 'xe';

% Create T0Edit
app.T0Edit = uieditfield(app.PFR4UIFigure, 'text');
app.T0Edit.ValueChangedFcn = createCallbackFcn(app, @T0EditValueChanged, true);
```

```
            app.T0Edit.Position = [360 460 45 22];

            % Create alfaEdit
            app.alfaEdit = uieditfield(app.PFR4UIFigure, 'text');
            app.alfaEdit.ValueChangedFcn = createCallbackFcn(app, @alfaEditValueChanged, true);
            app.alfaEdit.Position = [656 505 42 22];

            % Show the figure after all components are created
            app.PFR4UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr4

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.PFR4UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR4UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 5

pfr5.mlapp

```
classdef pfr5 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR5UIFigure                 matlab.ui.Figure
        TabGroup                     matlab.ui.container.TabGroup
        MolarflowTab                 matlab.ui.container.Tab
        UIAxes1                      matlab.ui.control.UIAxes
        TemperatureTab               matlab.ui.container.Tab
        UIAxes2                      matlab.ui.control.UIAxes
        EaSlider                     matlab.ui.control.Slider
        ActivationEnergykjmolSliderLabel  matlab.ui.control.Label
        T0Slider                     matlab.ui.control.Slider
        FeedtemperatureKSliderLabel  matlab.ui.control.Label
        F0Slider                     matlab.ui.control.Slider
        totalmolarflowratemolsSliderLabel  matlab.ui.control.Label
        rubrik                       matlab.ui.control.Label
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.PFR5UIFigure,'off');
                F0=app.F0Slider.Value;
                T0=app.T0Slider.Value;
                Ea=app.EaSlider.Value;
                k=5e8;
                Ta=625;
                Cpm=1;
                U=0.37;
                P=1.0;
                L=1.5;
                r=0.0125;
                R=8.206e-2;
                yxf=0.019;
                DeltaH=-1.28e6;
                R2=8.314e-3;
                MW=33.41;
                m=MW*F0/1000;
                A=pi*r.^2;
                Beta=DeltaH*A/m/Cpm;
                Gamma=2*pi*r*U/m/Cpm;
                [t,y]=ode15s(@model,[0 L],[yxf*F0/1000,T0],odeset('Abstol',1e-9,'RelTol',1e-4));
                plot(app.UIAxes1,t,y(:,1)*1e6,'k','LineWidth',1.5)
                plot(app.UIAxes2,t,y(:,2),'k','LineWidth',1.5)
                ylim=app.UIAxes1.YLim;
                app.UIAxes1.YLim=[0 ylim(2)];
                toogleUIC(app.PFR5UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function dy = model(~,y)
                dy=zeros(size(y));
                Fx=y(1);
                T=y(2);
                Rho=k*exp(-Ea/R2/T)*P/R/T*Fx/F0*1000;
                dy(1)=-Rho*A;
                dy(2)=-Beta*Rho+Gamma*(Ta-T);
            end
        end
    end


    % Callbacks that handle component events
    methods (Access = private)
```

```
    % Code that executes after component creation
    function startupFcn(app)
        draw(app);
    end

    % Value changed function: EaSlider, F0Slider, T0Slider
    function F0SliderValueChanged(app, event)
        draw(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create PFR5UIFigure and hide until all components are created
        app.PFR5UIFigure = uifigure('Visible', 'off');
        app.PFR5UIFigure.Color = [0.94 0.94 0.94];
        app.PFR5UIFigure.Position = [100 100 740 560];
        app.PFR5UIFigure.Name = 'PFR 5';
        app.PFR5UIFigure.Scrollable = 'on';

        % Create rubrik
        app.rubrik = uilabel(app.PFR5UIFigure);
        app.rubrik.FontSize = 22;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [48 518 693 27];
        app.rubrik.Text = 'Parametric Sensitivity of Plug Flow Reactor With Heat Exchange';

        % Create totalmolarflowratemolsSliderLabel
        app.totalmolarflowratemolsSliderLabel = uilabel(app.PFR5UIFigure);
        app.totalmolarflowratemolsSliderLabel.HorizontalAlignment = 'right';
        app.totalmolarflowratemolsSliderLabel.Position = [30 469 149 22];
        app.totalmolarflowratemolsSliderLabel.Text = {'total molar flow rate [mol/s]'; ''};

        % Create F0Slider
        app.F0Slider = uislider(app.PFR5UIFigure);
        app.F0Slider.Limits = [0.05 0.15];
        app.F0Slider.MajorTicks = [0.05 0.1 0.15];
        app.F0Slider.ValueChangedFcn = createCallbackFcn(app, @F0SliderValueChanged, true);
        app.F0Slider.MinorTicks = [0.075 0.125];
        app.F0Slider.Position = [200 478 150 3];
        app.F0Slider.Value = 0.1;

        % Create FeedtemperatureKSliderLabel
        app.FeedtemperatureKSliderLabel = uilabel(app.PFR5UIFigure);
        app.FeedtemperatureKSliderLabel.HorizontalAlignment = 'right';
        app.FeedtemperatureKSliderLabel.Position = [407 473 119 22];
        app.FeedtemperatureKSliderLabel.Text = 'Feed temperature [K]';

        % Create T0Slider
        app.T0Slider = uislider(app.PFR5UIFigure);
        app.T0Slider.Limits = [600 640];
        app.T0Slider.MajorTicks = [600 610 620 630 640];
        app.T0Slider.ValueChangedFcn = createCallbackFcn(app, @F0SliderValueChanged, true);
        app.T0Slider.MinorTicks = [];
        app.T0Slider.Position = [547 482 150 3];
        app.T0Slider.Value = 620;

        % Create ActivationEnergykjmolSliderLabel
        app.ActivationEnergykjmolSliderLabel = uilabel(app.PFR5UIFigure);
        app.ActivationEnergykjmolSliderLabel.HorizontalAlignment = 'right';
        app.ActivationEnergykjmolSliderLabel.Position = [385 406 140 22];
        app.ActivationEnergykjmolSliderLabel.Text = {'Activation Energy [kj/mol]'; ''};

        % Create EaSlider
        app.EaSlider = uislider(app.PFR5UIFigure);
        app.EaSlider.Limits = [83 125];
        app.EaSlider.MajorTicks = [83 95 105 115 125];
```

```
            app.EaSlider.ValueChangedFcn = createCallbackFcn(app, @F0SliderValueChanged, true);
            app.EaSlider.MinorTicks = [];
            app.EaSlider.Position = [546 415 150 3];
            app.EaSlider.Value = 90;

            % Create TabGroup
            app.TabGroup = uitabgroup(app.PFR5UIFigure);
            app.TabGroup.Position = [29 25 678 335];

            % Create MolarflowTab
            app.MolarflowTab = uitab(app.TabGroup);
            app.MolarflowTab.Title = 'Molar flow';

            % Create UIAxes1
            app.UIAxes1 = uiaxes(app.MolarflowTab);
            title(app.UIAxes1, 'Molar flow rate')
            xlabel(app.UIAxes1, 'L [m]')
            ylabel(app.UIAxes1, 'molar flow [mmol/s]')
            app.UIAxes1.XLim = [0 1.5];
            app.UIAxes1.XTick = [0.25 0.5 0.75 1 1.25 1.5];
            app.UIAxes1.XTickLabelRotation = 0;
            app.UIAxes1.YTickLabelRotation = 0;
            app.UIAxes1.ZTickLabelRotation = 0;
            app.UIAxes1.XGrid = 'on';
            app.UIAxes1.YGrid = 'on';
            app.UIAxes1.Position = [35 12 620 286];

            % Create TemperatureTab
            app.TemperatureTab = uitab(app.TabGroup);
            app.TemperatureTab.Title = 'Temperature';

            % Create UIAxes2
            app.UIAxes2 = uiaxes(app.TemperatureTab);
            title(app.UIAxes2, 'Temperature')
            xlabel(app.UIAxes2, 'L [m]')
            ylabel(app.UIAxes2, 'Temperature [K]')
            app.UIAxes2.XLim = [0 1.5];
            app.UIAxes2.XTick = [0 0.25 0.5 0.75 1 1.25 1.5];
            app.UIAxes2.XTickLabelRotation = 0;
            app.UIAxes2.YTickLabelRotation = 0;
            app.UIAxes2.ZTickLabelRotation = 0;
            app.UIAxes2.XGrid = 'on';
            app.UIAxes2.YGrid = 'on';
            app.UIAxes2.Position = [35 12 620 286];

            % Show the figure after all components are created
            app.PFR5UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr5

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.PFR5UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
```

```
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR5UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 6

pfr6.mlapp

```
classdef pfr6 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR6UIFigure                   matlab.ui.Figure
        TabGroup                       matlab.ui.container.TabGroup
        PressureTab                    matlab.ui.container.Tab
        UIAxes                         matlab.ui.control.UIAxes
        MoleFlowTab                    matlab.ui.container.Tab
        UIAxes_2                       matlab.ui.control.UIAxes
        ConversionTab                  matlab.ui.container.Tab
        UIAxes_3                       matlab.ui.control.UIAxes
        VolumetricflowTab              matlab.ui.container.Tab
        UIAxes_4                       matlab.ui.control.UIAxes
        DpSlider                       matlab.ui.control.Slider
        particlediametermmSliderLabel  matlab.ui.control.Label
        rubrik                         matlab.ui.control.Label
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.PFR6UIFigure,'off');
                Phi=0.4;
                Ac=0.002;
                k=0.017;
                Mu=0.000023;
                R=8.314;
                T=423;
                m1=0.0139;
                m2=0.0112;
                Ft1=0.257;
                Ft2=0.208;
                Dp=app.DpSlider.Value*1e-3;
                laminar=150*(1-Phi)*Mu/Dp;
                G=@(m) 1.75*m/Ac;
                [L1,Y1]=ode45(@(t,x) diffekv(t,x,G(m1),1,Ft1),[0 14],[Ft1 12e5]);
                [L2,Y2]=ode45(@(t,x) diffekv(t,x,G(m2),1,Ft2),[0 21],[Ft2 12e5]);
                [L3,Y3]=ode45(@(t,x) diffekv(t,x,G(m1),0,Ft1),[0 14],[Ft1 12e5]);
                [L4,Y4]=ode45(@(t,x) diffekv(t,x,G(m2),0,Ft2),[0 21],[Ft2 12e5]);
                plot(app.UIAxes,...
                    L1,Y1(:,2)/1e5,'b',...
                    L2,Y2(:,2)/1e5,'g',....
                    L3,Y3(:,2)/1e5,'b--',...
                    L4,Y4(:,2)/1e5,'g--');
                legend(app.UIAxes,...
                    'L= 14 m \Delta P <> 0',...
                    'L= 21 m \Delta P <> 0',...
                    'L= 14 m \Delta P = 0',...
                    'L= 21 m \Delta P = 0',"Location","best")
                plot(app.UIAxes_2,...
                    L1,Y1(:,1),'b',...
                    L2,Y2(:,1),'g',....
                    L3,Y3(:,1),'b--',...
                    L4,Y4(:,1),'g--');
                legend(app.UIAxes_2,...
                    'L= 14 m \Delta P <> 0',...
                    'L= 21 m \Delta P <> 0',...
                    'L= 14 m \Delta P = 0',...
                    'L= 21 m \Delta P = 0',"Location","best")
                plot(app.UIAxes_3,...
                    L1,(Y1(1,1)-Y1(:,1))/Y1(1,1),'b',...
                    L2,(Y2(1,1)-Y2(:,1))/Y2(1,1),'g',....
                    L3,(Y3(1,1)-Y3(:,1))/Y3(1,1),'b--',...
                    L4,(Y4(1,1)-Y4(:,1))/Y4(1,1),'g--');
```

```
            legend(app.UIAxes_3,...
                'L= 14 m \Delta P <> 0',...
                'L= 21 m \Delta P <> 0',...
                'L= 14 m \Delta P = 0',...
                'L= 21 m \Delta P = 0',"Location","best")
            plot(app.UIAxes_4,...
                L1,Ft1*R*T./Y1(:,2)*1e3,'b',...
                L2,Ft2*R*T./Y2(:,2)*1e3,'g',....
                L3,Ft1*R*T./Y3(:,2)*1e3,'b--',...
                L4,Ft2*R*T./Y4(:,2)*1e3,'g--');
            legend(app.UIAxes_4,...
                'L= 14 m \Delta P <> 0',...
                'L= 21 m \Delta P <> 0',...
                'L= 14 m \Delta P = 0',...
                'L= 21 m \Delta P = 0',"Location","best")
            toogleUIC(app.PFR6UIFigure,'on');
        catch ME
            errordlg(ME.message);
        end

        function dy = diffekv(~,y,p1,p2,p3)
            dy=zeros(size(y));
            FA=y(1);
            P=y(2);
            turbulent=p1;
            Ftot=p3;
            ca=FA/Ftot*P/R/T;
            dy(1)=-k*ca*Ac;
            if p2 == 1
                upsilon=Ftot*R*T/P;
                dy(2)=-(1-Phi)*upsilon/Dp/Phi^3/Ac*(laminar+turbulent);
            end

        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        simulera(app);
    end

    % Value changed function: DpSlider
    function DpSliderValueChanged(app, event)
        simulera(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create PFR6UIFigure and hide until all components are created
        app.PFR6UIFigure = uifigure('Visible', 'off');
        app.PFR6UIFigure.Color = [0.94 0.94 0.94];
        app.PFR6UIFigure.Position = [100 100 740 550];
        app.PFR6UIFigure.Name = 'PFR 6';

        % Create rubrik
        app.rubrik = uilabel(app.PFR6UIFigure);
        app.rubrik.FontSize = 24;
        app.rubrik.FontWeight = 'bold';
        app.rubrik.Position = [155 509 467 30];
        app.rubrik.Text = 'Pressure Drop in a Packed Bed Reactor';

        % Create particlediametermmSliderLabel
```

```
app.particlediametermmSliderLabel = uilabel(app.PFR6UIFigure);
app.particlediametermmSliderLabel.HorizontalAlignment = 'right';
app.particlediametermmSliderLabel.Position = [221 476 126 22];
app.particlediametermmSliderLabel.Text = 'particle diameter (mm)';

% Create DpSlider
app.DpSlider = uislider(app.PFR6UIFigure);
app.DpSlider.Limits = [1.2 5];
app.DpSlider.MajorTicks = [1.2 2 3 4 5 6];
app.DpSlider.ValueChangedFcn = createCallbackFcn(app, @DpSliderValueChanged, true);
app.DpSlider.MinorTicks = [];
app.DpSlider.Position = [368 485 150 3];
app.DpSlider.Value = 2;

% Create TabGroup
app.TabGroup = uitabgroup(app.PFR6UIFigure);
app.TabGroup.Position = [32 47 682 391];

% Create PressureTab
app.PressureTab = uitab(app.TabGroup);
app.PressureTab.Title = 'Pressure';

% Create UIAxes
app.UIAxes = uiaxes(app.PressureTab);
xlabel(app.UIAxes, 'reactor length [m]')
ylabel(app.UIAxes, 'pressure [bar]')
app.UIAxes.PlotBoxAspectRatio = [2.1841155234657 1 1];
app.UIAxes.XTickLabelRotation = 0;
app.UIAxes.YTickLabelRotation = 0;
app.UIAxes.ZTickLabelRotation = 0;
app.UIAxes.Position = [16 20 653 334];

% Create MoleFlowTab
app.MoleFlowTab = uitab(app.TabGroup);
app.MoleFlowTab.Title = 'Mole Flow';

% Create UIAxes_2
app.UIAxes_2 = uiaxes(app.MoleFlowTab);
xlabel(app.UIAxes_2, 'reactor length [m]')
ylabel(app.UIAxes_2, 'mole flow [mole/s]')
app.UIAxes_2.PlotBoxAspectRatio = [2.07191780821918 1 1];
app.UIAxes_2.XTickLabelRotation = 0;
app.UIAxes_2.YTick = [0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1];
app.UIAxes_2.YTickLabelRotation = 0;
app.UIAxes_2.ZTickLabelRotation = 0;
app.UIAxes_2.Position = [16 20 653 334];

% Create ConversionTab
app.ConversionTab = uitab(app.TabGroup);
app.ConversionTab.Title = 'Conversion';

% Create UIAxes_3
app.UIAxes_3 = uiaxes(app.ConversionTab);
xlabel(app.UIAxes_3, 'reactor length [m]')
ylabel(app.UIAxes_3, 'Conversion')
app.UIAxes_3.PlotBoxAspectRatio = [2.07191780821918 1 1];
app.UIAxes_3.XTickLabelRotation = 0;
app.UIAxes_3.YTick = [0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1];
app.UIAxes_3.YTickLabelRotation = 0;
app.UIAxes_3.ZTickLabelRotation = 0;
app.UIAxes_3.Position = [16 20 653 334];

% Create VolumetricflowTab
app.VolumetricflowTab = uitab(app.TabGroup);
app.VolumetricflowTab.Title = 'Volumetric flow';

% Create UIAxes_4
app.UIAxes_4 = uiaxes(app.VolumetricflowTab);
xlabel(app.UIAxes_4, 'reactor length [m]')
ylabel(app.UIAxes_4, 'volumetric flow [L/s]')
app.UIAxes_4.PlotBoxAspectRatio = [2.07534246575342 1 1];
```

```
            app.UIAxes_4.XTickLabelRotation = 0;
            app.UIAxes_4.YTick = [0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0.9 1];
            app.UIAxes_4.YTickLabelRotation = 0;
            app.UIAxes_4.ZTickLabelRotation = 0;
            app.UIAxes_4.Position = [16 20 653 334];

            % Show the figure after all components are created
            app.PFR6UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr6

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.PFR6UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR6UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 7

pfr7.mlapp

```
classdef pfr7 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        PFR7UIFigure             matlab.ui.Figure
        rEdit                    matlab.ui.control.EditField
        conversionideal          matlab.ui.control.Label
        conversion               matlab.ui.control.Label
        spacetime                matlab.ui.control.Label
        rSlider                  matlab.ui.control.Slider
        recycleratioSliderLabel  matlab.ui.control.Label
        rubrik                   matlab.ui.control.Label
        UIAxes                   matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.PFR7UIFigure,'off');
                vf=50;
                P=5.4;
                T=325;
                R=0.08314;
                Cf=P/R/T;
                Vreak=250;
                k=0.5;
                r=app.rSlider.Value;
                Ce=fzero(@findce,[eps Cf]);
                C0=(Cf+r*Ce)/(1+r);
                [V,C]=ode45(@diffekv,[0 Vreak],C0);
                plot(app.UIAxes,V,C);
                app.UIAxes.YLim=[0 1.1*C0];
                theta=Vreak/vf;
                app.spacetime.Text=sprintf('Space time %5.1f seconds',theta/(1+r)*60);
                app.conversion.Text=sprintf('Conversion %4.1f %%',(Cf-C(end))/Cf*100);
                app.conversionideal.Text=sprintf('PFR %4.1f %%  CSTR %4.1f %%',(1-exp(-theta*k))*100,(k*theta/(1+k*theta))*100);
                toogleUIC(app.PFR7UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

            function res = findce(C)
                res=log(C*(1+r)/(Cf+r*C))+Vreak/vf/(1+r)*k;
            end
            function dy = diffekv(~,C)
                dy=-k*C/vf/(1+r);
            end
        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            r=round(app.rSlider.Value);
            app.rEdit.Value=sprintf('%3.0f',r);
            simulera(app);
        end

        % Value changed function: rSlider
        function rSliderValueChanged(app, event)
            r=app.rSlider.Value;
            app.rEdit.Value=sprintf('%3.0f',r);
```

```
            simulera(app);
        end

        % Value changed function: rEdit
        function rEditValueChanged(app, event)
            value = app.rEdit.Value;
            if between(app,value,app.rSlider.Limits)
                r=round(str2double(value));
                app.rEdit.Value=sprintf('%3.0f',r);
                app.rSlider.Value=r;
                simulera(app);
            else
                app.rEdit.Value=num2str(app.rSlider.Value);
            end
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create PFR7UIFigure and hide until all components are created
            app.PFR7UIFigure = uifigure('Visible', 'off');
            app.PFR7UIFigure.Color = [0.94 0.94 0.94];
            app.PFR7UIFigure.Position = [100 100 640 500];
            app.PFR7UIFigure.Name = 'PFR 7';

            % Create UIAxes
            app.UIAxes = uiaxes(app.PFR7UIFigure);
            xlabel(app.UIAxes, 'Volume [L]')
            ylabel(app.UIAxes, 'Concentration [mol/L]')
            app.UIAxes.XLim = [0 250];
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [27 27 587 312];

            % Create rubrik
            app.rubrik = uilabel(app.PFR7UIFigure);
            app.rubrik.FontSize = 24;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [87 459 505 30];
            app.rubrik.Text = 'Isothermal Plug Flow Reactor with Recycle';

            % Create recycleratioSliderLabel
            app.recycleratioSliderLabel = uilabel(app.PFR7UIFigure);
            app.recycleratioSliderLabel.HorizontalAlignment = 'right';
            app.recycleratioSliderLabel.Position = [120 430 70 22];
            app.recycleratioSliderLabel.Text = 'recycle ratio';

            % Create rSlider
            app.rSlider = uislider(app.PFR7UIFigure);
            app.rSlider.MajorTicks = [0 5 10 20 30 40 50 60 70 80 90 100];
            app.rSlider.ValueChangedFcn = createCallbackFcn(app, @rSliderValueChanged, true);
            app.rSlider.MinorTicks = [1 2 3 4 5];
            app.rSlider.Position = [211 439 299 3];

            % Create spacetime
            app.spacetime = uilabel(app.PFR7UIFigure);
            app.spacetime.FontWeight = 'bold';
            app.spacetime.Position = [70 379 169 22];
            app.spacetime.Text = '';

            % Create conversion
            app.conversion = uilabel(app.PFR7UIFigure);
            app.conversion.FontWeight = 'bold';
            app.conversion.Position = [408 379 107 22];
            app.conversion.Text = '';

            % Create conversionideal
```

```
            app.conversionideal = uilabel(app.PFR7UIFigure);
            app.conversionideal.FontWeight = 'bold';
            app.conversionideal.Position = [408 346 174 22];
            app.conversionideal.Text = '';

            % Create rEdit
            app.rEdit = uieditfield(app.PFR7UIFigure, 'text');
            app.rEdit.ValueChangedFcn = createCallbackFcn(app, @rEditValueChanged, true);
            app.rEdit.Position = [523 429 44 22];

            % Show the figure after all components are created
            app.PFR7UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr7

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.PFR7UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.PFR7UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

PFR 8

pfr8.mlapp

```
classdef pfr8 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        UIFigure                     matlab.ui.Figure
        FirstreactorPanel            matlab.ui.container.Panel
        ProdLabel                    matlab.ui.control.Label
        xLabel                       matlab.ui.control.Label
        TLabel                       matlab.ui.control.Label
        SecondreactorPanel           matlab.ui.container.Panel
        totXLabel                    matlab.ui.control.Label
        V2                           matlab.ui.control.Slider
        VolumeSliderLabel            matlab.ui.control.Label
        T0Edit                       matlab.ui.control.EditField
        FA0                          matlab.ui.control.Spinner
        InletflowSpinnerLabel        matlab.ui.control.Label
        T0                           matlab.ui.control.Slider
        InlettemperatureSliderLabel  matlab.ui.control.Label
        Rubrik                       matlab.ui.control.Label
        TabGroup                     matlab.ui.container.TabGroup
        ConversionTab                matlab.ui.container.Tab
        UIAxes                       matlab.ui.control.UIAxes
        TemperatureTab               matlab.ui.container.Tab
        UIAxes_2                     matlab.ui.control.UIAxes
        kandKTab                     matlab.ui.container.Tab
        UIAxes_3                     matlab.ui.control.UIAxes
        XeTab                        matlab.ui.container.Tab
        UIAxes_4                     matlab.ui.control.UIAxes
        SecondreactorTab             matlab.ui.container.Tab
        UIAxes_5                     matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function k = rateConstant(~,T)
            E=1e4;
            R=1.987;
            k1=0.000035;
            T1=298;
            k=k1*exp(E/R*(1/T1-1./T));
        end

        function K = equilConstant(~,T)
            R=1.987;
            K1=75000;
            T1=298;
            deltaH=-14000;
            K=K1*exp(deltaH/R*(1/T1-1./T));
        end

        function plotChem(app)
            T=linspace(300,700);
            k=rateConstant(app,T);
            k1=k./rateConstant(app,298);
            yyaxis(app.UIAxes_3,'left');
            plot(app.UIAxes_3,T,log10(k1));
            app.UIAxes_3.YLabel.Interpreter='latex';
            ylabel(app.UIAxes_3,'$\log_{10}\left (\frac{k}{k_{298}}\right )$')
            K=equilConstant(app,T);
            K1=K./equilConstant(app,298);
            yyaxis(app.UIAxes_3,'right');
            plot(app.UIAxes_3,T,log10(K1));
            app.UIAxes_3.YLabel.Interpreter='latex';
            ylabel(app.UIAxes_3,'$\log_{10}\left (\frac{K}{K_{298}}\right )$')
            plot(app.UIAxes_4,T,K1./(1+K1));
        end
```

```
    function simulera(app)
        toogleUIC(app.UIFigure,'off')
        Tin=app.T0.Value;
        Fa0=5;
        Ca0=1;
        deltaH=-14000;
        Fin=Fa0*app.FA0.Value;
        [V,y]=ode45(@ode,[0 100],[Fin,Tin]);
        Keq=equilConstant(app,y(:,2));
        x=(Fin-y(:,1))./Fin;
        plot(app.UIAxes,V,x,V,Keq./(1+Keq));
        legend(app.UIAxes,'x','x_{eq}','location','best');
        plot(app.UIAxes_2,V,y(:,2))
        app.TLabel.Text=sprintf('Outlet temperatur %3.0f K',y(end,2));
        app.xLabel.Text=sprintf('Outlet conversion %4.2f %%',x(end));
        app.ProdLabel.Text=sprintf('Production B %3.2f mol/min',Fin*x(end));
        Volume=app.V2.Value;
        if Volume > 0
            [V,y]=ode45(@ode,[0,Volume],[y(end,1),Tin]);
            Keq=equilConstant(app,y(:,2));
            x=(Fin-y(:,1))./Fin;
            plot(app.UIAxes_5,V,x,V,Keq./(1+Keq));
            legend(app.UIAxes_5,'x','x_{eq}','location','best');
            app.totXLabel.Text=sprintf('Total conversion %4.2f %%',x(end));
        end
        toogleUIC(app.UIFigure,'on')
        function dy = ode(~,y)
            dy=zeros(size(y));
            CA=y(1)*Ca0/Fin;
            T=y(2);
            r=rateConstant(app,T)*(CA -(Ca0-CA)/equilConstant(app,T));
            dy(1)=-r;
            dy(2)=-r*deltaH/Fin/75;
        end

    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        plotChem(app);
        app.T0Edit.Value=sprintf('%3.0f',app.T0.Value);
        simulera(app);
    end

    % Value changed function: T0
    function T0ValueChanged(app, event)
        value = app.T0.Value;
        app.T0Edit.Value=sprintf('%3.0f',value);
        simulera(app);
    end

    % Value changed function: FA0
    function FA0ValueChanged(app, event)
        %value = app.FA0.Value;
        simulera(app);
    end

    % Value changed function: T0Edit
    function T0EditValueChanged(app, event)
        value = app.T0Edit.Value;
        if between(app,value,app.T0.Limits)
            app.T0.Value=str2double(value);
            simulera(app);
        else
            app.T0Edit.Value=num2str(app.T0.Value);
        end
```

```
        end

        % Value changed function: V2
        function V2ValueChanged(app, event)
            %value = app.V2.Value;
            simulera(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create UIFigure and hide until all components are created
            app.UIFigure = uifigure('Visible', 'off');
            app.UIFigure.Color = [0.94 0.94 0.94];
            app.UIFigure.Position = [100 100 740 675];
            app.UIFigure.Name = 'MATLAB App';

            % Create TabGroup
            app.TabGroup = uitabgroup(app.UIFigure);
            app.TabGroup.Position = [35 28 648 378];

            % Create ConversionTab
            app.ConversionTab = uitab(app.TabGroup);
            app.ConversionTab.Title = 'Conversion';

            % Create UIAxes
            app.UIAxes = uiaxes(app.ConversionTab);
            title(app.UIAxes, {'Conversion profile'; ''})
            xlabel(app.UIAxes, 'Volume [dm^3]')
            ylabel(app.UIAxes, {'Conversion'; ''})
            app.UIAxes.XTickLabelRotation = 0;
            app.UIAxes.YTickLabelRotation = 0;
            app.UIAxes.ZTickLabelRotation = 0;
            app.UIAxes.Position = [23 18 607 315];

            % Create TemperatureTab
            app.TemperatureTab = uitab(app.TabGroup);
            app.TemperatureTab.Title = 'Temperature';

            % Create UIAxes_2
            app.UIAxes_2 = uiaxes(app.TemperatureTab);
            title(app.UIAxes_2, 'Temperature profile')
            xlabel(app.UIAxes_2, 'Volume  [dm^3]')
            ylabel(app.UIAxes_2, 'Y')
            app.UIAxes_2.XTickLabelRotation = 0;
            app.UIAxes_2.YTickLabelRotation = 0;
            app.UIAxes_2.ZTickLabelRotation = 0;
            app.UIAxes_2.Position = [23 18 607 315];

            % Create kandKTab
            app.kandKTab = uitab(app.TabGroup);
            app.kandKTab.Title = 'k and K';

            % Create UIAxes_3
            app.UIAxes_3 = uiaxes(app.kandKTab);
            title(app.UIAxes_3, 'Title')
            xlabel(app.UIAxes_3, 'Temperature [K]')
            ylabel(app.UIAxes_3, 'Y')
            app.UIAxes_3.LabelFontSizeMultiplier = 1.5;
            app.UIAxes_3.XTickLabelRotation = 0;
            app.UIAxes_3.YTickLabelRotation = 0;
            app.UIAxes_3.ZTickLabelRotation = 0;
            app.UIAxes_3.Position = [23 18 607 315];

            % Create XeTab
            app.XeTab = uitab(app.TabGroup);
            app.XeTab.Title = 'Xe';
```

```
% Create UIAxes_4
app.UIAxes_4 = uiaxes(app.XeTab);
title(app.UIAxes_4, 'Equilibrium conversion')
xlabel(app.UIAxes_4, 'Temperature  [K]')
ylabel(app.UIAxes_4, 'Conversion')
app.UIAxes_4.XTickLabelRotation = 0;
app.UIAxes_4.YTickLabelRotation = 0;
app.UIAxes_4.ZTickLabelRotation = 0;
app.UIAxes_4.Position = [23 18 607 315];

% Create SecondreactorTab
app.SecondreactorTab = uitab(app.TabGroup);
app.SecondreactorTab.Title = 'Second reactor';

% Create UIAxes_5
app.UIAxes_5 = uiaxes(app.SecondreactorTab);
title(app.UIAxes_5, {'Conversion profile'; ''})
xlabel(app.UIAxes_5, 'Volume [dm^3]')
ylabel(app.UIAxes_5, {'Conversion'; ''})
app.UIAxes_5.XTickLabelRotation = 0;
app.UIAxes_5.YTickLabelRotation = 0;
app.UIAxes_5.ZTickLabelRotation = 0;
app.UIAxes_5.Position = [23 18 607 315];

% Create Rubrik
app.Rubrik = uilabel(app.UIFigure);
app.Rubrik.FontSize = 18;
app.Rubrik.FontWeight = 'bold';
app.Rubrik.Position = [193 622 388 22];
app.Rubrik.Text = 'Finding the Optimum Entering Temperature ';

% Create InlettemperatureSliderLabel
app.InlettemperatureSliderLabel = uilabel(app.UIFigure);
app.InlettemperatureSliderLabel.HorizontalAlignment = 'right';
app.InlettemperatureSliderLabel.Position = [29 561 114 22];
app.InlettemperatureSliderLabel.Text = 'Inlet temperature';

% Create T0
app.T0 = uislider(app.UIFigure);
app.T0.Limits = [300 700];
app.T0.MajorTicks = [300 400 500 600 700];
app.T0.ValueChangedFcn = createCallbackFcn(app, @T0ValueChanged, true);
app.T0.MinorTicks = [350 450 550 650];
app.T0.Position = [166 571 150 3];
app.T0.Value = 400;

% Create InletflowSpinnerLabel
app.InletflowSpinnerLabel = uilabel(app.UIFigure);
app.InletflowSpinnerLabel.HorizontalAlignment = 'right';
app.InletflowSpinnerLabel.Position = [514 551 53 22];
app.InletflowSpinnerLabel.Text = 'Inlet flow';

% Create FA0
app.FA0 = uispinner(app.UIFigure);
app.FA0.Step = 0.2;
app.FA0.Limits = [0.6 1.4];
app.FA0.ValueChangedFcn = createCallbackFcn(app, @FA0ValueChanged, true);
app.FA0.Position = [582 551 100 22];
app.FA0.Value = 1;

% Create T0Edit
app.T0Edit = uieditfield(app.UIFigure, 'text');
app.T0Edit.ValueChangedFcn = createCallbackFcn(app, @T0EditValueChanged, true);
app.T0Edit.Position = [334 561 49 22];

% Create SecondreactorPanel
app.SecondreactorPanel = uipanel(app.UIFigure);
app.SecondreactorPanel.Title = 'Second reactor';
app.SecondreactorPanel.FontAngle = 'italic';
app.SecondreactorPanel.FontWeight = 'bold';
app.SecondreactorPanel.Position = [396 426 287 106];
```

```
            % Create VolumeSliderLabel
            app.VolumeSliderLabel = uilabel(app.SecondreactorPanel);
            app.VolumeSliderLabel.HorizontalAlignment = 'right';
            app.VolumeSliderLabel.Position = [10 61 46 22];
            app.VolumeSliderLabel.Text = 'Volume';

            % Create V2
            app.V2 = uislider(app.SecondreactorPanel);
            app.V2.Limits = [0 200];
            app.V2.MajorTicks = [0 50 100 150 200];
            app.V2.ValueChangedFcn = createCallbackFcn(app, @V2ValueChanged, true);
            app.V2.MinorTicks = [25 75 125 175];
            app.V2.Position = [88 71 150 3];

            % Create totXLabel
            app.totXLabel = uilabel(app.SecondreactorPanel);
            app.totXLabel.FontWeight = 'bold';
            app.totXLabel.Position = [21 11 217 22];
            app.totXLabel.Text = '';

            % Create FirstreactorPanel
            app.FirstreactorPanel = uipanel(app.UIFigure);
            app.FirstreactorPanel.Title = 'First reactor';
            app.FirstreactorPanel.FontAngle = 'italic';
            app.FirstreactorPanel.FontWeight = 'bold';
            app.FirstreactorPanel.Position = [75 428 287 106];

            % Create TLabel
            app.TLabel = uilabel(app.FirstreactorPanel);
            app.TLabel.FontWeight = 'bold';
            app.TLabel.Position = [9 61 208 22];
            app.TLabel.Text = '';

            % Create xLabel
            app.xLabel = uilabel(app.FirstreactorPanel);
            app.xLabel.FontWeight = 'bold';
            app.xLabel.Position = [9 40 155 22];
            app.xLabel.Text = {''; ''};

            % Create ProdLabel
            app.ProdLabel = uilabel(app.FirstreactorPanel);
            app.ProdLabel.FontWeight = 'bold';
            app.ProdLabel.Position = [9 19 161 22];
            app.ProdLabel.Text = '';

            % Show the figure after all components are created
            app.UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = pfr8

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
```

```
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)
results=0;
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))
    results=1;
end
end
```

toogleUIC.m

```
function toogleUIC(h, state)
arguments
    h matlab.ui.Figure
    state char
end
% toggle the state of the uicontrols in the app
uics = findall(h);
for idx = 1:length(uics)
    uic = uics(idx);
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')
        uic.Enable = state;
    end
end
drawnow; % flush the ui events queue
end
```

Safety 1

safety1.mlapp

```
classdef safety1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Safety1UIFigure             matlab.ui.Figure
        FixaField                   matlab.ui.control.EditField
        StoppField                  matlab.ui.control.EditField
        FixaSlider                  matlab.ui.control.Slider
        LengthofstopminLabel        matlab.ui.control.Label
        StoppSlider                 matlab.ui.control.Slider
        StopafterminLabel           matlab.ui.control.Label
        AccidentMonsantoSaugetLabel  matlab.ui.control.Label
        UIAxes                      matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.Safety1UIFigure,'off');
                KylStop=app.StoppSlider.Value;
                KylStart=app.FixaSlider.Value+KylStop;
                Tstart=175;
                V1=3.26;NONCB=3.17;NNH3=43;NH2O=103.6;
                NCp=NONCB*40+NNH3*8.38+NH2O*18;V=V1;
                [t2,y2]=ode23s(@modell2,[0 3*60],[NONCB/V1 NNH3/V1 Tstart]);
                V2=5.119;NONCB=9.0448;NNH3=33;NH2O=103.7;
                NCp=NONCB*40+NNH3*8.38+NH2O*18;
                V=V2;
                [t4,y4]=ode23s(@modell2,[0 3*60],[NONCB/V2 NNH3/V2 Tstart]);
                plot(app.UIAxes,t2,y2(:,3),t4,y4(:,3),'LineWidth',1.5)
                app.UIAxes.YLim=[150 min([max(y4(:,3)),450])];
                legend(app.UIAxes,'Normal','Higher production','best')
                toogleUIC(app.Safety1UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function dy = modell2(t,y)
                UA=35.83;Ta=25;DeltaH=-5.9e5;
                CONCB=y(1);CNH3=y(2);T=y(3);
                dy=zeros(size(y));
                TK=T+273;
                k=0.00017*exp(11273/1.987*(1/461-1/TK));
                r=k*CONCB*CNH3;
                dy(1)=-r;
                dy(2)=-2*r;
                Qg=-r*DeltaH*V;
                if (t>KylStop) && (t<KylStart) || (T < Tstart)
                    Qr=0;
                else
                    Qr=UA*(T-Ta);
                end
                if t>KylStop || Qg > Qr
                    dy(3)=(Qg-Qr)/NCp;
                end
            end
        end
    end


    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            app.StoppField.Value=sprintf('%5.1f',app.StoppSlider.Value);
```

```
        app.FixaField.Value=sprintf('%4.1f',app.FixaSlider.Value);
        draw(app);
    end

    % Value changed function: StoppField
    function StoppFieldValueChanged(app, event)
        value = app.StoppField.Value;
        if between(app,value,app.StoppSlider.Limits)
            app.StoppSlider.Value=str2double(value);
            draw(app);
        else
            app.StoppField.Value=num2str(app.StoppSlider.Value);
        end
    end

    % Value changed function: FixaField
    function FixaFieldValueChanged(app, event)
        value = app.FixaField.Value;
        if between(app,value,app.FixaSlider.Limits)
            app.FixaSlider.Value=str2double(value);
            draw(app);
        else
            app.FixaField.Value=num2str(app.FixaSlider.Value);
        end
    end

    % Value changed function: StoppSlider
    function StoppSliderValueChanged(app, event)
        value = app.StoppSlider.Value;
        app.StoppField.Value=sprintf('%5.1f',value);
        draw(app);
    end

    % Value changed function: FixaSlider
    function FixaSliderValueChanged(app, event)
        value = app.FixaSlider.Value;
        app.FixaField.Value=sprintf('%4.1f',value);
        draw(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Safety1UIFigure and hide until all components are created
        app.Safety1UIFigure = uifigure('Visible', 'off');
        app.Safety1UIFigure.Color = [0.94 0.94 0.94];
        app.Safety1UIFigure.Position = [100 100 688 552];
        app.Safety1UIFigure.Name = 'Safety 1';
        app.Safety1UIFigure.Scrollable = 'on';

        % Create UIAxes
        app.UIAxes = uiaxes(app.Safety1UIFigure);
        xlabel(app.UIAxes, 'Time [min]')
        ylabel(app.UIAxes, {'Temperature [^0C]'; ''})
        app.UIAxes.FontName = 'Helvetica';
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.Position = [33 26 623 298];

        % Create AccidentMonsantoSaugetLabel
        app.AccidentMonsantoSaugetLabel = uilabel(app.Safety1UIFigure);
        app.AccidentMonsantoSaugetLabel.FontSize = 24;
        app.AccidentMonsantoSaugetLabel.FontWeight = 'bold';
        app.AccidentMonsantoSaugetLabel.Position = [181 494 318 30];
        app.AccidentMonsantoSaugetLabel.Text = 'Accident Monsanto Sauget';

        % Create StopafterminLabel
```

```
            app.StopafterminLabel = uilabel(app.Safety1UIFigure);
            app.StopafterminLabel.HorizontalAlignment = 'right';
            app.StopafterminLabel.Position = [111 442 87 22];
            app.StopafterminLabel.Text = 'Stop after [min]';

            % Create StoppSlider
            app.StoppSlider = uislider(app.Safety1UIFigure);
            app.StoppSlider.Limits = [10 150];
            app.StoppSlider.MajorTicks = [10 25 50 75 100 125 150];
            app.StoppSlider.ValueChangedFcn = createCallbackFcn(app, @StoppSliderValueChanged, true);
            app.StoppSlider.MinorTicks = [];
            app.StoppSlider.Position = [226 451 212 3];
            app.StoppSlider.Value = 45;

            % Create LengthofstopminLabel
            app.LengthofstopminLabel = uilabel(app.Safety1UIFigure);
            app.LengthofstopminLabel.HorizontalAlignment = 'right';
            app.LengthofstopminLabel.Position = [108 376 111 22];
            app.LengthofstopminLabel.Text = 'Length of stop [min]';

            % Create FixaSlider
            app.FixaSlider = uislider(app.Safety1UIFigure);
            app.FixaSlider.Limits = [0 20];
            app.FixaSlider.MajorTicks = [0 5 10 15 20];
            app.FixaSlider.ValueChangedFcn = createCallbackFcn(app, @FixaSliderValueChanged, true);
            app.FixaSlider.MinorTicks = [];
            app.FixaSlider.Position = [226 386 212 3];
            app.FixaSlider.Value = 10;

            % Create StoppField
            app.StoppField = uieditfield(app.Safety1UIFigure, 'text');
            app.StoppField.ValueChangedFcn = createCallbackFcn(app, @StoppFieldValueChanged, true);
            app.StoppField.Position = [460 441 36 22];

            % Create FixaField
            app.FixaField = uieditfield(app.Safety1UIFigure, 'text');
            app.FixaField.ValueChangedFcn = createCallbackFcn(app, @FixaFieldValueChanged, true);
            app.FixaField.Position = [460 376 36 22];

            % Show the figure after all components are created
            app.Safety1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = safety1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Safety1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Safety1UIFigure)
        end
    end
```

```
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Safety 2

safety2.mlapp

```
classdef safety2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Safety2UIFigure               matlab.ui.Figure
        timeField                     matlab.ui.control.EditField
        TabGroup                      matlab.ui.container.TabGroup
        TemperatureTab                matlab.ui.container.Tab
        UIAxes                        matlab.ui.control.UIAxes
        PressureTab                   matlab.ui.container.Tab
        UIAxes_2                      matlab.ui.control.UIAxes
        ConcentrationTab              matlab.ui.container.Tab
        UIAxes_3                      matlab.ui.control.UIAxes
        HeatProfileTab                matlab.ui.container.Tab
        UIAxes_4                      matlab.ui.control.UIAxes
        timeSlider                    matlab.ui.control.Slider
        TimeuntilcoolingworksminLabel  matlab.ui.control.Label
        rubtrk                        matlab.ui.control.Label
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Safety2UIFigure,'off');
                tspan = [0 4.]; % Range for the independent variable time
                y0 = [4.3; 5.1; 3.; 4.4; 422.]; % Initial values for the dependent variables i.e CA,CB,CS,P,and T
                E1A=128000;
                E2S=800000;
                DHRx1A=-45400;
                DHRx2S = -320000;
                UA=2.77e6;
                pset=28;
                VH=5000;
                Cv1=3360;
                Cv2=53600;
                V0 = 4000;
                Ta=373.15;
                SumNCp = 12600000;
                A1A = 4e14;
                A2S = 1E+84;
                [t,y]=ode23s(@diffekv,tspan,y0);
                plot(app.UIAxes,t,y(:,5));
                plot(app.UIAxes_2,t,y(:,4));
                plot(app.UIAxes_3,t,y(:,1:3));
                leg=legend(app.UIAxes_3,'C_A','C_B','C_S');
                leg.ItemTokenSize=[10,10];
                N=size(y,1);
                Qg=zeros(N,1);Qr=zeros(N,1);
                for i=1:N
                    k1A = A1A * exp(0 - (E1A / (8.31 * y(i,5))));
                    k2S = A2S * exp(0 - (E2S / (8.31 * y(i,5))));
                    r1A = 0 - (k1A * y(i,1) * y(i,2));
                    r2S = 0 - (k2S * y(i,3));
                    Qg(i)=V0 * (r1A * DHRx1A + r2S * DHRx2S);
                    Qr(i)= UA * (y(i,5) - Ta);
                end
                plot(app.UIAxes_4,t,Qg/3600/1e6,t,Qr/3600/1e6);
                leg=legend(app.UIAxes_4,'Q_g','Q_r');
                leg.ItemTokenSize=[10,10];
                toogleUIC(app.Safety2UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end
            function dy = diffekv(t,y)
                dy=zeros(size(y));
```

```
            CA = y(1);
            CB = y(2);
            CS = y(3);
            P = y(4);
            T = y(5);
            % Explicit equations

            k1A = A1A * exp(0 - (E1A / (8.31 * T)));
            k2S = A2S * exp(0 - (E2S / (8.31 * T)));
            if (T > 600 || P > 55)
                SW1 = 0;
            else
                SW1 = 1;
            end
            r1A = 0 - (k1A * CA * CB);
            r2S = 0 - (k2S * CS);
            FD = (-0.5 * r1A - (3 * r2S)) * V0;
            Qg=V0 * (r1A * DHRx1A + r2S * DHRx2S);
            if t < app.timeSlider.Value
                Qr=0;
            else
                Qr= UA * (T - Ta);
            end
            if (FD < 11400)
                Fvent = FD;
            else
                if (P < pset)
                    Fvent = (P - 1) * Cv1;
                else
                    Fvent = (P - 1) * (Cv1 + Cv2);
                end
            end
            % Differential equations
            dy(1) = SW1 * r1A;
            dy(2) = SW1 * r1A;
            dy(3) = SW1 * r2S;
            dy(4) = SW1 * (FD - Fvent) * 0.082 * T / VH;
            dy(5) = SW1 * (Qg-Qr) / SumNCp;
        end

    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        app.timeField.Value=sprintf('%4.1f',app.timeSlider.Value);
        simulera(app);
    end

    % Value changed function: timeSlider
    function timeSliderValueChanged(app, event)
        value = app.timeSlider.Value;
        app.timeField.Value=sprintf('%4.1f',value);
        simulera(app);
    end

    % Value changed function: timeField
    function timeFieldValueChanged(app, event)
        value = app.timeField.Value;
        if between(app,value,app.timeSlider.Limits)
            app.timeSlider.Value=str2double(value);
            simulera(app);
        else
            app.timeField.Value=num2str(app.timeSlider.Value);
        end
    end
end
```

```
% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Safety2UIFigure and hide until all components are created
        app.Safety2UIFigure = uifigure('Visible', 'off');
        app.Safety2UIFigure.Color = [0.94 0.94 0.94];
        app.Safety2UIFigure.Position = [100 100 740 590];
        app.Safety2UIFigure.Name = 'Safety 2';

        % Create rubtrk
        app.rubtrk = uilabel(app.Safety2UIFigure);
        app.rubtrk.FontSize = 18;
        app.rubtrk.FontWeight = 'bold';
        app.rubtrk.Position = [272 545 255 22];
        app.rubtrk.Text = 'Explosion at T2 Laboratories';

        % Create TimeuntilcoolingworksminLabel
        app.TimeuntilcoolingworksminLabel = uilabel(app.Safety2UIFigure);
        app.TimeuntilcoolingworksminLabel.Position = [53 492 194 22];
        app.TimeuntilcoolingworksminLabel.Text = 'Time until cooling works [min]';

        % Create timeSlider
        app.timeSlider = uislider(app.Safety2UIFigure);
        app.timeSlider.Limits = [0 10];
        app.timeSlider.ValueChangedFcn = createCallbackFcn(app, @timeSliderValueChanged, true);
        app.timeSlider.MinorTicks = [];
        app.timeSlider.Position = [263 502 183 3];

        % Create TabGroup
        app.TabGroup = uitabgroup(app.Safety2UIFigure);
        app.TabGroup.Position = [24 30 693 361];

        % Create TemperatureTab
        app.TemperatureTab = uitab(app.TabGroup);
        app.TemperatureTab.Title = 'Temperature';

        % Create UIAxes
        app.UIAxes = uiaxes(app.TemperatureTab);
        xlabel(app.UIAxes, 'Time [h]')
        ylabel(app.UIAxes, 'Temperatur [K]')
        app.UIAxes.XLim = [0 4];
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.Position = [24 15 647 297];

        % Create PressureTab
        app.PressureTab = uitab(app.TabGroup);
        app.PressureTab.Title = 'Pressure';

        % Create UIAxes_2
        app.UIAxes_2 = uiaxes(app.PressureTab);
        xlabel(app.UIAxes_2, 'Time [h]')
        ylabel(app.UIAxes_2, 'Pressure [atm]')
        app.UIAxes_2.XLim = [0 4];
        app.UIAxes_2.XTickLabelRotation = 0;
        app.UIAxes_2.YTickLabelRotation = 0;
        app.UIAxes_2.ZTickLabelRotation = 0;
        app.UIAxes_2.Position = [24 15 647 297];

        % Create ConcentrationTab
        app.ConcentrationTab = uitab(app.TabGroup);
        app.ConcentrationTab.Title = 'Concentration';

        % Create UIAxes_3
        app.UIAxes_3 = uiaxes(app.ConcentrationTab);
        xlabel(app.UIAxes_3, 'Time [h]')
        ylabel(app.UIAxes_3, 'Concentration [mol/dm^3]')
```

```
            app.UIAxes_3.XLim = [0 4];
            app.UIAxes_3.XTickLabelRotation = 0;
            app.UIAxes_3.YTickLabelRotation = 0;
            app.UIAxes_3.ZTickLabelRotation = 0;
            app.UIAxes_3.Position = [24 15 647 297];

            % Create HeatProfileTab
            app.HeatProfileTab = uitab(app.TabGroup);
            app.HeatProfileTab.Title = 'Heat Profile';

            % Create UIAxes_4
            app.UIAxes_4 = uiaxes(app.HeatProfileTab);
            xlabel(app.UIAxes_4, 'Time [h]')
            ylabel(app.UIAxes_4, 'Q [MJ/s]')
            app.UIAxes_4.XLim = [0 4];
            app.UIAxes_4.XTickLabelRotation = 0;
            app.UIAxes_4.YTickLabelRotation = 0;
            app.UIAxes_4.ZTickLabelRotation = 0;
            app.UIAxes_4.Position = [24 15 647 297];

            % Create timeField
            app.timeField = uieditfield(app.Safety2UIFigure, 'text');
            app.timeField.ValueChangedFcn = createCallbackFcn(app, @timeFieldValueChanged, true);
            app.timeField.Position = [463 492 37 22];

            % Show the figure after all components are created
            app.Safety2UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = safety2

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Safety2UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Safety2UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)
arguments
    h matlab.ui.Figure
    state char
end
% toggle the state of the uicontrols in the app
uics = findall(h);
for idx = 1:length(uics)
    uic = uics(idx);
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')
        uic.Enable = state;
    end
end
drawnow; % flush the ui events queue
end
```

Semibatch reactor 1

semibatch1.mlapp

```
classdef semibatch1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Semibatchreactor1UIFigure  matlab.ui.Figure
        TabGroup                   matlab.ui.container.TabGroup
        BatchTab                   matlab.ui.container.Tab
        UIAxes_1                   matlab.ui.control.UIAxes
        SemibatchTab               matlab.ui.container.Tab
        UIAxes_2                   matlab.ui.control.UIAxes
        SelectivityTab             matlab.ui.container.Tab
        UIAxes_3                   matlab.ui.control.UIAxes
        TemperatureTab             matlab.ui.container.Tab
        UIAxes_4                   matlab.ui.control.UIAxes
        H2Slider                   matlab.ui.control.Slider
        DeltaH2SliderLabel         matlab.ui.control.Label
        H1Slider                   matlab.ui.control.Slider
        DeltaH1SliderLabel         matlab.ui.control.Label
        UASlider                   matlab.ui.control.Slider
        UASliderLabel              matlab.ui.control.Label
        ypsilonSlider              matlab.ui.control.Slider
        FeedrateSliderLabel        matlab.ui.control.Label
        feedDropDown               matlab.ui.control.DropDown
        rubrik                     matlab.ui.control.Label
    end


    methods (Access = private)

        function draw(app)
            try
                toogleUIC(app.Semibatchreactor1UIFigure,'off');
                feed=app.feedDropDown.Value;
                ypsilon=app.ypsilonSlider.Value;
                UA=app.UASlider.Value;
                H1=app.H1Slider.Value;
                H2=app.H2Slider.Value;
                Cp=[30 60 20 35]';
                Tx=340;
                V0=10;
                tEnd=max([2*V0/ypsilon 2]);
                [t1,y1]=ode23s(@batch,[0 tEnd],[100 100 0 0 2*V0 340]);
                if feed == '1'
                    [t2,y2]=ode23s(@semibatch,[0 tEnd],[0 100 0 0 V0 340],odeset('Events',@fylld));
                else
                    [t2,y2]=ode23s(@semibatch,[0 tEnd],[100 0 0 0 V0 340],odeset('Events',@fylld));
                end
                plot(app.UIAxes_1,t1,y1(:,1),'b',...
                    t1,y1(:,2),'r',...
                    t1,y1(:,3),'g',...
                    t1,y1(:,4),'m','LineWidth',1.5);
                app.UIAxes_1.YLabel.String='Moles';
                app.UIAxes_1.Title.String='Batch';
                legend(app.UIAxes_1,'N_A','N_B','N_D','N_U','location','Best')
                plot(app.UIAxes_2,t2,y2(:,1),'b',...
                    t2,y2(:,2),'r',...
                    t2,y2(:,3),'g',...
                    t2,y2(:,4),'m','LineWidth',1.5);
                app.UIAxes_2.YLabel.String='Moles';
                app.UIAxes_2.Title.String='Semibatch';
                legend(app.UIAxes_2,'N_A','N_B','N_D','N_U','location','Best')
                S1=y1(:,3)./y1(:,4);S2=y2(:,3)./y2(:,4);
                tid=linspace(0,tEnd,50);
                S=interp1(t2,S2,tid)./interp1(t1,S1,tid);
                plot(app.UIAxes_3,tid,S,'k','LineWidth',1.5);
                legend(app.UIAxes_3,'off');
                app.UIAxes_3.YLabel.String='Selectivity ratio';
```

```
            app.UIAxes_3.Title.String='Semibatch S_{D/U} / Batch S_{D/U}';
            plot(app.UIAxes_4,t1,y1(:,6),'b',t2,y2(:,6),'r','LineWidth',1.5);
            app.UIAxes_4.YLabel.String='Temperature [K]';
            app.UIAxes_4.Title.String='';
            legend(app.UIAxes_4,'Batch','Semibatch','Location','Best')
            toogleUIC(app.Semibatchreactor1UIFigure,'on');
        catch ME
            errordlg(ME.message);
        end
        function dy = batch(~,y)
            dy=zeros(size(y));
            V=y(5);
            C=y(1:4)./V;
            T=y(6);
            k1=0.5*exp(6500/1.987*(1/320-1/T));
            k2=0.32*exp(7000/1.987*(1/300-1/T));
            r1=k1*C(1)*C(2)^2;
            r2=k2*C(1)*C(2);
            dy(1)=-V*(r1+r2);
            dy(2)=-V*(r1+r2);
            dy(3)=V*r1;
            dy(4)=V*r2;
            dy(6)=(UA*(Tx-T)+(r1*H1+r2*H2)*V)/sum(Cp.*y(1:4));
        end
        function dy = semibatch(~,y)
            dy=zeros(size(y));
            V=y(5);
            C=y(1:4)./V;
            T=y(6);
            Cf=10;
            Tf=340;
            k1=0.5*exp(6500/1.987*(1/320-1/T));
            k2=0.32*exp(7000/1.987*(1/300-1/T));
            r1=k1*C(1)*C(2)^2;
            r2=k2*C(1)*C(2);
            if V > 2*V0
                q=0;
            else
                q=ypsilon;
            end
            if feed == '1'
                Cpf=Cp(1);
                dy(1)=q*Cf-V*(r1+r2);
                dy(2)=-V*(r1+r2);
            else
                Cpf=Cp(2);
                dy(1)=-V*(r1+r2);
                dy(2)=q*Cf-V*(r1+r2);
            end
            dy(3)=V*r1;
            dy(4)=V*r2;
            dy(5)=q;
            dy(6)=(UA*(Tx-T)+q*Cf*Cpf*(Tf-T)+(r1*H1+r2*H2)*V)/sum(Cp.*y(1:4));
        end
        function [value,isterminal,direction]=fylld(~,y)
            value=y(5)-2*V0;
            isterminal=0;
            direction=0;
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        draw(app);
    end

    % Value changed function: H1Slider, H2Slider, UASlider,
```

```
        % ...and 2 other components
        function plotDropDownValueChanged(app, event)
            draw(app);
        end
    end

    % Component initialization
    methods (Access = private)

        % Create UIFigure and components
        function createComponents(app)

            % Create Semibatchreactor1UIFigure and hide until all components are created
            app.Semibatchreactor1UIFigure = uifigure('Visible', 'off');
            app.Semibatchreactor1UIFigure.Color = [0.94 0.94 0.94];
            app.Semibatchreactor1UIFigure.Position = [100 100 740 609];
            app.Semibatchreactor1UIFigure.Name = 'Semibatch reactor 1';
            app.Semibatchreactor1UIFigure.Scrollable = 'on';

            % Create rubrik
            app.rubrik = uilabel(app.Semibatchreactor1UIFigure);
            app.rubrik.FontSize = 20;
            app.rubrik.FontWeight = 'bold';
            app.rubrik.Position = [46 558 661 24];
            app.rubrik.Text = 'Parallel Nonisothermal Reactions in Batch and Semibatch Reactors';

            % Create feedDropDown
            app.feedDropDown = uidropdown(app.Semibatchreactor1UIFigure);
            app.feedDropDown.Items = {'Feed A to B', 'Feed B to A'};
            app.feedDropDown.ItemsData = {'1', '2', ''};
            app.feedDropDown.ValueChangedFcn = createCallbackFcn(app, @plotDropDownValueChanged, true);
            app.feedDropDown.Position = [331 520 100 22];
            app.feedDropDown.Value = '1';

            % Create FeedrateSliderLabel
            app.FeedrateSliderLabel = uilabel(app.Semibatchreactor1UIFigure);
            app.FeedrateSliderLabel.HorizontalAlignment = 'right';
            app.FeedrateSliderLabel.Position = [54 480 57 22];
            app.FeedrateSliderLabel.Text = 'Feed rate';

            % Create ypsilonSlider
            app.ypsilonSlider = uislider(app.Semibatchreactor1UIFigure);
            app.ypsilonSlider.Limits = [2 20];
            app.ypsilonSlider.MajorTicks = [0.2 2 4 6 8 10 12 14 16 18 20];
            app.ypsilonSlider.ValueChangedFcn = createCallbackFcn(app, @plotDropDownValueChanged, true);
            app.ypsilonSlider.MinorTicks = [];
            app.ypsilonSlider.Position = [132 489 150 3];
            app.ypsilonSlider.Value = 10;

            % Create UASliderLabel
            app.UASliderLabel = uilabel(app.Semibatchreactor1UIFigure);
            app.UASliderLabel.HorizontalAlignment = 'right';
            app.UASliderLabel.Position = [378 480 25 22];
            app.UASliderLabel.Text = {'UA'; ''};

            % Create UASlider
            app.UASlider = uislider(app.Semibatchreactor1UIFigure);
            app.UASlider.Limits = [0 60000];
            app.UASlider.MajorTicks = [0 10000 20000 30000 40000 50000 60000];
            app.UASlider.MajorTickLabels = {'0', '10000', '20000', '30000', '40000', '50000', '60000', ''};
            app.UASlider.ValueChangedFcn = createCallbackFcn(app, @plotDropDownValueChanged, true);
            app.UASlider.MinorTicks = [];
            app.UASlider.Position = [424 489 275 3];
            app.UASlider.Value = 10000;

            % Create DeltaH1SliderLabel
            app.DeltaH1SliderLabel = uilabel(app.Semibatchreactor1UIFigure);
            app.DeltaH1SliderLabel.HorizontalAlignment = 'right';
            app.DeltaH1SliderLabel.Position = [19 418 52 22];
            app.DeltaH1SliderLabel.Text = 'Delta H1';

            % Create H1Slider
```

```
app.H1Slider = uislider(app.Semibatchreactor1UIFigure);
app.H1Slider.Limits = [0 20000];
app.H1Slider.MajorTicks = [0 4000 8000 12000 16000 20000];
app.H1Slider.MajorTickLabels = {'0', '4000', '8000', '12000', '16000', '20000'};
app.H1Slider.ValueChangedFcn = createCallbackFcn(app, @plotDropDownValueChanged, true);
app.H1Slider.MinorTicks = [];
app.H1Slider.Position = [92 427 246 3];
app.H1Slider.Value = 4000;

% Create DeltaH2SliderLabel
app.DeltaH2SliderLabel = uilabel(app.Semibatchreactor1UIFigure);
app.DeltaH2SliderLabel.HorizontalAlignment = 'right';
app.DeltaH2SliderLabel.Position = [375 418 52 22];
app.DeltaH2SliderLabel.Text = 'Delta H2';

% Create H2Slider
app.H2Slider = uislider(app.Semibatchreactor1UIFigure);
app.H2Slider.Limits = [0 20000];
app.H2Slider.MajorTicks = [0 4000 8000 12000 16000 20000];
app.H2Slider.MajorTickLabels = {'0', '4000', '8000', '12000', '16000', '20000'};
app.H2Slider.ValueChangedFcn = createCallbackFcn(app, @plotDropDownValueChanged, true);
app.H2Slider.MinorTicks = [];
app.H2Slider.Position = [448 427 251 3];
app.H2Slider.Value = 4000;

% Create TabGroup
app.TabGroup = uitabgroup(app.Semibatchreactor1UIFigure);
app.TabGroup.Position = [30 16 686 357];

% Create BatchTab
app.BatchTab = uitab(app.TabGroup);
app.BatchTab.Title = 'Batch';

% Create UIAxes_1
app.UIAxes_1 = uiaxes(app.BatchTab);
xlabel(app.UIAxes_1, 'Time [h]')
app.UIAxes_1.XTickLabelRotation = 0;
app.UIAxes_1.YTickLabelRotation = 0;
app.UIAxes_1.ZTickLabelRotation = 0;
app.UIAxes_1.XGrid = 'on';
app.UIAxes_1.YGrid = 'on';
app.UIAxes_1.Position = [98 11 545 311];

% Create SemibatchTab
app.SemibatchTab = uitab(app.TabGroup);
app.SemibatchTab.Title = 'Semibatch';

% Create UIAxes_2
app.UIAxes_2 = uiaxes(app.SemibatchTab);
xlabel(app.UIAxes_2, 'Time [h]')
app.UIAxes_2.XTickLabelRotation = 0;
app.UIAxes_2.YTickLabelRotation = 0;
app.UIAxes_2.ZTickLabelRotation = 0;
app.UIAxes_2.XGrid = 'on';
app.UIAxes_2.YGrid = 'on';
app.UIAxes_2.Position = [98 11 545 311];

% Create SelectivityTab
app.SelectivityTab = uitab(app.TabGroup);
app.SelectivityTab.Title = 'Selectivity';

% Create UIAxes_3
app.UIAxes_3 = uiaxes(app.SelectivityTab);
xlabel(app.UIAxes_3, 'Time [h]')
app.UIAxes_3.XTickLabelRotation = 0;
app.UIAxes_3.YTickLabelRotation = 0;
app.UIAxes_3.ZTickLabelRotation = 0;
app.UIAxes_3.XGrid = 'on';
app.UIAxes_3.YGrid = 'on';
app.UIAxes_3.Position = [98 11 545 311];

% Create TemperatureTab
```

```
            app.TemperatureTab = uitab(app.TabGroup);
            app.TemperatureTab.Title = 'Temperature';

            % Create UIAxes_4
            app.UIAxes_4 = uiaxes(app.TemperatureTab);
            xlabel(app.UIAxes_4, 'Time [h]')
            app.UIAxes_4.XTickLabelRotation = 0;
            app.UIAxes_4.YTickLabelRotation = 0;
            app.UIAxes_4.ZTickLabelRotation = 0;
            app.UIAxes_4.XGrid = 'on';
            app.UIAxes_4.YGrid = 'on';
            app.UIAxes_4.Position = [98 11 545 311];

            % Show the figure after all components are created
            app.Semibatchreactor1UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = semibatch1

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.Semibatchreactor1UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.Semibatchreactor1UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);     ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```

Semibatch reactor 2

semibatch2.mlapp

```
classdef semibatch2 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        Semibatchreactor2UIFigure      matlab.ui.Figure
        T0                             matlab.ui.control.Slider
        CoolanttemperatureSliderLabel  matlab.ui.control.Label
        UA0                            matlab.ui.control.Slider
        initialcoolingcapacitySliderLabel  matlab.ui.control.Label
        DeltaH                         matlab.ui.control.Slider
        HeatofreactionSliderLabel      matlab.ui.control.Label
        Ea                             matlab.ui.control.Slider
        ActivationenergySliderLabel    matlab.ui.control.Label
        F0                             matlab.ui.control.Slider
        molarfeedrateSliderLabel       matlab.ui.control.Label
        TabGroup                       matlab.ui.container.TabGroup
        ReactantsTab                   matlab.ui.container.Tab
        UIAxes2                        matlab.ui.control.UIAxes
        TemperatureTab                 matlab.ui.container.Tab
        UIAxes                         matlab.ui.control.UIAxes
        rubrik                         matlab.ui.control.Label
    end


    methods (Access = private)

        function simulera(app)
            try
                toogleUIC(app.Semibatchreactor2UIFigure,'off');
                Tc=app.T0.Value;
                Fin=app.F0.Value;
                Tfeed=Tc;
                deltaH=app.DeltaH.Value;
                V0=100;
                Vfinal=2*V0;
                Cpa=60;
                Cpb=50;
                Cp=55;
                Cfeed=1;
                NB0=100;
                rho=1;
                rhoa=1;
                rhob=1;
                Rh=(rhoa*Cpa)/(rhob*Cpb);
                upsilon=Fin/Cfeed;
                tdos=(Vfinal-V0)/upsilon;
                eps=upsilon*tdos/V0;
                Wt=app.UA0.Value/upsilon/rhob/Cpb;
                %Wt=app.UA0.Value*tdos/eps/rhob/Cpb/V0
                tid=[linspace(0,tdos,50) linspace(tdos,2*tdos,50)];
                theta=[linspace(0,tdos,50)/tdos ones(1,50)];
                Tad=-deltaH*NB0/Cp/rho/V0;
                Tta=Tc+(1.05*Tad)./(eps*(Wt*(1+eps*theta)+Rh));
                [t,y]=ode15s(@diffekv,[0 2*tdos],[0 NB0 0 0 Tc V0]);
                plot(app.UIAxes2,t,y(:,1:3))
                legend(app.UIAxes2,'N_A','N_B','N_C & N_D',"Location","best")
                conv=(NB0-y(end,2))/NB0;
                app.UIAxes2.Title.String=sprintf("Conversion %5.2f%%",conv*100);
                plot(app.UIAxes,t,y(:,5),tid,Tta,'r--');
                legend(app.UIAxes,'Reactor','Target',"Location","best");
                app.UIAxes.Title.String=sprintf("Conversion %5.2f%%",conv*100);
                toogleUIC(app.Semibatchreactor2UIFigure,'on');
            catch ME
                errordlg(ME.message);
            end

            function dy = diffekv(~,y)
```

```
            dy=zeros(size(y));
            NA=y(1);
            NB=y(2);
            %NC=y(3);
            %ND=y(4);
            T=y(5);
            V=y(6);
            R=1.987;
            k=1.24e6*exp(-app.Ea.Value/R/T);
            rate=k*NA*NB/V/V;
            if V > 2*V0
                qf=0;
            else
                qf=upsilon;
            end
            dy(1)=qf*Cfeed-rate*V;
            dy(2)=-rate*V;
            dy(3)=rate*V;
            dy(4)=rate*V;
            UA=app.UA0.Value*V/V0;
            dy(5)=(UA*(Tc-T)+qf*Cpa*(Tfeed-T)-+rate*deltaH*V)/sum(y(1:4)*Cp);
            dy(6)=qf;
        end
    end
end


% Callbacks that handle component events
methods (Access = private)

    % Code that executes after component creation
    function startupFcn(app)
        simulera(app);
    end

    % Value changed function: F0
    function F0ValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: Ea
    function EaValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: DeltaH
    function DeltaHValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: UA0
    function UA0ValueChanged(app, event)
        simulera(app);
    end

    % Value changed function: T0
    function T0ValueChanged(app, event)
        simulera(app);
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create Semibatchreactor2UIFigure and hide until all components are created
        app.Semibatchreactor2UIFigure = uifigure('Visible', 'off');
        app.Semibatchreactor2UIFigure.Color = [0.94 0.94 0.94];
        app.Semibatchreactor2UIFigure.Position = [100 100 740 723];
```

```
app.Semibatchreactor2UIFigure.Name = 'Semibatch reactor 2';

% Create rubrik
app.rubrik = uilabel(app.Semibatchreactor2UIFigure);
app.rubrik.FontSize = 18;
app.rubrik.FontWeight = 'bold';
app.rubrik.Position = [217 680 344 22];
app.rubrik.Text = 'Safe Operation of a Semibatch Reactor';

% Create TabGroup
app.TabGroup = uitabgroup(app.Semibatchreactor2UIFigure);
app.TabGroup.Position = [33 52 677 398];

% Create ReactantsTab
app.ReactantsTab = uitab(app.TabGroup);
app.ReactantsTab.Title = 'Reactants';

% Create UIAxes2
app.UIAxes2 = uiaxes(app.ReactantsTab);
xlabel(app.UIAxes2, 'Time')
ylabel(app.UIAxes2, 'Moles')
app.UIAxes2.PlotBoxAspectRatio = [2.05714285714286 1 1];
app.UIAxes2.FontWeight = 'bold';
app.UIAxes2.XTickLabelRotation = 0;
app.UIAxes2.YTickLabelRotation = 0;
app.UIAxes2.ZTickLabelRotation = 0;
app.UIAxes2.Position = [13 22 623 335];

% Create TemperatureTab
app.TemperatureTab = uitab(app.TabGroup);
app.TemperatureTab.Title = 'Temperature';

% Create UIAxes
app.UIAxes = uiaxes(app.TemperatureTab);
xlabel(app.UIAxes, 'Time')
ylabel(app.UIAxes, 'Temperature')
app.UIAxes.PlotBoxAspectRatio = [1.87043189368771 1 1];
app.UIAxes.FontWeight = 'bold';
app.UIAxes.XTickLabelRotation = 0;
app.UIAxes.YTickLabelRotation = 0;
app.UIAxes.ZTickLabelRotation = 0;
app.UIAxes.Position = [13 0 610 356];

% Create molarfeedrateSliderLabel
app.molarfeedrateSliderLabel = uilabel(app.Semibatchreactor2UIFigure);
app.molarfeedrateSliderLabel.HorizontalAlignment = 'right';
app.molarfeedrateSliderLabel.Position = [65 625 86 22];
app.molarfeedrateSliderLabel.Text = 'molar feed rate';

% Create F0
app.F0 = uislider(app.Semibatchreactor2UIFigure);
app.F0.Limits = [20 200];
app.F0.ValueChangedFcn = createCallbackFcn(app, @F0ValueChanged, true);
app.F0.MinorTicks = [];
app.F0.Position = [172 634 150 3];
app.F0.Value = 50;

% Create ActivationenergySliderLabel
app.ActivationenergySliderLabel = uilabel(app.Semibatchreactor2UIFigure);
app.ActivationenergySliderLabel.HorizontalAlignment = 'right';
app.ActivationenergySliderLabel.Position = [433 625 98 22];
app.ActivationenergySliderLabel.Text = 'Activation energy';

% Create Ea
app.Ea = uislider(app.Semibatchreactor2UIFigure);
app.Ea.Limits = [6000 10000];
app.Ea.ValueChangedFcn = createCallbackFcn(app, @EaValueChanged, true);
app.Ea.MinorTicks = [];
app.Ea.Position = [552 634 151 3];
app.Ea.Value = 8000;

% Create HeatofreactionSliderLabel
```

```
        app.HeatofreactionSliderLabel = uilabel(app.Semibatchreactor2UIFigure);
        app.HeatofreactionSliderLabel.HorizontalAlignment = 'right';
        app.HeatofreactionSliderLabel.Position = [440 560 90 22];
        app.HeatofreactionSliderLabel.Text = 'Heat of reaction';

        % Create DeltaH
        app.DeltaH = uislider(app.Semibatchreactor2UIFigure);
        app.DeltaH.Limits = [-80000 -10000];
        app.DeltaH.ValueChangedFcn = createCallbackFcn(app, @DeltaHValueChanged, true);
        app.DeltaH.MinorTicks = [];
        app.DeltaH.Position = [551 569 150 3];
        app.DeltaH.Value = -20000;

        % Create initialcoolingcapacitySliderLabel
        app.initialcoolingcapacitySliderLabel = uilabel(app.Semibatchreactor2UIFigure);
        app.initialcoolingcapacitySliderLabel.HorizontalAlignment = 'right';
        app.initialcoolingcapacitySliderLabel.Position = [49 508 122 22];
        app.initialcoolingcapacitySliderLabel.Text = 'initial cooling capacity';

        % Create UA0
        app.UA0 = uislider(app.Semibatchreactor2UIFigure);
        app.UA0.Limits = [5000 50000];
        app.UA0.ValueChangedFcn = createCallbackFcn(app, @UA0ValueChanged, true);
        app.UA0.MinorTicks = [];
        app.UA0.Position = [192 517 150 3];
        app.UA0.Value = 30000;

        % Create CoolanttemperatureSliderLabel
        app.CoolanttemperatureSliderLabel = uilabel(app.Semibatchreactor2UIFigure);
        app.CoolanttemperatureSliderLabel.HorizontalAlignment = 'right';
        app.CoolanttemperatureSliderLabel.Position = [46 568 115 22];
        app.CoolanttemperatureSliderLabel.Text = 'Coolant temperature';

        % Create T0
        app.T0 = uislider(app.Semibatchreactor2UIFigure);
        app.T0.Limits = [275 370];
        app.T0.MajorTicks = [275 300 325 350 370];
        app.T0.ValueChangedFcn = createCallbackFcn(app, @T0ValueChanged, true);
        app.T0.MinorTicks = [];
        app.T0.Position = [182 577 150 3];
        app.T0.Value = 365;

        % Show the figure after all components are created
        app.Semibatchreactor2UIFigure.Visible = 'on';
    end
end

% App creation and deletion
methods (Access = public)

    % Construct app
    function app = semibatch2

        % Create UIFigure and components
        createComponents(app)

        % Register the app with App Designer
        registerApp(app, app.Semibatchreactor2UIFigure)

        % Execute the startup function
        runStartupFcn(app, @startupFcn)

        if nargout == 0
            clear app
        end
    end

    % Code that executes before app deletion
    function delete(app)

        % Delete UIFigure when app is deleted
```

```
            delete(app.Semibatchreactor2UIFigure)
        end
    end
end
```

toogleUIC.m

```
function toogleUIC(h, state)
arguments
    h matlab.ui.Figure
    state char
end
% toggle the state of the uicontrols in the app
uics = findall(h);
for idx = 1:length(uics)
    uic = uics(idx);
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')
        uic.Enable = state;
    end
end
drawnow; % flush the ui events queue
end
```

Semibatch reactor 3

semibatch3.mlapp

```
classdef semibatch3 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        UIFigure                  matlab.ui.Figure
        HeatExchangerPanel        matlab.ui.container.Panel
        MassFlowField             matlab.ui.control.EditField
        MassFlowSlider            matlab.ui.control.Slider
        MassFlowSliderLabel       matlab.ui.control.Label
        InletflowPanel            matlab.ui.container.Panel
        RateField                 matlab.ui.control.EditField
        RateSlider                matlab.ui.control.Slider
        RateSliderLabel           matlab.ui.control.Label
        TemperatureField          matlab.ui.control.EditField
        TemperatureSlider         matlab.ui.control.Slider
        TemperatureSliderLabel    matlab.ui.control.Label
        ConcentrationField        matlab.ui.control.EditField
        ConcentrationSlider       matlab.ui.control.Slider
        ConcentrationSliderLabel  matlab.ui.control.Label
        rubrik                    matlab.ui.control.Label
        TabGroup                  matlab.ui.container.TabGroup
        TemperatureTab            matlab.ui.container.Tab
        UIAxes                    matlab.ui.control.UIAxes
        ConcentrationTab          matlab.ui.container.Tab
        UIAxes_2                  matlab.ui.control.UIAxes
        ConversionATab            matlab.ui.container.Tab
        UIAxes_3                  matlab.ui.control.UIAxes
    end


    methods (Access = private)

        function simulera(app)
            toogleUIC(app.UIFigure,'off');
            [t,y]=ode45(@diffekv,[0 350],[5*0.2 0 0 0 30.7*0.2 0.2 300]);
            plot(app.UIAxes_2,t,y(:,1:3)./y(:,6));
            legend(app.UIAxes_2,'C_a','C_b','C_c & C_d','location','best')
            plot(app.UIAxes,t,y(:,7),[0 350],[315 315],'r--');
            CA=y(:,1)./y(:,6);
            plot(app.UIAxes_3,t,(5-CA)/5)
            toogleUIC(app.UIFigure,'on');
            function dy = diffekv(~,y)
                dy=zeros(size(y));
                T=y(7);
                N=y(1:5);
                V=y(6);
                C=N./V;
                upsilon=app.RateSlider.Value/1000;
                CBin=app.ConcentrationSlider.Value;
                T0=app.TemperatureSlider.Value;
                mc=app.MassFlowSlider.Value;
                k=0.39175*exp(5472.7*(1/273-1/T));
                Kc=10.^(3885.44/T);
                r=k*(C(1)*C(2)-C(3)*C(4)/Kc);
                dy(1)=-r;
                dy(2)=-r+upsilon*CBin;
                dy(3)=r;
                dy(4)=r;
                dy(5)=upsilon*55;
                dy(6)=upsilon;
                DeltaH=-79076e3;
                Qg=-r*V*DeltaH;
                Qr1=upsilon*(CBin+55)*75240*(T-T0);
                UA=3000;
                Qr2=mc*(T-285)*(1-exp(-UA/mc/18));
                dy(7)=(Qg-Qr1-Qr2)./(sum(y(2:5))*75240+y(1)*170700);
            end
```

```
        end
    end

    % Callbacks that handle component events
    methods (Access = private)

        % Code that executes after component creation
        function startupFcn(app)
            app.ConcentrationField.Value=sprintf('%3.1f',app.ConcentrationSlider.Value);
            app.TemperatureField.Value=sprintf('%4.0f',app.TemperatureSlider.Value);
            app.MassFlowField.Value=sprintf('%4.0f',app.MassFlowSlider.Value);
            app.RateField.Value=sprintf('%3.1f',app.RateSlider.Value);
            simulera(app);
        end

        % Value changed function: ConcentrationSlider
        function ConcentrationSliderValueChanged(app, event)
            value = app.ConcentrationSlider.Value;
            app.ConcentrationField.Value=sprintf('%3.1f',value);
            simulera(app);
        end

        % Value changed function: TemperatureSlider
        function TemperatureSliderValueChanged(app, event)
            value = app.TemperatureSlider.Value;
            app.TemperatureField.Value=sprintf('%4.0f',value);
            simulera(app);
        end

        % Value changed function: RateSlider
        function RateSliderValueChanged(app, event)
            value = app.RateSlider.Value;
            app.RateField.Value=sprintf('%3.1f',value);
            simulera(app);
        end

        % Value changed function: MassFlowSlider
        function MassFlowSliderValueChanged(app, event)
            value = app.MassFlowSlider.Value;
            app.MassFlowField.Value=sprintf('%4.0f',value);
            simulera(app);
        end

        % Value changed function: ConcentrationField
        function ConcentrationFieldValueChanged(app, event)
            value = app.ConcentrationField.Value;
            if between(app,value,app.ConcentrationSlider.Limits)
                app.ConcentrationSlider.Value=str2double(value);
                simulera(app);
            else
                app.ConcentrationField.Value=num2str(app.ConcentrationSlider.Value);
            end
        end

        % Value changed function: TemperatureField
        function TemperatureFieldValueChanged(app, event)
            value = app.TemperatureField.Value;
            if between(app,value,app.TemperatureSlider.Limits)
                app.TemperatureSlider.Value=str2double(value);
                simulera(app);
            else
                app.TemperatureField.Value=num2str(app.TemperatureSlider.Value);
            end
        end

        % Value changed function: RateField
        function RateFieldValueChanged(app, event)
            value = app.RateField.Value;
            if between(app,value,app.RateSlider.Limits)
                app.RateSlider.Value=str2double(value);
```

```
            simulera(app);
        else
            app.RateField.Value=num2str(app.RateSlider.Value);
        end
    end

    % Value changed function: MassFlowField
    function MassFlowFieldValueChanged(app, event)
        value = app.MassFlowField.Value;
        if between(app,value,app.MassFlowSlider.Limits)
            app.MassFlowSlider.Value=str2double(value);
            simulera(app);
        else
            app.MassFlowField.Value=num2str(app.MassFlowSlider.Value);
        end
    end
end

% Component initialization
methods (Access = private)

    % Create UIFigure and components
    function createComponents(app)

        % Create UIFigure and hide until all components are created
        app.UIFigure = uifigure('Visible', 'off');
        app.UIFigure.Color = [0.94 0.94 0.94];
        app.UIFigure.Position = [100 100 740 640];
        app.UIFigure.Name = 'MATLAB App';

        % Create TabGroup
        app.TabGroup = uitabgroup(app.UIFigure);
        app.TabGroup.Position = [35 17 676 373];

        % Create TemperatureTab
        app.TemperatureTab = uitab(app.TabGroup);
        app.TemperatureTab.Title = 'Temperature';

        % Create UIAxes
        app.UIAxes = uiaxes(app.TemperatureTab);
        xlabel(app.UIAxes, 'Time [s]')
        ylabel(app.UIAxes, 'Temperature [K]')
        app.UIAxes.XLim = [0 350];
        app.UIAxes.YLim = [300 400];
        app.UIAxes.XTickLabelRotation = 0;
        app.UIAxes.YTickLabelRotation = 0;
        app.UIAxes.ZTickLabelRotation = 0;
        app.UIAxes.XGrid = 'on';
        app.UIAxes.YGrid = 'on';
        app.UIAxes.Position = [23 18 606 312];

        % Create ConcentrationTab
        app.ConcentrationTab = uitab(app.TabGroup);
        app.ConcentrationTab.Title = 'Concentration';

        % Create UIAxes_2
        app.UIAxes_2 = uiaxes(app.ConcentrationTab);
        xlabel(app.UIAxes_2, 'Time [s]')
        ylabel(app.UIAxes_2, 'Concentration [mol/L]')
        app.UIAxes_2.XLim = [0 350];
        app.UIAxes_2.YLim = [0 5];
        app.UIAxes_2.XTickLabelRotation = 0;
        app.UIAxes_2.YTickLabelRotation = 0;
        app.UIAxes_2.ZTickLabelRotation = 0;
        app.UIAxes_2.XGrid = 'on';
        app.UIAxes_2.YGrid = 'on';
        app.UIAxes_2.Position = [23 18 606 312];

        % Create ConversionATab
        app.ConversionATab = uitab(app.TabGroup);
        app.ConversionATab.Title = 'Conversion A';
```

```
% Create UIAxes_3
app.UIAxes_3 = uiaxes(app.ConversionATab);
xlabel(app.UIAxes_3, 'Time [s]')
ylabel(app.UIAxes_3, 'Conversion')
app.UIAxes_3.XLim = [0 350];
app.UIAxes_3.YLim = [0 1];
app.UIAxes_3.XTickLabelRotation = 0;
app.UIAxes_3.YTickLabelRotation = 0;
app.UIAxes_3.ZTickLabelRotation = 0;
app.UIAxes_3.XGrid = 'on';
app.UIAxes_3.YGrid = 'on';
app.UIAxes_3.Position = [23 18 606 312];

% Create rubrik
app.rubrik = uilabel(app.UIFigure);
app.rubrik.FontSize = 18;
app.rubrik.FontWeight = 'bold';
app.rubrik.Position = [231 604 318 22];
app.rubrik.Text = 'Heat Effects in a Semibatch Recator';

% Create InletflowPanel
app.InletflowPanel = uipanel(app.UIFigure);
app.InletflowPanel.Title = 'Inlet flow';
app.InletflowPanel.Position = [32 405 335 189];

% Create ConcentrationSliderLabel
app.ConcentrationSliderLabel = uilabel(app.InletflowPanel);
app.ConcentrationSliderLabel.HorizontalAlignment = 'right';
app.ConcentrationSliderLabel.Position = [11 137 80 22];
app.ConcentrationSliderLabel.Text = 'Concentration';

% Create ConcentrationSlider
app.ConcentrationSlider = uislider(app.InletflowPanel);
app.ConcentrationSlider.Limits = [0.5 5];
app.ConcentrationSlider.MajorTicks = [0.5 1.5 2.5 3.5 4.5 5];
app.ConcentrationSlider.ValueChangedFcn = createCallbackFcn(app, @ConcentrationSliderValueChanged, true);
app.ConcentrationSlider.MinorTicks = [1 2 3 4];
app.ConcentrationSlider.Position = [112 146 150 3];
app.ConcentrationSlider.Value = 1;

% Create ConcentrationField
app.ConcentrationField = uieditfield(app.InletflowPanel, 'text');
app.ConcentrationField.ValueChangedFcn = createCallbackFcn(app, @ConcentrationFieldValueChanged, true);
app.ConcentrationField.Position = [277 135 41 22];

% Create TemperatureSliderLabel
app.TemperatureSliderLabel = uilabel(app.InletflowPanel);
app.TemperatureSliderLabel.HorizontalAlignment = 'right';
app.TemperatureSliderLabel.Position = [14 85 73 22];
app.TemperatureSliderLabel.Text = 'Temperature';

% Create TemperatureSlider
app.TemperatureSlider = uislider(app.InletflowPanel);
app.TemperatureSlider.Limits = [280 370];
app.TemperatureSlider.MajorTicks = [280 300 320 340 370];
app.TemperatureSlider.ValueChangedFcn = createCallbackFcn(app, @TemperatureSliderValueChanged, true);
app.TemperatureSlider.MinorTicks = [290 310 330 350 360];
app.TemperatureSlider.Position = [108 94 150 3];
app.TemperatureSlider.Value = 310;

% Create TemperatureField
app.TemperatureField = uieditfield(app.InletflowPanel, 'text');
app.TemperatureField.ValueChangedFcn = createCallbackFcn(app, @TemperatureFieldValueChanged, true);
app.TemperatureField.Position = [277 83 41 22];

% Create RateSliderLabel
app.RateSliderLabel = uilabel(app.InletflowPanel);
app.RateSliderLabel.HorizontalAlignment = 'right';
app.RateSliderLabel.Position = [53 31 31 22];
app.RateSliderLabel.Text = 'Rate';

% Create RateSlider
```

```
            app.RateSlider = uislider(app.InletflowPanel);
            app.RateSlider.Limits = [1 8];
            app.RateSlider.MajorTicks = [1 2 3 4 5 6 7 8];
            app.RateSlider.ValueChangedFcn = createCallbackFcn(app, @RateSliderValueChanged, true);
            app.RateSlider.MinorTicks = [300 325 375 425 475];
            app.RateSlider.Position = [105 40 150 3];
            app.RateSlider.Value = 4;

            % Create RateField
            app.RateField = uieditfield(app.InletflowPanel, 'text');
            app.RateField.ValueChangedFcn = createCallbackFcn(app, @RateFieldValueChanged, true);
            app.RateField.Position = [277 30 41 22];

            % Create HeatExchangerPanel
            app.HeatExchangerPanel = uipanel(app.UIFigure);
            app.HeatExchangerPanel.Title = 'Heat Exchanger';
            app.HeatExchangerPanel.Position = [387 405 324 189];

            % Create MassFlowSliderLabel
            app.MassFlowSliderLabel = uilabel(app.HeatExchangerPanel);
            app.MassFlowSliderLabel.HorizontalAlignment = 'right';
            app.MassFlowSliderLabel.Position = [19 136 63 22];
            app.MassFlowSliderLabel.Text = 'Mass Flow';

            % Create MassFlowSlider
            app.MassFlowSlider = uislider(app.HeatExchangerPanel);
            app.MassFlowSlider.Limits = [50 500];
            app.MassFlowSlider.ValueChangedFcn = createCallbackFcn(app, @MassFlowSliderValueChanged, true);
            app.MassFlowSlider.MinorTicks = [75 100 150 175 225 250 300 325 375 400 450 475];
            app.MassFlowSlider.Position = [103 145 150 3];
            app.MassFlowSlider.Value = 100;

            % Create MassFlowField
            app.MassFlowField = uieditfield(app.HeatExchangerPanel, 'text');
            app.MassFlowField.ValueChangedFcn = createCallbackFcn(app, @MassFlowFieldValueChanged, true);
            app.MassFlowField.Position = [276 135 38 22];

            % Show the figure after all components are created
            app.UIFigure.Visible = 'on';
        end
    end

    % App creation and deletion
    methods (Access = public)

        % Construct app
        function app = semibatch3

            % Create UIFigure and components
            createComponents(app)

            % Register the app with App Designer
            registerApp(app, app.UIFigure)

            % Execute the startup function
            runStartupFcn(app, @startupFcn)

            if nargout == 0
                clear app
            end
        end

        % Code that executes before app deletion
        function delete(app)

            % Delete UIFigure when app is deleted
            delete(app.UIFigure)
        end
    end
end
```

between.m

```
function results = between(~,value,item)■
results=0;■
if (str2double(value) >= min(item)) && (str2double(value) <= max(item))■
    results=1;■
end■
end■
```

toogleUIC.m

```
function toogleUIC(h, state)■
arguments■
    h matlab.ui.Figure■
    state char■
end■
% toggle the state of the uicontrols in the app■
uics = findall(h);■
for idx = 1:length(uics)■
    uic = uics(idx);    ■
    if isprop(uic,'Enable')&& ~isa(uic,'matlab.ui.control.Label')■
        uic.Enable = state;■
    end■
end■
drawnow; % flush the ui events queue■
end■
```